Common Medical Procedures

A Comprehensive Reference

Invasive and Non-Invasive Clinical Procedures

Medical Reference Team

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To the clinicians, laboratorians, and pathologists
who turn numbers into knowledge and data into diagnoses.

Preface

Modern clinical practice relies heavily on both diagnostic testing and interventional care. This book focuses on the clinical procedures themselves — the invasive and non-invasive interventions performed at the bedside, in the endoscopy suite, in the operating room, and in the interventional radiology lab.

Scope: This book covers medical procedures organized alphabetically by procedure name. Laboratory tests and diagnostic imaging studies are outside the scope of this volume.

Organization: Individual procedures follow a patient-facing format covering: Overview, Alternative Names, Why It Is Done, Preparation, How It Is Performed, Contraindications and Precautions, Risks and Limitations, Aftercare and Recovery, Outcome and Follow-up, When to Seek Medical Care, and References. Principle, History, and Clinical Role may be included when they add useful context.

Medical Reference Team

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List of Abbreviations

Abbreviation	Meaning
ACL	Anterior cruciate ligament
BAL	Bronchoalveolar lavage
BMT	Bone marrow transplant
CABG	Coronary artery bypass grafting
CPR	Cardiopulmonary resuscitation
CSF	Cerebrospinal fluid
CVC	Central venous catheter
D&C	Dilation and curettage
ECT	Electroconvulsive therapy
EGD	Esophagogastroduodenoscopy (upper GI endoscopy)
ERCP	Endoscopic retrograde cholangiopancreatography
ET	Endotracheal
EVAR	Endovascular aneurysm repair
FFP	Fresh frozen plasma
FNA	Fine needle aspiration
I&D	Incision and drainage
ICD	Implantable cardioverter-defibrillator
ICU	Intensive care unit
ILR	Implantable loop recorder
IUD	Intrauterine device
IV	Intravenous
NG	Nasogastric
NPO	Nothing by mouth (nil per os)
OR	Operating room
PACU	Post-anesthesia care unit
PCNL	Percutaneous nephrolithotomy
PD	Peritoneal dialysis
PEG	Percutaneous endoscopic gastrostomy
PICC	Peripherally inserted central catheter
PRBC	Packed red blood cells
SCT	Stem cell transplant
SLNB	Sentinel lymph node biopsy
TAVR	Transcatheter aortic valve replacement
TIPS	Transjugular intrahepatic portosystemic shunt
UAE	Uterine artery embolization
VATS	Video-assisted thoracoscopic surgery
VP	Ventriculoperitoneal

Chapter 1

Acupuncture

Overview

Acupuncture is a component of traditional Chinese medicine in which thin sterile needles are inserted into specific points on the body to modulate symptoms. It is most commonly used in the West for pain relief and for nausea and vomiting.

Alternative Names

Traditional acupuncture; needling therapy; acupressure (needleless variant); auricular acupuncture

Principle

According to traditional theory, acupuncture restores the flow of vital energy (qi) along meridians to rebalance the body. From a biomedical perspective, needle insertion at acupuncture points stimulates local and central nervous system pathways, releasing endogenous opioids, serotonin, and other neuromodulators that reduce pain and influence autonomic function.

History

Acupuncture originated in ancient China more than two millennia ago and was systematized during the Han dynasty, spreading to Europe and the Americas from the 17th century onward.

Why It Is Done

Ordered or recommended for chronic pain conditions such as low back pain, osteoarthritis, and neck pain, and for postoperative and chemotherapy-related nausea and vomiting. It is also used for migraine, fibromyalgia, and as an adjunct in other conditions in which patients seek nonpharmacologic therapy.

Is This a Primary or Secondary Test or Procedure

A secondary or complementary therapy used alongside, or when conventional treatment is insufficient.

How It Is Performed

After a history and examination, the practitioner inserts sterile single-use needles at selected points to a depth of a few millimeters to a few centimeters and may briefly twist or electrically stimulate them. The needles are left in place for roughly 10 to 30 minutes while the patient lies still, and the session ends with removal and disposal of the needles.

Preparation

No special preparation is needed. The patient should eat lightly and wear loose clothing, and should inform the practitioner of bleeding disorders, anticoagulant use, and pregnancy.

Contraindications and Precautions

Acupuncture is generally avoided at sites of skin infection, scar tissue, or lymphedema, and needle use is avoided over a pacemaker site or during electrical stimulation in patients with implanted devices. Patients with severe bleeding disorders or on anticoagulation should be treated cautiously, and those with needle phobia may not tolerate the procedure.

Risks

Serious complications are rare when performed by a qualified practitioner. Mild bruising, soreness, or bleeding at the site may occur, and rare complications include infection, pneumothorax, and organ puncture.

Aftercare and Recovery

The patient may feel relaxed or mildly sore and can resume normal activities immediately. Repeat sessions are scheduled according to the treatment plan, with symptom response reviewed periodically.

Outcome and Follow-up

Improvement is judged by symptom scores, pain reduction, and functional gains over a course of several sessions. If no benefit is seen after a reasonable number of treatments, further therapy is generally not recommended.

Expected Results

Relief or meaningful reduction in the presenting symptom without significant adverse effects.

Follow-up

Symptom diaries and pain scales are used to track response between visits. If the response is incomplete or symptoms persist, the treatment plan is revised or additional evaluation for an underlying cause is pursued.

References

1. MedlinePlus. Acupuncture. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 2

Allergen Immunotherapy (Allergy Shots)

Overview

Allergen immunotherapy, commonly known as allergy shots, is a treatment that gradually desensitizes a patient to specific allergens through repeated injections of gradually increasing doses. It is used to modify the immune response in allergic rhinitis, asthma, and stinging-insect hypersensitivity.

Alternative Names

Allergy shots; allergen desensitization; subcutaneous immunotherapy (SCIT); hyposensitization therapy

Principle

Repeated exposure to escalating allergen doses shifts the immune response from a type 1 hypersensitivity reaction driven by allergen-specific IgE toward tolerance, characterized by IgG-blocking antibodies, reduced mast cell and basophil reactivity, and induction of regulatory T cells. Over time, symptoms to natural allergen exposure are diminished. The treatment is administered in a buildup phase followed by a maintenance phase.

History

Allergen immunotherapy was pioneered by Leonard Noon and John Freeman in 1911, who demonstrated that repeated grass pollen injections reduced allergic symptoms, and it has remained a cornerstone of allergy treatment for over a century.

Why It Is Done

Ordered for patients with allergic rhinitis or allergic asthma who have persistent symptoms despite optimal pharmacologic therapy and environmental control, and for those who cannot tolerate or wish to avoid long-term medication. It is also the treatment of choice for severe stinging-insect venom allergy and may be considered for atopic dermatitis triggered by aeroallergens.

Is This a Primary or Secondary Test or Procedure

A secondary, disease-modifying therapeutic procedure used when allergen avoidance and pharmacotherapy are insufficient.

How It Is Performed

After skin or blood testing identifies the relevant allergens, the patient receives subcutaneous injections of a standardized allergen extract. Doses are increased weekly during a buildup phase of several months and then given at longer intervals during maintenance, typically every 2 to 4 weeks. Injections are always administered in a clinic where anaphylaxis can be treated.

Preparation

The patient should have no acute illness or unstable asthma on the day of injection. Antihistamines may need to be held before skin testing, and the patient should remain in the clinic for at least 30 minutes after each injection for observation.

Contraindications and Precautions

Unstable or poorly controlled asthma, significant cardiac disease, and inability to tolerate severe systemic reactions are relative contraindications. Immunotherapy should not be started during pregnancy, although maintenance may be continued, and beta-blockers are generally avoided because they impair treatment of anaphylaxis.

Risks

Local swelling and redness at the injection site are common. Systemic reactions range from rhinitis, urticaria, and wheezing to rare, potentially life-threatening anaphylaxis, which is why injections require clinic observation.

Aftercare and Recovery

The patient is observed for at least 30 minutes after each injection and is advised on recognizing delayed reactions. Injections continue on a scheduled basis, with symptom diaries and medication use reviewed at each visit to guide dosing.

Outcome and Follow-up

Improvement is measured by reduced symptom scores, decreased medication use, and improved quality of life, typically beginning within months of maintenance therapy. Venom immunotherapy may be confirmed by demonstrating tolerance through repeat sting challenge or by measuring venom-specific antibody levels.

Expected Results

A significant reduction in allergic symptoms and medication requirement without systemic reactions during treatment.

Follow-up

Symptoms, peak expiratory flow, and medication use are monitored at each visit, and skin or serum testing may be repeated to reassess sensitivity. After a successful course of 3 to 5 years, therapy is often discontinued and the patient followed for relapse.

References

1. MedlinePlus. Allergy shots. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 3

Allergy Skin Testing

Overview

Allergy skin testing is an in-vivo diagnostic procedure that observes the skin's reaction to small amounts of suspected allergens. It helps identify triggers for allergic symptoms.

Why It Is Done

It may evaluate allergic rhinitis, asthma, eczema, food allergy, medication allergy, or reactions to insect venom. The results are interpreted together with the person's history because a positive reaction does not always mean that an allergen causes symptoms.

How It Is Performed

For a skin-prick test, drops of allergen extracts are placed on the skin, usually the forearm or back, and the surface is lightly scratched or pricked. For selected situations, a small amount may be injected into the skin for intradermal testing. The clinician measures raised, itchy wheals after a short observation period.

Preparation

Some antihistamines and other medicines can interfere with results and may need to be stopped only as directed by the clinician. Tell the allergy specialist about medicines, asthma control, pregnancy, and previous severe reactions. Do not stop essential medicines without advice.

Risks and Results

Itching and local swelling are common and usually temporary. Rarely, a more serious allergic reaction can occur, so testing should be performed where appropriate treatment is available. Results help guide avoidance, medication choices, or allergen immunotherapy; negative testing does not exclude every type of allergy.

References

1. MedlinePlus. Allergy skin tests. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 4

Amniocentesis

Overview

Amniocentesis is a prenatal test where a small amount of amniotic fluid is removed from the sac surrounding the fetus to test for genetic abnormalities or lung maturity.

Alternative Names

Amniotic fluid test, Genetic amniocentesis

Principle

A thin needle is passed through the maternal abdomen into the amniotic sac under ultrasound guidance and a small volume of amniotic fluid is aspirated. The fluid contains fetal cells and biochemical products that are cultured and analyzed for chromosomes, gene mutations, proteins, and other markers of fetal health.

History

First performed in the late 19th century to relieve polyhydramnios. Evolved in the 1970s with ultrasound guidance and cytogenetic culture.

Why It Is Done

Ordered for pregnant women with an increased risk of fetal genetic abnormalities (due to age or abnormal screening tests) or to check fetal lung maturity.

Is This a Primary or Secondary Test or Procedure

Primary prenatal diagnostic test for chromosomal abnormalities.

How It Is Performed

Under ultrasound guidance, a long, thin needle is inserted through the abdomen into the amniotic sac to withdraw about 20 mL of amniotic fluid.

Preparation

Signed informed consent; NPO is not required; full or empty bladder depending on gestational age.

Contraindications and Precautions

None in high-risk pregnancies; relative includes maternal blood-borne infections (HIV, Hepatitis B/C) due to transmission risk.

Risks

Miscarriage (0.1-0.3%), leakage of amniotic fluid, cramping, or uterine infection.

Aftercare and Recovery

Rest for the remainder of the day; avoid heavy lifting or strenuous activity for 24-48 hours.

Outcome and Follow-up

Cells from the fluid are cultured and analyzed for chromosomal karyotype, microarrays, or specific gene mutations.

Expected Results

Normal fetal karyotype (46,XX or 46,XY); no chromosomal abnormalities detected.

Follow-up

Genetic counseling; prenatal care planning.

References

1. MedlinePlus. Amniocentesis. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 5

Amniotomy (Artificial Rupture of Membranes)

Overview

Amniotomy is the deliberate artificial rupture of the amniotic membranes performed during labor. It is used to augment labor, assess the amniotic fluid, and permit internal fetal monitoring.

Alternative Names

Artificial rupture of membranes, ARM, Breaking the waters

Principle

Using a sterile instrument such as an amnihook, the clinician creates a controlled opening in the amniotic membranes during a vaginal examination. Release of amniotic fluid may strengthen uterine contractions and accelerate labor. The procedure also allows inspection of the fluid and placement of internal fetal monitors.

History

Amniotomy has been performed since the late 19th century and became a common intervention for labor augmentation during the 20th century.

Why It Is Done

Amniotomy is ordered to induce or augment labor when progress is slow, to allow internal fetal heart rate or intrauterine pressure monitoring, and to examine the amniotic fluid for meconium. It may also shorten the first stage of labor.

Is This a Primary or Secondary Test or Procedure

Secondary procedure used adjunctively during labor management rather than as a primary intervention.

How It Is Performed

The clinician performs a sterile vaginal examination to confirm vertex presentation and intact membranes. A sterile instrument is used to rupture the membranes with a controlled release of fluid. Fetal heart rate is monitored before and after the procedure to detect cord compression.

Preparation

Fetal presentation, gestational age, and the absence of active bleeding or infection are confirmed. Maternal blood type and group B streptococcus status should be known, and the fetal heart rate tracing is reviewed before the procedure.

Contraindications and Precautions

Amniotomy is avoided when the presentation is not vertex, or with active genital herpes, placenta previa, or vasa previa. Relative contraindications include an unstable fetal lie and non-reassuring fetal status. Caution is needed when the presenting part is not well applied to the cervix because cord prolapse may follow.

Risks

Umbilical cord prolapse, fetal heart rate decelerations, and ascending infection (chorioamnionitis) are the main risks. The amnihook may rarely cause a scalp injury or cephalohematoma.

Aftercare and Recovery

Labor is closely monitored, and the duration of membrane rupture is tracked to guide the timing of delivery. The mother is observed for fever or other signs of infection, and group B streptococcus prophylaxis is given when indicated.

Outcome and Follow-up

The outcome is assessed by labor progress, cervical dilation, and fetal status rather than by a laboratory value. Effective labor augmentation with a reassuring fetal heart rate tracing indicates success.

Expected Results

Regular contractions with progressive cervical dilation and a reassuring fetal heart rate; amniotic fluid that is clear, without meconium.

Follow-up

Continuous or intermittent fetal monitoring with reassessment of labor progress is continued, and delivery generally occurs within 12–24 hours of rupture. Cesarean delivery is performed if labor fails to progress or fetal compromise develops.

References

1. MedlinePlus. Amniotomy. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 6

Anoscopy

Overview

Anoscopy is an examination of the anus and distal rectum using a short, rigid, lighted instrument called an anoscope. It provides direct visualization of the anal canal and anorectal junction.

Alternative Names

Anal examination, Anal canal inspection, Proctoscopy

Principle

The anoscope is a short, rigid tube with an obturator and a light source that is gently inserted through the anal sphincter. It holds the anal canal open, allowing direct inspection of the mucosa, hemorrhoidal cushions, and anorectal junction. Biopsy forceps and therapeutic instruments can be passed through the scope to sample or treat lesions.

History

Direct speculum examination of the rectum dates to the 19th century. Modern anoscopes with integrated lighting became standard in the mid-20th century.

Why It Is Done

Anoscopy is ordered to evaluate anorectal complaints such as rectal bleeding, anal pain, itching, discharge, or a palpable mass. It is used to diagnose internal and external hemorrhoids, anal fissures, polyps, abscesses, fistulas, and anorectal tumors, and to obtain biopsies of suspicious lesions.

Is This a Primary or Secondary Test or Procedure

Primary test for evaluating symptoms and signs localized to the distal anal canal.

How It Is Performed

The patient lies on the side with knees drawn toward the chest, and the perianal skin is inspected first. The lubricated anoscope is gently inserted through the anus and advanced into the rectum, then withdrawn slowly while the examiner inspects the mucosa. Biopsies or therapeutic interventions can be performed through the scope.

Preparation

No special preparation is required; the patient may be asked to empty the bowel beforehand. A small enema may be used to clear the anal canal. Sedation is usually not needed.

Contraindications and Precautions

Use caution in patients with recent anorectal surgery, severe anal stenosis, or an acutely thrombosed, very painful hemorrhoid. Acute anal fissure with severe spasm may make instrumentation difficult. The examination is generally avoided when anorectal infection is suspected to require imaging first.

Risks

Minor bleeding, transient discomfort, or mild pain after the examination. Injury to the anal mucosa and infection are uncommon.

Aftercare and Recovery

Most patients resume normal activities immediately after the examination. Minor spotting or discomfort may persist for a few hours. Persistent bleeding, severe pain, or fever should be reported promptly.

Outcome and Follow-up

Findings are interpreted by direct visual inspection, with suspicious lesions sampled for histologic confirmation. Results help distinguish conditions such as hemorrhoids, fissures, and malignancy.

Expected Results

Normal appearance of the anal mucosa without lesions, inflammation, masses, or abnormal discharge.

Follow-up

Colonoscopy is recommended if proximal colonic disease is suspected or screening is indicated. Repeat anoscopy may be used to monitor healing or the response of known lesions to treatment.

References

1. MedlinePlus. Anoscopy. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 7

Arterial Line Placement and Arterial Blood Gas Sampling

Overview

Arterial line placement is the insertion of a small catheter into a peripheral or central artery for continuous blood pressure monitoring and repeated blood sampling. Arterial blood gas (ABG) sampling is the withdrawal of arterial blood to assess oxygenation, ventilation, and acid-base status.

Alternative Names

A-line, arterial cannulation, radial artery catheterization, ABG sampling

Principle

An arterial catheter transduces the phasic pressure waveform of arterial blood to a monitor, providing beat-to-beat blood pressure and enabling repeated sampling without repeated needle sticks. ABG analysis measures pH, partial pressures of oxygen and carbon dioxide, and derived parameters such as bicarbonate and base excess, reflecting gas exchange and metabolic status.

History

Arterial cannulation was described in the 20th century and became widespread in intensive care during the mid-1900s. The Allen test was historically used to assess collateral ulnar circulation before radial cannulation, though its predictive value is debated.

Why It Is Done

Ordered for continuous blood pressure monitoring in hemodynamically unstable patients, frequent ABG or lab sampling, and titration of vasoactive drugs or mechanical ventilation. ABG sampling is obtained to diagnose respiratory failure, evaluate acid-base disorders, and guide oxygenation and ventilation targets.

Is This a Primary or Secondary Test or Procedure

A primary monitoring and diagnostic procedure in critically ill patients requiring real-time hemodynamic or blood gas assessment.

How It Is Performed

The artery, most often the radial, is palpated or visualized with ultrasound, and the overlying skin is prepared and anesthetized. The catheter is advanced over a guidewire or directly into the artery until pulsatile blood return is seen, then secured and connected to a pressurized, heparinized transducer system. For isolated ABG sampling, a small volume of blood is drawn anaerobically with a heparinized syringe and processed immediately.

Preparation

Informed consent, assessment of distal perfusion and collateral flow, and sterile technique. The Allen test or ultrasound-guided confirmation of vessel patency is often performed before radial cannulation.

Contraindications and Precautions

Relative contraindications include inadequate collateral circulation, Raynaud phenomenon, peripheral vascular disease, coagulopathy, and local infection. Precautions include avoiding overly aggressive blood sampling, maintaining sterile technique, and prompt removal when the catheter is no longer needed.

Risks

Hematoma, bleeding, arterial thrombosis with distal ischemia, infection, pseudoaneurysm, and accidental intra-arterial injection of medications.

Aftercare and Recovery

The line is kept patent with a continuous flush solution and secured to prevent dislodgement; the site is inspected regularly for ischemia or infection. The catheter is removed with firm pressure when monitoring is no longer required, and the site is observed for bleeding.

Outcome and Follow-up

ABG results show oxygenation (PaO2), ventilation (PaCO2), and pH; low PaO2 with normal PaCO2 suggests hypoxemic respiratory failure, while elevated PaCO2 with acidosis indicates ventilatory failure. Continuous arterial pressure readings guide vasopressor and fluid therapy in shock states.

Expected Results

Normal ABG values include a pH of 7.35–7.45, PaO2 of 80–100 mm Hg on room air, and PaCO2 of 35–45 mm Hg, with mean arterial pressure near 70–100 mm Hg.

Follow-up

Repeat ABG and other labs are drawn through the line as indicated by the clinical course. When the line is removed, assessment of distal perfusion and, if clinically indicated, pulse oximetry or vascular ultrasound follow.

References

1. MedlinePlus. Blood gases. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 8

Arthrocentesis

Overview

Arthrocentesis is the aspiration of synovial fluid from a joint using a needle, performed for diagnosis, symptom relief, and intra-articular therapy.

Alternative Names

Joint aspiration; joint tap; knee tap; intra-articular injection

Principle

A needle is passed through the skin and joint capsule into the synovial space. Fluid is withdrawn for analysis (cell count, crystals, culture), and medication such as corticosteroid or local anesthetic can be instilled.

History

Joint aspiration has been practiced since the 19th century for diagnosis of joint effusions. Aspiration became a cornerstone of gout, septic arthritis, and inflammatory arthritis diagnosis.

Why It Is Done

Performed to evaluate an acute monoarticular effusion (to exclude septic arthritis), diagnose crystal arthropathies, relieve large symptomatic effusions, inject corticosteroids or viscosupplements, and monitor therapy in inflammatory arthritis.

Is This a Primary or Secondary Test or Procedure

Primary diagnostic and therapeutic procedure for joint effusions; a screening and monitoring tool in rheumatic disease.

How It Is Performed

The joint is identified and prepped sterilely, the skin and capsule may be anesthetized with lidocaine, and a needle (often 18-20 gauge) is advanced into the joint space. Fluid is aspirated, a portion is sent for analysis, and medication may be injected. The knee is the most common site.

Preparation

Usually none beyond a targeted history and examination. Aspirin and anticoagulants are generally continued, but the physician assesses bleeding risk; septic joint precautions apply.

Contraindications and Precautions

Absolute contraindications: overlying skin infection or cellulitis. Relative precautions: severe coagulopathy or anticoagulation, and a prosthetic joint (higher concern for introduced infection).

Risks

Pain, bleeding, infection (rare with sterile technique), damage to cartilage or nerves, and post-injection flare.

Aftercare and Recovery

Pressure dressing, ice, and rest for 24-48 hours. Analgesia as needed. Results guide further workup.

Outcome and Follow-up

Cloudy fluid suggests infection or inflammation; high white count with neutrophils suggests septic arthritis; monosodium urate crystals confirm gout; calcium pyrophosphate crystals confirm pseudogout; and a positive culture confirms septic arthritis.

Expected Results

Clear or slightly yellow, low-viscosity fluid with low white blood cell count (usually under $200/\mu L$) and no crystals or organisms.

Follow-up

Culture and sensitivity if infection is suspected; repeat aspiration or imaging may be needed. Response to intra-articular injection is assessed clinically.

References

1. MedlinePlus. Joint aspiration. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 9

Arthroscopy

Overview

Arthroscopy is a minimally invasive surgical procedure on a joint in which an examination and sometimes treatment of damage is performed using an arthroscope.

Alternative Names

Joint scope, Keyhole joint surgery

Principle

A narrow fiberoptic scope with a camera is inserted into a joint through a small incision and the joint is distended with fluid. The interior is viewed on a monitor while instruments passed through additional small ports diagnose and treat damage, minimizing trauma to surrounding structures.

History

Kenji Takagi performed the first arthroscopy on a cadaver knee in 1918. Masaki Watanabe developed modern arthroscopic instruments and techniques in the 1950s.

Why It Is Done

Ordered to diagnose and treat joint pain, swelling, or instability. Commonly performed on knees, shoulders, elbows, ankles, hips, and wrists.

Is This a Primary or Secondary Test or Procedure

Primary surgical standard for joint-preserving procedures (e.g., meniscus repair, ACL reconstruction).

How It Is Performed

Performed under general, regional, or local anesthesia. Small incisions (portals) are made around the joint. The scope and surgical instruments are inserted to inspect and repair the joint.

Preparation

Fast for 8 hours before surgery. Arrange for a ride home and assistance at home.

Contraindications and Precautions

Active skin infection over the joint, or severe joint stiffness (ankylosis) preventing scope maneuverability.

Risks

Joint infection (septic arthritis), hemarthrosis (bleeding in the joint), nerve damage, or blood clots (DVT).

Aftercare and Recovery

Keep incisions clean and dry. Apply ice and elevate the joint to reduce swelling. Follow weight-bearing instructions.

Outcome and Follow-up

Visualizes cartilage tears, ligament tears, loose bodies, or synovitis. Surgical repair is performed during the same procedure.

Expected Results

Intact articular cartilage; normal ligaments and meniscus; no loose bodies or inflammation.

Follow-up

Physical therapy, wound check in 7-10 days, and gradual return to activity.

References

1. MedlinePlus. Arthroscopy. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 10

Balloon Enteroscopy

Overview

Balloon enteroscopy is an endoscopic examination of the small intestine using an overtube and one or two balloons to advance the scope beyond the reach of standard endoscopy.

Why It Is Done

It evaluates obscure gastrointestinal bleeding, suspected small-bowel tumors, Crohn disease, abnormal imaging, and lesions requiring biopsy or treatment.

How It Is Performed

Under sedation or anesthesia, the scope is introduced through the mouth or anus. Balloon inflation anchors the bowel while the scope and overtube are advanced. Biopsy, polyp removal, dilation, or bleeding treatment can be performed.

Risks and Recovery

Pancreatitis, bleeding, perforation, aspiration, infection, and sedation complications are possible. Patients are observed until alert and should follow instructions about eating, driving, and warning symptoms.

References

1. MedlinePlus. Enteroscopy. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 11

Balloon Valvuloplasty

Overview

Balloon valvuloplasty is a catheter-based procedure that widens a narrowed heart valve by inflating a balloon across the stenotic orifice. It is most commonly used for mitral and aortic valve stenosis.

Alternative Names

Balloon valvotomy, percutaneous balloon valvuloplasty, mitral balloon commissurotomy, aortic balloon valvuloplasty

Principle

A balloon catheter is advanced across the stenotic valve and inflated to high pressure, splitting fused commissures and stretching the valve leaflets. This increases the effective orifice area and reduces the transvalvular pressure gradient. The result improves forward blood flow and relieves symptoms of obstruction.

History

Rubio and Limón Lason performed the first balloon valvuloplasty for pulmonic stenosis in 1953, and Inoue developed the modern mitral balloon technique in 1982. It became an alternative to open surgical commissurotomy in selected patients.

Why It Is Done

Performed for symptomatic severe mitral stenosis with favorable valve morphology, and for severe aortic stenosis in patients who are poor surgical candidates, such as bridging therapy before surgical or transcatheter valve replacement, or for critical aortic stenosis in pregnancy. In aortic stenosis it provides only temporary symptomatic relief.

Is This a Primary or Secondary Test or Procedure

Primary treatment for selected patients with mitral stenosis; a palliative or bridge procedure for aortic stenosis in those ineligible for valve replacement.

How It Is Performed

Performed in the cardiac catheterization laboratory under conscious sedation or general anesthesia. Access is obtained via the femoral vein (mitral, using transseptal puncture) or femoral artery (aortic). Under fluoroscopic and echocardiographic guidance, the balloon is positioned across the valve, inflated, and withdrawn.

Preparation

Transthoracic and often transesophageal echocardiography for valve scoring and left atrial thrombus exclusion. Blood typing and crossmatch, and withholding of anticoagulants as directed.

Contraindications and Precautions

Contraindicated with severe valvular regurgitation, left atrial thrombus, or severe mitral subvalvular and leaflet calcification (for mitral). Precautions include moderate-to-severe aortic regurgitation and coexisting coronary artery disease (for aortic).

Risks

Valvular regurgitation, systemic embolism from calcium or thrombus, vascular access complications, cardiac perforation, and arrhythmias. Restenosis is common, particularly after aortic balloon valvuloplasty.

Aftercare and Recovery

Patients are monitored in a cardiac observation unit with echocardiography to assess valve area and regurgitation. Most are discharged within 24-48 hours and may resume normal activities shortly thereafter.

Outcome and Follow-up

Success is defined by an increase in valve area of at least 50% and a reduction in the mean gradient without significant regurgitation. Residual symptoms or high gradients after aortic valvuloplasty predict early recurrence.

Expected Results

An effective valve orifice area near normal (mitral >2.0 cm^2; aortic >1.5 cm^2) with a low mean gradient and no more than mild regurgitation.

Follow-up

Serial echocardiography to monitor valve area, gradient, and restenosis. Definitive valve replacement should be planned for patients with aortic stenosis, and repeat valvuloplasty or surgery may be needed for mitral restenosis.

References

1. MedlinePlus. Balloon valvuloplasty. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 12

Barium Enema

Overview

A barium enema, also called a lower gastrointestinal (GI) series, is a fluoroscopic X-ray examination of the colon and rectum. Barium contrast outlines the inside of the large intestine so that its shape and lining can be evaluated.

Alternative Names

Lower GI series, lower GI tract radiography, double-contrast barium enema

Principle

Barium sulfate is introduced through the rectum and coats the colonic lining. X-ray images are obtained as the patient changes position; in a double-contrast study, air or carbon dioxide is also introduced to expand the colon and improve visualization of the mucosa.

History

Barium examinations of the gastrointestinal tract became established after the introduction of radiopaque contrast in the early twentieth century. Modern fluoroscopy and double-contrast techniques provide more detailed images, although colonoscopy and CT-based studies are preferred for many indications.

Why It Is Done

It may be ordered to evaluate changes in bowel habits, abdominal pain, rectal bleeding, suspected narrowing or obstruction, diverticular disease, inflammatory changes, or an abnormality that cannot be fully assessed by another study.

Is This a Primary or Secondary Test or Procedure

It is a diagnostic imaging test and is generally a secondary or alternative examination when colonoscopy is incomplete, unsuitable, unavailable, or not required for the clinical question. It does not allow tissue biopsy or polyp removal.

How It Is Performed

You lie on an X-ray table, usually on your left side. A lubricated enema tube is gently inserted into the rectum, and barium is allowed to flow into the colon. The radiologist or technologist takes fluoroscopic images while you change position. The tube is then removed and you may be asked to expel the contrast in a bathroom before additional images are obtained.

Preparation

The colon must be empty. Follow the facility's instructions for a low-residue or clear-liquid diet, laxatives, and possibly cleansing enemas. Tell the radiology team if you may be pregnant, have severe abdominal pain, suspected bowel obstruction or perforation, recent bowel surgery, or difficulty controlling your bowel movements. Do not change prescribed medicines unless instructed.

Contraindications and Precautions

Avoid a barium enema when bowel perforation is suspected or when there is severe acute colitis, toxic megacolon, or a high risk of perforation. A water-soluble iodinated contrast agent may be chosen instead when perforation is a concern. Pregnancy requires individualized assessment because the examination uses ionizing radiation.

Risks

Temporary cramping, bloating, and an urgent need to evacuate are common. Rare complications include contrast impaction, bowel perforation, infection, or inflammation if barium escapes through an unrecognized hole. The examination also involves a small amount of ionizing radiation. Allergic reactions to barium are extremely uncommon.

Aftercare and Recovery

You can usually return home soon after the examination. Drink fluids as advised and follow instructions about resuming food and medicines. Stools may be pale or white for a short time. Contact a clinician promptly for severe or persistent abdominal pain, marked swelling, fever, vomiting, dizziness, or inability to pass stool or gas.

Outcome and Follow-up

The images are reviewed for the contour and caliber of the colon, the appearance of the mucosal lining, diverticula, strictures, obstruction, masses, or other abnormalities. An abnormal result may require colonoscopy, CT, biopsy, or another test for confirmation.

Expected Results

The colon and rectum have normal caliber and position, with an intact, smooth mucosal outline and no evidence of mass, obstruction, significant inflammation, or diverticular abnormality.

Follow-up

Follow-up depends on the findings and the reason for testing. Colonoscopy is commonly used when direct inspection or biopsy is needed; CT or other imaging may be recommended for suspected complications or disease outside the bowel wall.

References

1. MedlinePlus. Barium enema. U.S. National Library of Medicine; reviewed January 1, 2025.

2. American College of Radiology and Radiological Society of North America. Lower GI X-ray (Barium Enema). RadiologyInfo.org; reviewed April 22, 2025.

Chapter 13

Basic Airway Management

Overview

Basic airway management is the use of simple maneuvers and devices to open and maintain a patent airway in an unconscious or obstructed patient. It includes head-tilt and jaw-thrust positioning, oropharyngeal and nasopharyngeal airway insertion, and bag-valve-mask ventilation.

Alternative Names

Airway opening maneuvers; basic life support airway; BVM ventilation; oropharyngeal airway insertion

Principle

In the unconscious patient the tongue and soft tissues fall backward and obstruct the pharynx. Head-tilt and chin-lift or jaw-thrust lift the tongue forward, and oropharyngeal or nasopharyngeal airways hold the pharynx open. Bag-valve-mask ventilation then delivers breaths through the maintained airway.

History

Airway management evolved from the anesthesia techniques of the late 19th century, and the bag-valve-mask device and standardized airway adjuncts became central to resuscitation and emergency care in the mid-20th century.

Why It Is Done

Performed in any patient with an obstructed airway, apnea, or impaired consciousness, during cardiopulmonary resuscitation, and as the first step before advanced airway placement such as endotracheal intubation.

Is This a Primary or Secondary Test or Procedure

Primary lifesaving procedure for airway obstruction and a bridge to advanced airway management.

How It Is Performed

The patient is positioned supine, and the head-tilt/chin-lift (or jaw-thrust if cervical injury is suspected) is applied to open the airway. Secretions and foreign bodies are cleared, an appropriately sized airway adjunct is inserted, and ventilation is delivered with a bag-valve-mask at an appropriate rate and volume, checking for chest rise.

Preparation

No specific preparation beyond suction, airway adjuncts of several sizes, a bag-valve-mask with reservoir, and supplementary oxygen. Cervical spine protection is maintained when trauma is suspected.

Contraindications and Precautions

Oropharyngeal airways are avoided in patients with an intact gag reflex, and nasopharyngeal airways are avoided with facial fractures or suspected basilar skull fracture. Precautions include correct sizing and reassessment of effectiveness.

Risks

Gastric insufflation with resultant aspiration, inadequate ventilation, oropharyngeal injury, and nasal bleeding from nasopharyngeal insertion.

Aftercare and Recovery

The patient is continuously monitored with pulse oximetry, capnography, and clinical assessment, and ventilation is continued until spontaneous breathing or definitive airway management is established.

Outcome and Follow-up

Effective airway management is confirmed by chest rise, adequate oxygen saturation, and end-tidal carbon dioxide on capnography. Failure to ventilate prompts repositioning, adjunct adjustment, or advanced airway placement.

Expected Results

A patent airway with visible chest expansion, audible breath sounds, and stable or improving oxygenation.

Follow-up

Patients who remain unconscious or fail basic airway management proceed to endotracheal intubation, and those who recover are observed for aspiration, airway injury, and the underlying cause.

References

1. MedlinePlus. Basic life support and CPR. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 14

Biliary Drainage

Overview

Biliary drainage relieves obstruction of the bile ducts by placing a catheter or stent through an endoscopic, percutaneous, or surgical approach.

Why It Is Done

It may be required for obstructive jaundice, cholangitis, gallstones, postoperative injury, strictures, or tumors that block bile flow.

How It Is Performed

During ERCP, a stent may be passed through the papilla into the bile duct. If endoscopic access is not possible, interventional radiology can place a transhepatic drain through the skin and liver. The approach depends on anatomy and cause.

Risks and Follow-up

Pancreatitis, bleeding, infection, perforation, catheter blockage, and electrolyte or fluid losses are possible. Drain and stent care, laboratory monitoring, and planned exchange or removal are required.

References

1. MedlinePlus. Biliary Drainage. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 15

Blepharoplasty

Overview

Blepharoplasty is surgery of the eyelids that removes or repositions excess skin, muscle, or fat to improve eyelid function, visual-field obstruction, or appearance.

Why It Is Done

It may be performed for upper-eyelid skin that obstructs vision, lower-eyelid bags, eyelid asymmetry, or cosmetic concerns after appropriate eye evaluation.

How It Is Performed

Through carefully placed eyelid incisions, the surgeon removes or redistributes tissue and closes the skin. Local anesthesia with sedation or general anesthesia may be used.

Risks and Recovery

Bruising, swelling, dry eyes, asymmetry, infection, bleeding, scarring, difficulty closing the eyes, and rare vision-threatening complications are possible. Cold compresses, head elevation, and follow-up protect healing.

References

1. MedlinePlus. Eyelid Surgery. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 16

Blood Culture Collection

Overview

Blood culture collection is the sterile withdrawal of blood for culture to detect bacteremia, fungemia, and bloodstream infection. Specimens may be obtained from a peripheral venepuncture or from an existing central venous catheter.

Alternative Names

Blood cultures; bacteremia testing; blood sampling for culture; central line blood culture

Principle

A small volume of blood is inoculated into culture bottles containing media that support bacterial and fungal growth. Because blood is normally sterile, any growth indicates a true infection once skin contaminants are excluded, and the organism's antibiotic susceptibility can guide therapy.

History

The blood culture was introduced in the late 19th century and became a cornerstone of the diagnosis of sepsis. Modern automated culture systems continuously monitor bottles for growth and have greatly improved speed and yield.

Why It Is Done

Ordered for suspected sepsis, fever of unknown origin, endocarditis, pneumonia with bacteremia, and monitoring of patients with central venous catheters at risk of catheter-related bloodstream infection.

Is This a Primary or Secondary Test or Procedure

Primary diagnostic procedure for bloodstream infection and a key driver of antimicrobial therapy.

How It Is Performed

The skin is disinfected with chlorhexidine and allowed to dry. Blood is drawn from a peripheral vein or through a central catheter after disinfecting the access hub, and each bottle is filled with the recommended volume. Two to three sets are typically collected from different sites, and the bottles are labelled with the site and time.

Preparation

No patient preparation is required, but cultures should be drawn before antibiotics when possible, using strict aseptic technique. Ideal volumes are collected from separate sites rather than a single draw.

Contraindications and Precautions

There are no absolute contraindications; precautions include drawing from a central catheter only when peripheral access is not possible, and avoiding cultures drawn through an obviously infected or freshly accessed line.

Risks

Pain and bruising at the venepuncture site, bleeding, and contamination of the sample with skin flora that produces false-positive results.

Aftercare and Recovery

Bottles are transported promptly to the laboratory and incubated in automated systems, typically for up to five days. Positive growth triggers Gram stain and identification, with susceptibility testing reported to guide therapy.

Outcome and Follow-up

A positive culture identifies the causative organism, and susceptibility results guide targeted antibiotics. Growth in multiple sets from different sites strongly supports true bacteremia, whereas growth in a single set may represent contamination.

Expected Results

No growth of organisms from the blood cultures after the full incubation period.

Follow-up

Patients with positive cultures may need repeat cultures to confirm clearance, echocardiography for suspected endocarditis, and imaging or source control when a focus of infection is identified.

References

1. MedlinePlus. Blood culture. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 17

Blood Transfusion

Overview

Blood transfusion is the intravenous administration of whole blood or a blood component, such as packed red blood cells, platelets, plasma, or cryoprecipitate, to replace deficient blood elements. It is a common supportive therapy for anemia, hemorrhage, and coagulation disorders.

Alternative Names

Red blood cell transfusion, packed red cell transfusion, platelet transfusion, fresh frozen plasma transfusion, cryoprecipitate transfusion

Principle

Transfusion replaces circulating blood volume and deficient cellular or protein components, restoring oxygen-carrying capacity, hemostasis, and oncotic pressure. Compatibility is ensured by ABO and Rh typing with crossmatching, so that transfused cells and proteins are not destroyed by recipient antibodies. Hemovigilance systems monitor for transfusion reactions and infectious complications.

History

The first human blood transfusion was recorded in the 17th century, and safe, compatible transfusion became practical after Karl Landsteiner described the ABO blood groups in 1900. Component therapy and modern blood banking evolved during the 20th century.

Why It Is Done

Ordered for symptomatic anemia, acute hemorrhage with hemodynamic instability, severe thrombocytopenia or platelet dysfunction with bleeding, coagulation factor deficiencies, and exchange transfusion in severe neonatal jaundice or sickle cell crisis. It is also used to support patients undergoing surgery, chemotherapy, or bone marrow transplantation.

Is This a Primary or Secondary Test or Procedure

A primary therapeutic procedure when specific blood components are required to treat acute or life-threatening deficiency.

How It Is Performed

Blood is collected into bags containing anticoagulant-preservative solutions and processed into components. After type and crossmatch verification, the unit is administered through a dedicated intravenous line with a filter, usually over one to four hours. Vital signs are monitored before, during, and shortly after the transfusion, which is stopped immediately if a reaction is suspected.

Preparation

Informed consent, ABO and Rh typing, antibody screening, and crossmatch of donor and recipient blood. Premedication with antipyretics is sometimes considered for patients with prior febrile reactions.

Contraindications and Precautions

Transfusion is avoided when alternative treatments are effective, such as iron, folate, or vitamin B12 therapy for nutritional anemia, and in patients with antibody-mediated incompatibility. Precautions include strict patient identification to prevent mistransfusion, leukoreduction and irradiation for immunocompromised recipients, and caution in heart failure or volume overload states.

Risks

Febrile nonhemolytic and allergic reactions are most common; more serious risks include acute and delayed hemolytic reactions, transfusion-related acute lung injury (TRALI), transfusion-associated circulatory overload (TACO), and transmission of infectious agents.

Aftercare and Recovery

The patient is observed for several hours for signs of a transfusion reaction, and follow-up complete blood counts are checked to document response. Recurring needs may prompt a search for the cause of ongoing blood loss or marrow failure.

Outcome and Follow-up

The goal is clinical improvement, reflected by a rising hemoglobin, platelet count, or correction of coagulopathy to a target range. Results that fail to improve suggest ongoing losses, dilution, or refractoriness to transfusion.

Expected Results

Hemoglobin concentration, platelet count, and coagulation parameters improve toward age- and sex-appropriate reference ranges after transfusion.

Follow-up

Repeat blood counts and coagulation studies guide further transfusions. Patients with repeated reactions or platelet refractoriness may require antibody testing and HLA-matched components.

References

1. MedlinePlus. Blood transfusion. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 18

Bone Marrow Biopsy

Overview

Bone marrow biopsy and aspiration are procedures that obtain bone marrow tissue and cells, usually from the posterior iliac crest, to evaluate blood-forming disorders.

Alternative Names

Bone marrow aspirate; trephine biopsy; bone marrow examination

Principle

A needle is passed through the cortex into the medullary cavity. Aspiration draws liquid marrow for smears and flow cytometry; a trephine needle captures a solid core for histology, preserving architecture for staging and fibrosis assessment.

History

Bone marrow aspiration was introduced in the 1920s and trephine biopsy in the 1950s. They became standard for diagnosing hematologic malignancy and marrow failure.

Why It Is Done

Performed for unexplained cytopenias, suspected leukemia, lymphoma, myeloma, or myelodysplasia, to stage hematologic malignancy, to evaluate iron stores and metastatic disease, and to monitor response to therapy.

Is This a Primary or Secondary Test or Procedure

Primary diagnostic and staging procedure for many hematologic disorders.

How It Is Performed

Performed at the bedside or in clinic with local anesthesia. The patient lies prone or lateral. A Jamshidi-type needle is advanced into the posterior superior iliac spine, aspiration is performed, then a core is obtained with a twisting motion. Specimens are sent for smears, flow cytometry, cytogenetics, and histology.

Preparation

Usually none. Informing the clinician about anticoagulants, aspirin, and bleeding disorders is important. Local anesthetic is infiltrated at the site.

Contraindications and Precautions

Severe coagulopathy or profound thrombocytopenia increases bleeding risk and may require correction. Osteomyelitis or cellulitis over the site is a contraindication. The sternum is used only for aspiration, never trephine.

Risks

Bleeding, infection, and pain are uncommon. Sternal injury is avoided by using the iliac crest. Hematoma may occur at the site.

Aftercare and Recovery

Pressure is held over the site for 5-10 minutes, and the patient rests briefly. Mild soreness is treated with acetaminophen or ice.

Outcome and Follow-up

Results reveal cellularity, myeloid-to-erythroid ratio, blast percentage, the presence of malignant cells, fibrosis, and iron stores. Flow cytometry and cytogenetics add immunophenotypic and karyotypic detail.

Expected Results

A normocellular marrow (approximately 30-70 percent by age) with normal trilineage maturation and no abnormal infiltrate or fibrosis.

Follow-up

Results guide further workup such as cytogenetics, FISH, or molecular panels, and serial biopsies may monitor treatment response.

References

1. MedlinePlus. Bone marrow biopsy. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 19

Botulinum Toxin Injection

Overview

Botulinum toxin injection is the delivery of botulinum neurotoxin into specific muscles or glands to treat conditions of excessive muscle contraction, glandular hypersecretion, pain, or overactive smooth muscle. The toxin produces a temporary, dose-dependent weakening of the injected target.

Alternative Names

Botox injection; botulinum toxin type A; botulinum toxin type B; BTX-A; BTX-B

Principle

Botulinum toxin blocks the release of acetylcholine at the neuromuscular junction by cleaving SNARE proteins, producing localized muscle paralysis. The effect develops over several days and wears off as new nerve terminals form over three to six months.

History

Botulinum toxin type A was approved for strabismus and blepharospasm in the 1980s, and its therapeutic and cosmetic applications have since expanded dramatically.

Why It Is Done

Ordered for blepharospasm, cervical dystonia, spasticity from stroke or cerebral palsy, chronic migraine, hyperhidrosis, overactive bladder, strabismus, and other disorders of excessive muscle tone, movement, or secretion.

Is This a Primary or Secondary Test or Procedure

A primary treatment for several focal dystonias and spasticity; it is adjunctive in conditions such as migraine and spasticity where oral therapies are also used.

How It Is Performed

The target muscles are identified by palpation, electromyography, or ultrasound guidance. A fine needle delivers small volumes of reconstituted toxin into the selected sites, with dosing adjusted to the muscle size and the intended effect.

Preparation

Informed consent, review of concurrent aminoglycoside or neuromuscular-blocking medications, and identification of the injection sites are needed. No fasting is required for most indications.

Contraindications and Precautions

Known hypersensitivity to botulinum toxin or albumin, infection at the injection site, and concurrent use of aminoglycoside antibiotics that may potentiate weakness are contraindications. Precautions include pregnancy and neuromuscular disorders such as myasthenia gravis, and care to avoid unintended spread to adjacent muscles.

Risks

Risks include local pain, bruising, weakness of nearby muscles, dysphagia or respiratory compromise when treating cervical or laryngeal muscles, and rarely distant spread of toxin.

Aftercare and Recovery

The full effect develops over several days, and functional improvement is often assessed at two to four weeks. Relief lasts approximately three to six months, after which repeat injection can be scheduled.

Outcome and Follow-up

Improvement in spasm, pain, or hyperhidrosis indicates a response, while insufficient benefit may prompt dose adjustment, change of target site, or switching between type A and type B preparations. Loss of effect over time may reflect antibody-mediated resistance.

Expected Results

Marked reduction of the targeted involuntary contraction, pain, or sweating with preserved function of unaffected muscles indicates a successful treatment outcome.

Follow-up

Outcomes are reviewed at follow-up visits and the dose is titrated for subsequent cycles. Physical or occupational therapy, oral spasmolytics, or surgical options may be added if response is inadequate.

References

1. MedlinePlus. Botulinum toxin injection. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 20

Brachytherapy

Overview

Brachytherapy is internal radiation therapy in which a radioactive source is placed inside or next to a tumor for a planned period.

Why It Is Done

It treats selected cancers, including prostate, cervical, uterine, breast, skin, and other tumors, either alone or with external-beam radiation or surgery.

How It Is Performed

Applicators, needles, or small seeds are positioned using imaging and treatment planning. The source may be temporary and removed after treatment or permanently implanted at a low dose rate.

Risks and Follow-up

Side effects depend on the treatment site and include bleeding, infection, urinary or bowel symptoms, fatigue, and damage to nearby tissue. Radiation-oncology follow-up monitors tumor response and late effects.

References

1. National Cancer Institute. Brachytherapy. 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 21

Breast Augmentation

Overview

Breast augmentation increases breast size or restores volume using implants or, in selected cases, transferred fat.

Why It Is Done

It is an elective reconstructive or cosmetic procedure for volume loss, asymmetry, or patient preference after informed counseling.

How It Is Performed

Under anesthesia, an implant is placed behind the breast tissue or chest muscle through an incision. Implant type, pocket, incision, and technique are selected after shared decision-making.

Risks and Follow-up

Pain, infection, bleeding, altered sensation, capsular contracture, implant rupture, asymmetry, reoperation, and rare implant-associated malignancy are possible. Implants are not lifetime devices and require ongoing clinical follow-up.

References

1. U.S. Food and Drug Administration. Breast Implants. 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 22

Breast Biopsy

Overview

Breast biopsy is the removal of a small sample of breast tissue for microscopic examination to determine whether a lump or suspicious imaging finding is benign or malignant. It is the definitive diagnostic test for breast lesions.

Alternative Names

Core needle biopsy; vacuum-assisted biopsy; stereotactic breast biopsy; ultrasound-guided breast biopsy; excisional breast biopsy

Principle

A needle or surgical instrument obtains breast tissue from a suspicious area, which is then examined by a pathologist. Sampling may be guided by ultrasound, mammography (stereotaxis), or MRI to target the abnormality precisely, and the presence of cancer cells in the sample establishes the diagnosis.

History

Breast biopsy developed with the rise of surgical pathology in the late 19th and early 20th centuries, and image-guided core needle biopsy became the standard approach in the 1990s, replacing most open surgical biopsies.

Why It Is Done

Ordered for a palpable breast mass, suspicious calcifications or masses on mammography, abnormal ultrasound or MRI findings, nipple discharge, or skin changes that raise concern for malignancy.

Is This a Primary or Secondary Test or Procedure

Primary diagnostic procedure for breast lesions and essential to confirming or excluding breast cancer.

How It Is Performed

The lesion is localized by palpation or imaging. After local anesthesia, a core needle or vacuum-assisted device is inserted through a small skin nick and multiple samples are taken from the lesion, often under ultrasound or stereotactic guidance. A small metal clip may be placed to mark the biopsy site.

Preparation

Informed consent, review of anticoagulant use, and avoidance of aspirin or blood thinners before the procedure. No fasting is required for core needle biopsy.

Contraindications and Precautions

Precautions include bleeding disorders or anticoagulation, and the procedure is deferred in patients with active breast infection. The patient should be able to remain still during image-guided sampling.

Risks

Pain, bruising and hematoma, bleeding, infection, and rarely pneumothorax when sampling near the chest wall. Core needle biopsy has a small rate of false-negative results.

Aftercare and Recovery

The site is bandaged and the patient applies ice and avoids heavy lifting for a day or two. The specimens are sent for histopathologic analysis, with results typically available within several days.

Outcome and Follow-up

The pathology report classifies the lesion as benign, atypical, or malignant, and reports tumor type, grade, and hormone receptor status when cancer is found. A benign result that does not match the imaging appearance may require repeat biopsy or surgical excision.

Expected Results

Benign findings such as fibroadenoma, fibrocystic change, or benign ductal tissue with no malignant cells.

Follow-up

Patients with benign results may require imaging follow-up, while those with cancer proceed to staging and treatment planning, including surgery, radiation, and systemic therapy.

References

1. MedlinePlus. Breast biopsy. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 23

Bronchoalveolar Lavage

Overview

Bronchoalveolar lavage (BAL) is a procedure performed during bronchoscopy in which sterile saline is instilled into a segment of the lung and then suctioned back out. This washes the alveoli and collects cells, microorganisms, and other material for laboratory analysis.

Alternative Names

BAL, bronchoalveolar lavage, lung lavage, bronchoscopic lavage

Principle

BAL fluid is analyzed for total cell count and differential (macrophages, lymphocytes, neutrophils, eosinophils) by cytocentrifugation and Wright-Giemsa staining. Flow cytometry subsets lymphocytes. Microbiology includes Gram stain, fungal and AFB smears, cultures, and PCR. Cytology screens for malignant cells.

History

Bronchoalveolar lavage was developed in the 1970s with the advent of flexible bronchoscopy. It enabled sampling of the lower respiratory tract for diagnosing infections, interstitial lung disease, and malignancy.

Why It Is Done

Your doctor orders this test when they need to examine the cells and organisms deep within your lungs. It is performed during a bronchoscopy, where a thin camera is passed through your mouth or nose into your lungs. It answers the question: what is causing your lung disease — an infection, an autoimmune condition, or an occupational exposure?

Is This a Primary or Secondary Test or Procedure

Secondary. It is performed when less invasive tests like sputum samples or imaging have not provided a clear answer.

How It Is Performed

During bronchoscopy, a thin flexible tube with a camera is passed through your mouth or nose, down your throat, and into your lungs. Saline is injected into a small section of your lung and then suctioned back out. This washes the alveoli (tiny air sacs) and collects cells, organisms, and other material for analysis. You receive sedation during the procedure, so you should not feel pain.

Preparation

You must not eat or drink for at least six hours before the procedure. You must arrange for someone to drive you home afterward. Your doctor will review your medications and may ask you to stop blood thinners temporarily.

Contraindications and Precautions

This procedure has the following contraindications and precautions:

- **Relative contraindications**: Severe hypoxemia (PaO2 below 60 mmHg or significant oxygen requirement), unstable asthma with active bronchospasm, uncontrolled coagulopathy or severe thrombocytopenia, severe pulmonary hypertension, and hemodynamic instability.
- **Precautions**: Correct coagulopathy when feasible before the procedure; use continuous pulse oximetry and supplemental oxygen during and after the procedure; limit the volume of saline instilled in patients with marginal respiratory reserve; hold anticoagulants if clinically safe.
- **Special populations**: Children require smaller lavage volumes and shorter procedure time; pregnant patients should undergo the procedure only when the diagnostic benefit outweighs fetal risk.

Risks

Transient hypoxemia, localized fever, minor bleeding, or bronchospasm.

Aftercare and Recovery

Vital signs are monitored for 1-2 hours; avoid eating or drinking until your gag reflex returns.

Outcome and Follow-up

The lavage fluid is analyzed for cell count and differential, microbiology, and cytology. A lymphocytic pattern (often with a low CD4:CD8 ratio) suggests sarcoidosis or hypersensitivity pneumonitis, a neutrophilic predominance suggests bacterial infection, and an eosinophilic pattern suggests eosinophilic lung disease or drug reaction. Positive cultures or PCR identify the causative organism.

Expected Results

Normal: No predominant pathogen. Mixed respiratory flora. WBC differential varies by lavage site. No malignant cells.

Population	Reference Range
Adult Male	No predominant pathogen; mixed flora; no malignant cells
Adult Female	Same as male
Children	Same as adult
Elderly	Same as adult
Pregnancy	Same as non-pregnant

Follow-up

Follow-up depends on the findings:

- **Antimicrobial therapy:** When infection is identified, targeted treatment is guided by culture and sensitivity results.
- **Repeat bronchoscopy:** May be repeated if the initial lavage is non-diagnostic or if symptoms persist.
- **Surgical lung biopsy:** Considered when the BAL remains non-diagnostic and interstitial lung disease is suspected.
- **Specialist consultation:** Pulmonology referral for integrated interpretation and management planning.

References

1. MedlinePlus. Bronchoalveolar Lavage. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 24

Bronchoscopy

Overview

Bronchoscopy is a procedure that examines the airways (trachea and bronchi) with a thin, flexible tube passed through the nose or mouth into the lungs.

Alternative Names

Flexible bronchoscopy; rigid bronchoscopy; bronchoscopic biopsy

Principle

A bronchoscope with a camera is advanced into the airways to visualize the mucosa. Instruments passed through the working channel allow sampling (biopsy, brushing, bronchoalveolar lavage), aspiration, stent placement, or removal of foreign bodies.

History

Rigid bronchoscopy was introduced in the 1890s by Gustav Killian, and the flexible fiberoptic bronchoscope was developed by Shigeto Ikeda in the 1960s, transforming pulmonary medicine.

Why It Is Done

Performed to evaluate persistent cough, hemoptysis, abnormal imaging findings, suspected infection or malignancy, and to obtain biopsy specimens, clear mucus plugs, or remove aspirated foreign bodies.

Is This a Primary or Secondary Test or Procedure

Primary diagnostic and therapeutic procedure for airway and lung disease.

How It Is Performed

Performed under topical anesthesia with sedation, or general anesthesia for rigid bronchoscopy. The scope is passed through the nose or mouth into the trachea and bronchi; samples are taken as needed. The patient is monitored throughout.

Preparation

NPO before sedation, review of anticoagulants, and pre-procedure imaging. Patients with significant respiratory compromise may require extra precautions.

Contraindications and Precautions

Severe refractory hypoxemia, uncontrolled bleeding, recent myocardial infarction, and unstable arrhythmia are relative contraindications. Precaution: bleeding risk increases with biopsy in uremia or coagulopathy.

Risks

Bleeding, pneumothorax, hypoxemia, fever, laryngospasm, and transient sore throat.

Aftercare and Recovery

The patient is observed until sedation wears off and is asked to fast until the gag reflex returns. Specimens are sent for culture, cytology, and histology.

Outcome and Follow-up

Cytology or biopsy may reveal malignancy, infection, inflammation, or granulomas; lavage culture identifies pathogens.

Expected Results

Patent airways with intact mucosa and no abnormal masses, secretions, or foreign bodies.

Follow-up

Results guide treatment; repeat bronchoscopy may be needed for surveillance or therapy.

References

1. MedlinePlus. Bronchoscopy. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 25

Bursal Aspiration and Injection

Overview

Bursal aspiration and injection is a procedure in which fluid is removed from an inflamed bursa and, when appropriate, a corticosteroid is injected into the bursa to relieve inflammation and pain. It is most commonly performed for bursitis of the shoulder, elbow, hip, or knee.

Alternative Names

Bursa aspiration; bursa injection; aspiration and injection of a bursa; corticosteroid bursal injection

Principle

Aspiration removes excess fluid and provides material for analysis or symptom relief, while corticosteroid injection reduces local inflammation of the bursal lining, decreasing swelling and pain.

History

Aspiration and injection of inflamed bursae has been practiced since the early 20th century, and became routine with the availability of injectable corticosteroids after the 1950s.

Why It Is Done

Ordered for painful bursitis with significant effusion, to relieve symptoms, to obtain fluid for testing when septic bursitis or gout is suspected, and to deliver corticosteroid directly to the inflamed bursa.

Is This a Primary or Secondary Test or Procedure

A primary therapeutic and diagnostic procedure for bursitis.

How It Is Performed

The bursa is palpated and the skin prepared aseptically. A needle is inserted into the bursal cavity, fluid is aspirated, and corticosteroid may then be injected through the same needle. Ultrasound guidance may be used for deeper bursae.

Preparation

Informed consent and review of the patient's anticoagulation status. The injection is deferred if there is overlying cellulitis or suspected infection.

Contraindications and Precautions

Contraindications include infection in the area and bleeding diathesis. Precautions include repeated corticosteroid injections, which can weaken tendons, and use near the Achilles or patellar tendon.

Risks

Pain, transient flare of symptoms, infection, bleeding, skin atrophy at the injection site, and tendon rupture after repeated injections.

Aftercare and Recovery

The site is bandaged and the patient rests the area for a day. Pain relief from the corticosteroid typically begins within days, while fluid aspirated for testing is sent to the laboratory.

Outcome and Follow-up

Improvement in pain and swelling indicates a good response. Fluid analysis results determine whether the bursitis is inflammatory, crystalline (gout or pseudogout), or infectious.

Expected Results

Relief of symptoms and a benign fluid profile with no crystals or organisms.

Follow-up

Patients are reassessed within weeks; persistent symptoms may prompt imaging, repeat aspiration, or further treatment including physical therapy.

References

1. MedlinePlus. Bursitis. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 26

Canalith Repositioning (Epley Maneuver)

Overview

Canalith repositioning, commonly known as the Epley maneuver, is a sequence of head and body movements that moves displaced calcium crystals out of the semicircular canals of the inner ear. It is the standard bedside treatment for benign paroxysmal positional vertigo (BPPV).

Alternative Names

Epley maneuver; canalith repositioning procedure; particle repositioning maneuver; Dix-Hallpike-guided repositioning

Principle

In BPPV, calcium carbonate debris (canaliths) from the utricle falls into a semicircular canal, usually the posterior canal, causing vertigo with head movement. Sequential head turns and positional changes use gravity to guide the debris back into the utricle, where it no longer stimulates the canal during movement.

History

The Epley maneuver was described by John Epley in 1980. The associated Dix-Hallpike test, used to diagnose and position the patient, was introduced in 1952.

Why It Is Done

Ordered for classic BPPV presenting as brief episodes of rotational vertigo triggered by rolling over in bed, looking up, or bending forward, confirmed by the Dix-Hallpike test.

Is This a Primary or Secondary Test or Procedure

Primary definitive treatment for posterior canal BPPV, and used for other canal variants with modified positioning sequences.

How It Is Performed

The patient sits upright with the head turned 45 degrees toward the affected ear, then lies back with the head hanging below the bed. The head is turned 90 degrees to the opposite side, the body is rolled onto the same side, and the patient finally sits up. Each position is held about 30 seconds. The sequence may be repeated until the provoking nystagmus resolves.

Preparation

No specific preparation is required. A diagnosis of BPPV should be confirmed by the Dix-Hallpike test, and the maneuver is often performed in the clinic with the patient able to recline.

Contraindications and Precautions

Contraindications include cervical spine instability, cervical radiculopathy, severe kyphosis, and high-grade carotid stenosis. Precautions include a history of recent neck trauma or surgery and limited neck mobility.

Risks

Transient worsening of vertigo and nausea during the maneuver, vomiting, and rarely conversion of posterior canal BPPV to a horizontal canal variant. Falls may occur if dizziness persists after standing.

Aftercare and Recovery

The patient is advised to avoid rapid head movements and extreme head extension for 24-48 hours and may sleep propped up. Symptoms usually resolve immediately but may recur.

Outcome and Follow-up

Resolution of positional vertigo and a negative repeat Dix-Hallpike test indicate success. Persistent vertigo suggests canal conversion, incomplete repositioning, or an alternative diagnosis.

Expected Results

A negative Dix-Hallpike test with no nystagmus or vertigo after repositioning is considered a normal, successful result.

Follow-up

Patients are reassessed in days to weeks; recurrent BPPV may be treated by repeating the maneuver, and patients may be taught to perform the maneuver at home. Imaging and vestibular testing are reserved for atypical or refractory cases.

References

1. MedlinePlus. Benign positional vertigo. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 27

Capsule Endoscopy

Overview

Capsule endoscopy is a procedure used to record images of the gastrointestinal tract for use in medicine. It utilizes a pill-sized capsule containing a wireless camera.

Alternative Names

Pill cam, Wireless capsule endoscopy

Principle

The patient swallows a small capsule containing a miniature camera, light source, battery, and transmitter. As peristalsis moves it through the digestive tract, it captures thousands of images that are transmitted to a recorder worn on the body, allowing visualization of the small intestine, which is hard to reach with standard endoscopes.

History

Gavriel Iddan and Paul Swain co-invented the capsule endoscope in the late 1990s, and it was approved by the FDA in 2001.

Why It Is Done

Ordered to find the source of bleeding in the small intestine (which cannot be reached by standard endoscopy), diagnose Crohn's disease, or detect small bowel tumors.

Is This a Primary or Secondary Test or Procedure

Primary diagnostic tool for small bowel mucosal imaging and obscure GI bleeding.

How It Is Performed

You swallow a vitamin-sized capsule containing a camera. As it travels naturally through your digestive tract, it transmits thousands of images to a recorder worn on your belt.

Preparation

Fast for 12 hours before swallowing the capsule. A laxative bowel prep is usually required the night before.

Contraindications and Precautions

Known or suspected gastrointestinal obstruction, stricture, or fistula. Severe swallowing disorders.

Risks

Capsule retention (<2

Aftercare and Recovery

You can eat a light snack after 4 hours. The capsule is disposable and passes naturally in your stool within 24 to 72 hours.

Outcome and Follow-up

Images are downloaded from the recorder and reviewed by a gastroenterologist for small bowel ulcers, bleeding, or polyps.

Expected Results

Normal small bowel mucosa; no active bleeding, ulcers, erosions, or vascular malformations.

Follow-up

Double-balloon enteroscopy, CT enterography, or medical management.

References

1. MedlinePlus. Capsule Endoscopy. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 28

Cardiac Ablation

Overview

Cardiac ablation uses heat (radiofrequency) or cold (cryoablation) energy to scar tiny areas in the heart that are creating or conducting irregular electrical signals, thereby restoring normal sinus rhythm.

Alternative Names

Catheter ablation, Radiofrequency ablation, Cryoablation

Principle

Catheters with electrode tips are advanced into the heart to map abnormal electrical pathways, then deliver radiofrequency energy or extreme cold to create small scars. The scars interrupt the reentrant circuits that cause arrhythmias, restoring normal rhythm.

History

Pioneered in the 1980s using direct current shocks. The technology evolved to radiofrequency catheter ablation in the 1990s, becoming the primary curative treatment for many arrhythmias.

Why It Is Done

Ordered to treat symptomatic arrhythmias like atrial fibrillation, atrial flutter, or supraventricular tachycardia (SVT) when medications are ineffective or not tolerated.

Is This a Primary or Secondary Test or Procedure

Primary curative treatment for specific tachyarrhythmias (SVT, WPW, atrial flutter).

How It Is Performed

Performed under sedation or general anesthesia. Diagnostic catheters are inserted through blood vessels in the groin and threaded to the heart. Mapping identifies abnormal pathways, and the ablation catheter delivers energy to isolate or destroy the targeted tissue.

Preparation

Fast for 6-8 hours before the procedure. Stop antiarrhythmic drugs 3-5 half-lives before the procedure as directed by your electrophysiologist.

Contraindications and Precautions

Active infection, intracardiac thrombus (must be ruled out via transesophageal echo), or severe uncorrected bleeding disorders.

Risks

Cardiac tamponade, groin hematoma, femoral pseudoaneurysm, stroke, transient ischemic attack (TIA), heart block requiring a permanent pacemaker, or phrenic nerve injury.

Aftercare and Recovery

Lie flat for 4-6 hours post-procedure to prevent groin bleeding. Monitor the puncture site. Avoid strenuous activity and heavy lifting for 1 week.

Outcome and Follow-up

Success is defined by the inability to induce the target arrhythmia in the electrophysiology lab post-ablation.

Expected Results

Termination of the abnormal pathway; restoration and maintenance of normal sinus rhythm.

Follow-up

ECG monitoring, Holter monitoring, and cardiology check-up in 1 month.

References

1. MedlinePlus. Cardiac Ablation. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 29

Cardiac Catheterization

Overview

Cardiac catheterization involves passing a thin flexible tube (catheter) through a blood vessel in the groin or arm and guiding it into the heart to diagnose or treat cardiovascular conditions.

Alternative Names

Heart cath, Coronary catheterization, Cardiac cath

Principle

A catheter is threaded through a blood vessel into the heart under X-ray guidance, permitting measurement of pressures and oxygen content in the chambers, injection of contrast to image the coronary arteries, and passage of interventional tools such as balloons, stents, and valves.

History

Werner Forssmann performed the first human cardiac catheterization on himself in 1929; Andre Cournand and Dickinson Richards developed its clinical use, sharing the 1956 Nobel Prize.

Why It Is Done

Ordered to evaluate chest pain, assess heart valve function, measure intracardiac pressures, or locate blockages in coronary arteries.

Is This a Primary or Secondary Test or Procedure

Primary diagnostic and guiding intervention for coronary and valvular heart disease.

How It Is Performed

Under local anesthesia and mild sedation, a catheter is inserted into the femoral or radial artery and threaded to the heart under fluoroscopic guidance.

Preparation

Fast for 6 hours; check coagulation profile; stop blood thinners as directed.

Contraindications and Precautions

Active local infection over puncture site; severe uncorrected bleeding disorders.

Risks

Bleeding, hematoma at puncture site, cardiac arrhythmias, coronary artery dissection, or contrast allergy.

Aftercare and Recovery

Lie flat for 4-6 hours if femoral access was used; monitor puncture site for bleeding; drink fluids to flush contrast.

Outcome and Follow-up

Provides detailed information about coronary artery blockages and heart chamber pressures.

Expected Results

Normal intracardiac pressures; no significant coronary artery blockages.

Follow-up

Coronary angioplasty, CABG, or medical management.

References

1. MedlinePlus. Cardiac Catheterization. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 30

Cardiopulmonary Resuscitation (CPR)

Overview

CPR is an emergency procedure that combines chest compressions and artificial ventilation to manually preserve brain function and circulation in a person in cardiac arrest.

Alternative Names

CPR, Basic Life Support, BLS

Principle

Rhythmic chest compressions squeeze the heart against the spine, generating artificial blood flow to the brain and vital organs, while rescue breaths provide oxygenation. Maintaining perfusion during cardiac arrest bridges the patient until defibrillation and advanced care restore a spontaneous rhythm.

History

Modern CPR was developed in the late 1950s. Kouwenhoven, Knickerbocker, and Jude discovered the efficacy of external chest compressions in 1960.

Why It Is Done

Performed immediately on any individual who is unresponsive, pulseless, and not breathing (cardiac arrest).

Is This a Primary or Secondary Test or Procedure

Primary emergency resuscitation method for out-of-hospital and in-hospital cardiac arrest.

How It Is Performed

Consists of cycles of 30 high-quality chest compressions (100-120 per minute, at least 2 inches deep) followed by 2 rescue breaths. An Automated External Defibrillator (AED) is applied as soon as available.

Preparation

Ensure scene safety. Assess responsiveness and breathing. Call for emergency help immediately.

Contraindications and Precautions

Active Do Not Resuscitate (DNR) order, or obvious signs of death (rigor mortis, decapitation).

Risks

Rib fractures, sternal fractures, gastric inflation, or aspiration of gastric contents.

Aftercare and Recovery

If Return of Spontaneous Circulation (ROSC) is achieved, immediately transition the patient to advanced post-cardiac arrest care (ventilator, targeted temperature management).

Outcome and Follow-up

ROSC is identified by a palpable pulse, measurable blood pressure, and organized electrical activity on the cardiac monitor.

Expected Results

Return of Spontaneous Circulation (ROSC); return of spontaneous respiratory effort; hemodynamically stable.

Follow-up

Intensive care unit admission, coronary angiography, or therapeutic hypothermia.

References

1. MedlinePlus. Cardiopulmonary Resuscitation (CPR). U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 31

Cardioversion

Overview

Cardioversion restores a normal heart rhythm in people with certain types of abnormal heartbeats (arrhythmias), usually by delivering a synchronized electrical shock through electrode pads on the chest.

Alternative Names

Electrical cardioversion, Direct current cardioversion, DCCV

Principle

A synchronized electrical shock delivered through pads or paddles depolarizes the heart muscle simultaneously, abruptly terminating reentrant electrical circuits. The pause lets the sinoatrial node resume control and a coordinated, normal rhythm be restored.

History

Bernard Lown developed the direct current cardioverter-defibrillator in 1961, introducing synchronized cardioversion for atrial fibrillation and flutter, which was much safer than chemical methods.

Why It Is Done

Ordered to treat atrial fibrillation or atrial flutter when medications fail, or when the patient is hemodynamically unstable (e.g., severe hypotension, pulmonary edema).

Is This a Primary or Secondary Test or Procedure

Primary treatment for unstable tachyarrhythmias; elective treatment for persistent symptomatic atrial fibrillation.

How It Is Performed

Performed under brief general anesthesia or heavy sedation. Electrode pads are placed on the chest (anterior-posterior or anterior-lateral). The defibrillator delivers a shock synchronized to the R wave of the QRS complex to avoid triggering ventricular fibrillation.

Preparation

Fast for 6-8 hours. Ensure adequate anticoagulation for at least 3 weeks prior, or perform a transesophageal echocardiogram (TEE) to rule out left atrial thrombus.

Contraindications and Precautions

Known left atrial thrombus, digitalis toxicity-induced arrhythmia, or uncorrected hypokalemia.

Risks

Thromboembolism (stroke), transient arrhythmias, skin irritation/burns from pads, or respiratory depression from sedation.

Aftercare and Recovery

Monitor vital signs and airway until fully awake. Verify rhythm with a 12-lead ECG. Continue oral anticoagulation as directed.

Outcome and Follow-up

Conversion of the arrhythmia to sinus rhythm is immediately visible on the monitor.

Expected Results

Restoration and maintenance of normal sinus rhythm.

Follow-up

ECG check-up; maintenance of antiarrhythmic and anticoagulant therapy as indicated.

References

1. MedlinePlus. Cardioversion. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 32

Carotid Angioplasty and Stenting

Overview

Carotid angioplasty and stenting (CAS) is a minimally invasive procedure that opens a narrowed carotid artery using a balloon and stent, usually placed through a transfemoral approach.

Alternative Names

Carotid artery stenting; CAS; percutaneous carotid revascularization

Principle

A balloon catheter is advanced into the carotid stenosis, inflated to dilate the vessel, and a self-expanding stent is deployed to scaffold the artery open. An embolic protection device is often used to capture debris released during dilation.

History

Carotid stenting was developed in the 1990s as an alternative to endarterectomy. It proved particularly useful for patients at high surgical risk, and randomized trials established its role in the 2000s.

Why It Is Done

Performed for symptomatic carotid stenosis (greater than 50 percent) or asymptomatic stenosis (greater than 60-80 percent) in patients at high risk for open surgery, or for restenosis after endarterectomy.

Is This a Primary or Secondary Test or Procedure

Primary or alternative revascularization for carotid stenosis; secondary after failed endarterectomy.

How It Is Performed

Performed under local or general anesthesia in an angiography suite. A sheath is placed in the femoral artery, a guidewire crosses the stenosis, an embolic protection device is deployed, the balloon is inflated, and the stent is placed. Angiography confirms patency.

Preparation

Preoperative carotid ultrasound and often CT or MR angiography, dual antiplatelet therapy (aspirin plus clopidogrel) started beforehand, and assessment of renal function and contrast allergy.

Contraindications and Precautions

Severe aortic or carotid tortuosity, heavily calcified or thrombotic lesions, or recent large stroke are relative contraindications. Advanced age and unfavorable arch anatomy increase stroke risk.

Risks

Periprocedural stroke or transient ischemic attack, dissection, stent thrombosis or restenosis, bleeding or pseudoaneurysm at the puncture site, and contrast nephropathy.

Aftercare and Recovery

Observation with neurologic checks, dual antiplatelet therapy continued for weeks to months, and puncture-site care. Most patients are discharged within 1-2 days.

Outcome and Follow-up

Post-procedure angiography and ultrasound confirm stent patency and residual stenosis. Results guide antiplatelet duration and risk-factor management.

Expected Results

A patent stent with less than 30 percent residual stenosis, no new neurologic deficit, and no access-site complication.

Follow-up

Carotid ultrasound at 1, 6, and 12 months and then annually to detect in-stent restenosis, along with aggressive risk-factor control.

References

1. MedlinePlus. Carotid artery stenting. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 33

Cast Application

Overview

A cast circumferentially immobilizes an injured limb (usually a fracture) using plaster or fiberglass bandages over cotton padding. This maintains bone alignment during healing.

Alternative Names

Orthopedic casting, Plaster cast

Principle

Wet plaster or fiberglass is wrapped around the limb and molded to its contours, then hardens into a rigid shell. The cast immobilizes the bone ends or joint, holds the fracture in correct alignment, and prevents the movement that would disrupt healing.

History

Antonius Mathijsen, a Dutch military surgeon, invented plaster-of-Paris bandages in 1851. Fiberglass casts were introduced in the 1970s.

Why It Is Done

Ordered to immobilize fractures, severe sprains, or post-surgical repairs to prevent movement and facilitate bone and tissue healing.

Is This a Primary or Secondary Test or Procedure

Primary conservative treatment for stable, reduced bone fractures.

How It Is Performed

A stockinette is applied to the limb, followed by layers of soft cotton padding. Plaster or fiberglass rolls are wet and wrapped around the limb, then molded by the clinician.

Preparation

Clean and dry the skin. Ensure the fracture is adequately reduced and aligned.

Contraindications and Precautions

Severe acute swelling (relative; a splint should be used initially to avoid compartment syndrome).

Risks

Compartment syndrome (pressure buildup), skin pressure ulcers/necrosis, joint stiffness, or thermal burns during plaster setting.

Aftercare and Recovery

Elevate the limb above the heart for 24-48 hours. Monitor for '5 Ps' of compartment syndrome: Pain, Pulselessness, Pallor, Paresthesia, Paralysis. Keep the cast dry.

Outcome and Follow-up

Signs of a successful application include bone stability, comfortable fit, and distal limb warmth and pink color.

Expected Results

Rigid immobilization of the joint/bone; warm distal extremities; capillary refill <2 seconds; intact sensation.

Follow-up

Cast check in 1 week; serial X-rays to ensure stable alignment; cast removal in 4-8 weeks.

References

1. MedlinePlus. Cast Application. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 34

Cataract Surgery

Overview

Cataract surgery removes a cloudy natural lens from the eye and replaces it with an artificial intraocular lens (IOL) to improve vision.

Why It Is Done

It is performed when a cataract causes visual impairment that interferes with reading, driving, work, or daily activities and glasses no longer provide adequate improvement.

How It Is Performed

After local anesthesia, the surgeon makes a small corneal incision, breaks up and removes the cloudy lens, and inserts a foldable IOL. The incision usually seals without stitches.

Risks and Recovery

Infection, bleeding, inflammation, retinal problems, increased eye pressure, lens displacement, and posterior capsule clouding are possible. Use prescribed eye drops, avoid contamination or heavy strain as instructed, and attend follow-up visits.

References

1. MedlinePlus. Cataract Removal. U.S. National Library of Medicine; 2025.
2. American Academy of Ophthalmology. Cataract Surgery; 2025.

Chapter 35

Central Venous Catheter Placement

Overview

A central venous catheter (CVC) is a catheter placed into a large vein (internal jugular, subclavian, or femoral) to administer fluids, medications, or monitor hemodynamics.

Alternative Names

Central line insertion, CVC placement, Triple lumen catheter

Principle

A long catheter is inserted into a large vein such as the internal jugular, subclavian, or femoral, and advanced so its tip lies in the superior vena cava near the right atrium. The high blood flow at this site permits reliable delivery of drugs and nutrition, pressure monitoring, and blood sampling.

History

Werner Forssmann inserted a catheter into his own heart in 1929. Sven-Ivar Seldinger introduced the guidewire technique (Seldinger technique) in 1953.

Why It Is Done

Ordered for administration of vasopressors, hypertonic solutions, total parenteral nutrition (TPN), long-term chemotherapy, or hemodialysis access.

Is This a Primary or Secondary Test or Procedure

Primary vascular access method for hemodynamically unstable patients requiring vasoactive infusions.

How It Is Performed

Under sterile conditions and ultrasound guidance. The skin is numbed, the target vein is punctured with a needle, a guidewire is advanced, and the catheter is threaded over the wire.

Preparation

Verify coagulation profile (INR and platelets). Obtain sterile gown, gloves, mask, and catheter kit.

Contraindications and Precautions

Active infection at the insertion site, or thrombosis of the target vein.

Risks

Pneumothorax (especially subclavian), arterial puncture, air embolism, cardiac arrhythmias (if wire enters the atrium), or catheter-related bloodstream infection (CRBSI).

Aftercare and Recovery

Obtain a chest X-ray immediately to confirm correct catheter tip position (cavoatrial junction) and rule out pneumothorax before using the line.

Outcome and Follow-up

catheter is patent, flushes easily, draws blood, and the tip is correctly positioned.

Expected Results

Catheter tip located in the lower third of the superior vena cava; no pneumothorax; patent lumens.

Follow-up

Daily site inspection for signs of infection; immediate removal when no longer clinically necessary.

References

1. MedlinePlus. Central Venous Catheter Placement. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 36

Cerumen Removal

Overview

Cerumen removal is the extraction of impacted earwax from the external auditory canal to relieve symptoms or improve visualization of the tympanic membrane. It is performed with instruments, suction, irrigation, or cerumenolytic agents.

Alternative Names

Earwax removal; ear irrigation; cerumen disimpaction; aural toilet

Principle

Impacted cerumen is softened or displaced so it can be extracted from the ear canal. Instrumentation uses curettes or loops under direct visualization, irrigation uses a controlled water stream to flush wax outward, and cerumenolytics chemically soften the wax before removal. The procedure restores canal patency and auditory conduction.

History

Cerumen removal has been practiced since antiquity, with ear syringing documented for centuries and modern suction and micro-instrumentation developed in the 20th century.

Why It Is Done

Performed for symptoms of cerumen impaction such as hearing loss, fullness, tinnitus, itching, or dizziness, and to allow examination of the tympanic membrane when wax obscures the view. It is also done to permit hearing aid fitting and to monitor the ear after surgery.

Is This a Primary or Secondary Test or Procedure

A primary therapeutic procedure for symptomatic cerumen impaction.

How It Is Performed

The canal is inspected with an otoscope under magnification. The clinician removes wax using a loop or curette, suction, or irrigation with body-temperature water after confirming an intact tympanic membrane. A cerumenolytic may be instilled several days beforehand to soften hard wax.

Preparation

No routine preparation is needed. Patients are asked about otorrhea, otalgia, prior ear surgery, and hearing aid use. An intact tympanic membrane is confirmed before irrigation.

Contraindications and Precautions

Irrigation is contraindicated when the tympanic membrane is perforated, when a cholesteatoma or myringotomy tube is present, and in patients with a history of otitis media, because of the risk of infection or vertigo. Instrumentation is performed cautiously in anticoagulated patients and in those with a narrow or tortuous canal.

Risks

Canal abrasion, bleeding, otitis externa, tinnitus, and tympanic membrane perforation; irrigation may rarely cause tympanic perforation, acute otitis media, or caloric-induced vertigo.

Aftercare and Recovery

The ear is re-examined to confirm complete removal and to assess the tympanic membrane. Patients are advised to keep the ear dry and to avoid inserting cotton swabs, which can push wax deeper.

Outcome and Follow-up

Complete removal with an intact tympanic membrane and improved hearing confirms success. Persisting symptoms or inability to remove wax safely may indicate referral to an ENT specialist.

Expected Results

A clear external auditory canal with a visible, intact tympanic membrane and no residual obstructing wax.

Follow-up

If hearing loss persists despite clearing, audiometry is performed to evaluate for other causes. Patients with recurrent impaction may be advised to use cerumenolytics or periodic cleaning, and referral is considered for suspected underlying canal pathology.

References

1. MedlinePlus. Ear wax removal. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 37

Cervical Biopsy

Overview

A cervical biopsy is the removal of a small sample of tissue from the cervix for histologic examination. It is used to diagnose cervical abnormalities, particularly precancerous or cancerous lesions.

Alternative Names

Colposcopy-directed biopsy, Cervical punch biopsy, Cone biopsy (conization), Loop electrosurgical excision procedure (LEEP) biopsy

Principle

Using colposcopy, the cervix is inspected after applying acetic acid, which makes abnormal epithelium visible, and biopsy forceps sample the affected areas. The tissue is fixed and examined microscopically for dysplasia or malignancy. Larger excisional techniques such as LEEP or cone biopsy remove a conically shaped specimen that includes the transformation zone.

History

Cervical biopsy developed with the introduction of the colposcope in the 1920s, and loop electrosurgical excision came into use in the 1980s.

Why It Is Done

Cervical biopsy is ordered when cervical cytology (Pap smear) is abnormal, when the cervix appears abnormal on examination, or to evaluate suspicious lesions or unexplained bleeding. It confirms the presence and grade of cervical dysplasia or cancer before treatment.

Is This a Primary or Secondary Test or Procedure

Primary diagnostic test for confirming cervical pathology identified by screening.

How It Is Performed

With the patient in the lithotomy position, a speculum is placed and the cervix is visualized under colposcopy after application of acetic acid. Punch forceps remove small samples of abnormal tissue, or an electrosurgical loop removes a larger cone specimen. Bleeding is controlled with pressure, silver nitrate, or Monsel's solution.

Preparation

Informed consent is obtained, and the procedure is ideally scheduled when the patient is not menstruating. The patient is advised to expect mild cramping, and analgesics may be offered.

Contraindications and Precautions

Biopsy is avoided during active pelvic infection or acute cervicitis and in bleeding disorders that increase hemorrhagic risk. In pregnancy, biopsy of the transformation zone is generally deferred. Patients should avoid tampons, douching, and intercourse before the procedure.

Risks

Bleeding, infection, and cramping are the main risks; cone biopsy and LEEP carry additional risks of cervical stenosis and adverse pregnancy outcomes. Minor bleeding and dark discharge may persist for several days.

Aftercare and Recovery

The patient may resume activity but should avoid tampons, douching, and intercourse for one to two weeks, longer after LEEP or cone biopsy. Results are usually available within one to two weeks.

Outcome and Follow-up

Results report the presence and grade of dysplasia (CIN 1, 2, or 3) or invasive carcinoma, including the margin status of excised specimens. Abnormal results guide treatment and follow-up, while normal results confirm no pathologic change in the sampled tissue.

Expected Results

No dysplastic or malignant cells are present; the cervical tissue is benign.

Follow-up

Management follows the grade of dysplasia and ranges from surveillance with repeat cytology and HPV testing to treatment with excision or ablation. Post-treatment surveillance continues for years because of the risk of recurrence.

References

1. MedlinePlus. Cervical biopsy. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 38

Cervical Cerclage

Overview

Cervical cerclage is a procedure that places a suture around the cervix during pregnancy to hold it closed and prevent preterm birth in women with cervical insufficiency.

Alternative Names

Cervical stitch; cervical suture; cervical incompetence treatment

Principle

A non-absorbable suture encircles the cervix at the level of the internal os, providing mechanical support that keeps the cervix closed against progressive dilation during pregnancy.

History

Cervical cerclage was introduced in the 1950s (Shirodkar 1955, McDonald 1957) and remains the standard treatment for cervical insufficiency.

Why It Is Done

Performed for a history of painless second-trimester loss, a short cervix on ultrasound, or premature cervical dilation in the current pregnancy.

Is This a Primary or Secondary Test or Procedure

Primary preventive procedure for cervical insufficiency.

How It Is Performed

Performed under regional or general anesthesia. Through the vagina, the cervix is exposed and a suture is placed (McDonald purse-string or Shirodkar technique) and tied to close the internal os.

Preparation

Ultrasound evaluation of fetal anatomy and cervical length, and exclusion of infection or preterm labor.

Contraindications and Precautions

Chorioamnionitis, preterm premature rupture of membranes, fetal anomaly, and active labor are contraindications.

Risks

Infection, membrane rupture, bleeding, cervical injury, and preterm labor.

Aftercare and Recovery

The patient is monitored and activity may be limited; the stitch is removed at 36-37 weeks or at delivery.

Outcome and Follow-up

Serial ultrasound checks cervical length; the stitch is assessed for displacement.

Expected Results

A closed cervix and a pregnancy carried to term.

Follow-up

Serial cervical length ultrasound; removal of the stitch near term.

References

1. MedlinePlus. Cervical cerclage. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 39

Chemotherapy

Overview

Chemotherapy uses chemical substances (antineoplastic drugs) to kill rapidly dividing cancer cells. It can be systemic, affecting cells throughout the body.

Alternative Names

Chemo, Cancer chemotherapy

Principle

Cytotoxic drugs are given orally or intravenously and circulate through the body, where they act on rapidly dividing cells by damaging DNA, blocking cell division, or interrupting signaling pathways. The goal is to kill cancer cells while normal tissues recover between treatment cycles.

History

Began in the 1940s with the use of nitrogen mustard (derived from chemical warfare research) to treat lymphoma. Sidney Farber pioneered folic acid antagonists for leukemia.

Why It Is Done

Ordered to cure cancer, control cancer growth, shrink tumors before surgery (neoadjuvant), or kill remaining cancer cells after surgery (adjuvant).

Is This a Primary or Secondary Test or Procedure

Primary treatment for hematological malignancies and advanced solid tumors.

How It Is Performed

Administered in cycles, usually intravenously through a central line or port-a-cath, or orally as pills. Treatment is given over several hours or days, followed by a rest period.

Preparation

Baseline blood tests (CBC, liver/kidney function) and cardiac evaluation (echocardiogram) are required before each cycle. Hydration is key.

Contraindications and Precautions

Severe bone marrow suppression (absolute neutrophil count <1,000 or platelets <50,000), severe active infection, or end-organ dysfunction depending on the drug.

Risks

Neutropenia (infection risk), thrombocytopenia (bleeding), anemia, nausea, hair loss, fatigue, cardiotoxicity, or peripheral neuropathy.

Aftercare and Recovery

Monitor for fever (>100.4 F) which indicates neutropenic fever, an oncology emergency. Manage side effects with antiemetics and supportive care.

Outcome and Follow-up

Monitored using imaging scans (CT, MRI) and tumor markers to assess tumor shrinkage or remission.

Expected Results

Stable or shrinking tumor size; normal bone marrow recovery between cycles.

Follow-up

CBC monitoring, imaging scans, and regular oncology follow-up.

References

1. MedlinePlus. Chemotherapy. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 40

Chest Tube Insertion

Overview

Chest tube insertion, or tube thoracostomy, is placement of a drain into the pleural space to evacuate air, blood, pus, or fluid and to re-expand the lung.

Alternative Names

Tube thoracostomy; intercostal drain placement; pleural drain

Principle

A drainage catheter is inserted into the pleural space through the chest wall and connected to a sealed drainage system with an underwater seal. The seal allows one-way outflow of air or fluid while the lung expands.

History

Chest drainage has been practiced for empyema since the 19th century, popularized by surgeons during World War I. Closed underwater-seal drainage became standard during the Korean and Vietnam wars.

Why It Is Done

Performed for pneumothorax (especially tension or large spontaneous), hemothorax, pleural effusion or empyema, post-thoracotomy drainage, and after cardiothoracic surgery.

Is This a Primary or Secondary Test or Procedure

Primary emergency and therapeutic procedure for pleural air and fluid collections.

How It Is Performed

Performed at the bedside or in the operating room. The patient is positioned, the skin over the fourth or fifth intercostal space in the midaxillary line is prepped and anesthetized, a small incision is made, the tract is bluntly dissected, and the tube is guided into the pleural space toward the apex (for air) or base (for fluid). It is secured and connected to suction or an underwater seal.

Preparation

Consent, coagulation assessment, and in elective cases imaging to localize the collection. Emergency insertion may proceed without delay.

Contraindications and Precautions

Relative contraindications include coagulopathy, bleeding diathesis, and local infection at the site. Precaution: do not insert above the level of the nipple line without imaging guidance to avoid injury to the subclavian vessels.

Risks

Injury to lung, intercostal vessels, liver, spleen, or diaphragm; bleeding; infection; malposition; and re-expansion pulmonary edema.

Aftercare and Recovery

The tube is connected to a drainage system, serial chest radiographs confirm position and lung expansion, and the tube is removed when the air leak or drainage resolves and the lung remains expanded.

Outcome and Follow-up

Chest radiograph confirms tube position and lung re-expansion. Output volume and air leak guide removal timing; laboratory analysis of drained fluid may guide diagnosis.

Expected Results

Complete lung re-expansion, cessation of air leak, and drainage of the collection, with tube removal after 24-48 hours of stability.

Follow-up

Chest radiograph after removal and clinical follow-up for recurrence of pneumothorax or effusion.

References

1. MedlinePlus. Chest tube insertion. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 41

Chorionic Villus Sampling

Overview

Chorionic villus sampling (CVS) is a prenatal test that obtains a small sample of placental tissue, the chorionic villi, to detect chromosomal and genetic abnormalities in the fetus. It is typically performed between 10 and 13 weeks of gestation.

Alternative Names

CVS, Chorionic villi biopsy, Placental biopsy

Principle

A small sample of chorionic villi, which derive from the placenta and share the genetic makeup of the fetus, is obtained through the cervix or the abdominal wall under ultrasound guidance. The villi are cultured and analyzed for chromosomal karyotype, molecular genetic, or enzymatic studies. Because the villi and the fetus arise from the same zygote, results reflect fetal genetic material.

History

Chorionic villus sampling was developed in the 1960s and 1970s and came into clinical use in the 1980s as an earlier alternative to amniocentesis.

Why It Is Done

CVS is ordered for women at increased risk of fetal chromosomal or genetic abnormalities, such as those with advanced maternal age, abnormal prenatal screening results, or a family history of a genetic disorder. It provides earlier diagnosis than amniocentesis, allowing earlier decisions about the pregnancy.

Is This a Primary or Secondary Test or Procedure

Primary invasive prenatal diagnostic test for genetic evaluation in the first trimester.

How It Is Performed

Under continuous ultrasound guidance, a catheter or needle is introduced into the placenta either transvaginally or through the abdominal wall, and a small amount of chorionic villi is aspirated. The sample is examined under a microscope to confirm adequate tissue. The procedure takes several minutes and generally does not require anesthesia, though local anesthesia may be used for the abdominal route.

Preparation

Informed consent is obtained and maternal blood type is reviewed so that anti-D immune globulin can be given to Rh-negative women. No fasting is required, and a full bladder may aid the transcervical approach.

Contraindications and Precautions

The transcervical route is avoided with active vaginal bleeding, cervical infection, or cervical lesions. Relative contraindications include a posterior placenta, uterine fibroids, or an approach that is technically difficult. Vaginal spotting is common afterward and should be reported if heavy.

Risks

Miscarriage occurs in approximately 0.5–1% of cases, slightly higher than with amniocentesis. Cramping, vaginal spotting, leakage of fluid, and infection may occur, and limb reduction defects were reported when sampling was performed before 10 weeks.

Aftercare and Recovery

The patient rests briefly and usually returns to normal activity the same day. Results are reported after culture or molecular analysis is complete, typically within one to two weeks.

Outcome and Follow-up

Results report the fetal chromosomal pattern or the status of tested genes. Abnormal results are reviewed with a genetic counselor, while normal results reduce but do not eliminate the risk of birth defects.

Expected Results

Normal karyotype (46,XX or 46,XY) with no evidence of the tested genetic disorders.

Follow-up

Genetic counseling is provided for abnormal results, and a detailed fetal ultrasound evaluates for structural anomalies. Confirmatory amniocentesis may be offered when mosaicism or uncertain results are found.

References

1. MedlinePlus. Chorionic villus sampling. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 42

Circumcision

Overview

Circumcision is surgical removal of the foreskin covering the glans of the penis.

Why It Is Done

It may be performed for cultural or personal reasons or to treat selected medical problems such as recurrent balanitis, scarring, or pathologic phimosis. It is not required for every person with a foreskin.

How It Is Performed

After appropriate local or general anesthesia, the foreskin is separated, removed, and the edges are closed with absorbable sutures or allowed to heal according to the technique used.

Risks and Recovery

Pain, bleeding, infection, swelling, excessive or inadequate skin removal, and altered sensation are possible. Keep the wound clean, follow dressing instructions, and avoid sexual activity until healing is complete.

References

1. MedlinePlus. Circumcision. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 43

Colonoscopy

Overview

A colonoscopy uses a long, flexible scope with a camera to examine the entire lining of the large intestine (colon) and rectum to detect inflammation, polyps, or cancer.

Alternative Names

Lower GI endoscopy, Colonic evaluation

Principle

A flexible endoscope with a camera is advanced through the anus into the colon, which is distended with gas to allow inspection of the entire large intestine. Instruments passed through the scope permit biopsy, polypectomy, and treatment of bleeding.

History

Developed by William Wolff and Hiromi Shinya in 1969, introducing the technique of colonoscopic polypectomy which revolutionized colon cancer prevention.

Why It Is Done

Ordered for colorectal cancer screening starting at age 45, or to investigate chronic diarrhea, rectal bleeding, unexplained weight loss, or abdominal pain.

Is This a Primary or Secondary Test or Procedure

Primary screening and preventive tool for colorectal cancer.

How It Is Performed

You lie on your side under sedation. The colonoscope is inserted through the rectum and slowly advanced to the cecum. Air or CO2 is used to inflate the colon.

Preparation

Follow a clear liquid diet the day before and consume a strong laxative solution (bowel prep) to completely clear the colon of stool.

Contraindications and Precautions

Toxic megacolon, fulminant colitis, or suspected bowel perforation.

Risks

Bowel perforation (<1 in 1,000), bleeding post-polypectomy (<1 in 200), or sedation-related complications.

Aftercare and Recovery

You will remain in recovery until the sedative wears off. Mild gas and bloating are normal. Avoid driving for 24 hours.

Outcome and Follow-up

If polyps are found, they are removed (polypectomy) and sent for biopsy. Normal results mean no polyps or lesions were detected.

Expected Results

Clean mucosal lining of the colon and rectum; no polyps, diverticula, inflammation, or masses.

Follow-up

Repeat screening in 10 years if normal, or sooner (3-5 years) depending on the size and pathology of polyps removed.

References

1. MedlinePlus. Colonoscopy. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 44

Colposcopy

Overview

Colposcopy uses a specialized magnifying instrument (colposcope) to inspect the cervix, vagina, and vulva for abnormal cells, typically guided by applying acetic acid.

Alternative Names

Colposcopic evaluation

Principle

The cervix is viewed under magnification with a colposcope after acetic acid and iodine solutions are applied to highlight abnormal epithelial areas. The enhanced view of cellular patterns guides targeted biopsies of the most abnormal-looking regions.

History

Hans Hinselmann invented the colposcope in Germany in 1925 as a screening tool for early cervical cancer detection.

Why It Is Done

Ordered after an abnormal Pap smear result (e.g., LSIL, HSIL, persistent high-risk HPV) to visualize the cervix and guide biopsies.

Is This a Primary or Secondary Test or Procedure

Primary diagnostic follow-up test for abnormal cervical cytology screening.

How It Is Performed

You lie on the exam table. A speculum is inserted, and a 3-5

Preparation

Avoid vaginal intercourse, douching, and using vaginal medications or tampons for 24-48 hours before the test. It is best scheduled when you are not menstruating.

Contraindications and Precautions

None. Biopsy may be deferred or performed with caution in pregnancy.

Risks

Mild cramping, spotting, or light vaginal bleeding for a few days. Very low risk of pelvic infection.

Aftercare and Recovery

A dark vaginal discharge (due to Monsel's solution used for hemostasis) is normal. Avoid tampons, douching, and vaginal intercourse for 1 week.

Outcome and Follow-up

Biopsies are graded as CIN 1 (mild dysplasia), CIN 2 (moderate), or CIN 3 (severe dysplasia/carcinoma in situ).

Expected Results

Normal squamocolumnar junction; no acetowhite changes, abnormal vascular patterns, or suspicious lesions.

Follow-up

Review of biopsy pathology; treatment (LEEP, cryotherapy) or surveillance depending on CIN grade.

References

1. MedlinePlus. Colposcopy. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 45

Contraceptive Implant Insertion

Overview

Contraceptive implant insertion places a small, flexible rod (typically etonogestrel) under the skin of the inner upper arm. It provides long-acting, reversible contraception lasting up to 5 years.

Alternative Names

Etonogestrel implant placement, Implanon/Nexplanon insertion, subcutaneous contraceptive implant placement

Principle

The implant releases a steady dose of the progestin etonogestrel into the circulation. Progestin suppresses ovulation, thickens cervical mucus to impede sperm transport, and thins the endometrium. Contraceptive effect is established within 24 hours if inserted during the first 5 days of menses.

History

The first contraceptive implant, a levonorgestrel system, was approved in 1991. The single-rod etonogestrel implant replaced earlier multi-rod devices and was approved in the United States in 2006.

Why It Is Done

Patients request contraceptive implant insertion for highly effective, long-acting reversible contraception without daily adherence. It is also used in appropriate candidates as an alternative to other hormonal methods and may help manage heavy or painful menstruation.

Is This a Primary or Secondary Test or Procedure

It is a primary contraceptive procedure providing long-acting reversible contraception.

How It Is Performed

The clinician numbs the inner upper arm with local anesthetic, makes a small incision (2 mm) with the proprietary applicator, and inserts the rod just under the skin. Placement is confirmed by palpating the rod and, when indicated, by ultrasound or radiography. No sutures are typically required; a pressure dressing and bandage are applied.

Preparation

A pregnancy test is performed to exclude pregnancy before insertion. Timing should be within the first 5 days of the menstrual cycle or per product labeling. No fasting or other specific preparation is required.

Contraindications and Precautions

Contraindications include known or suspected pregnancy, active thromboembolic disease, and hypersensitivity to the implant components. Caution is warranted in women with unexplained vaginal bleeding, liver disease, or a history of venous thromboembolism.

Risks

Risks include insertion-site pain, bruising, infection, and scarring. Implant migration, breakage, and difficult removal are rare.

Aftercare and Recovery

The bandage is removed after 24 hours and the insertion site kept clean and dry. Irregular bleeding is common in the first months; fertility returns promptly after removal.

Outcome and Follow-up

The implant is considered effective if it remains palpable subcutaneously and the patient has no new bleeding pattern concerns. Removal is required if the rod is no longer palpable or if contraception is no longer desired.

Expected Results

Successful placement of a single, palpable implant in the subcutaneous tissue of the upper arm with no signs of infection or migration.

Follow-up

Follow-up at 4-6 weeks is offered to confirm the site is healing and to address bleeding concerns. The implant should be removed and replaced by 5 years.

References

1. MedlinePlus. Contraceptive Implant Insertion. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 46

Coronary Angioplasty

Overview

Coronary angioplasty is a minimally invasive procedure used to open clogged heart arteries. It utilizes a balloon-tipped catheter to widen the lumen and place a drug-eluting stent.

Alternative Names

Percutaneous Coronary Intervention, PCI, Balloon angioplasty

Principle

A balloon-tipped catheter is positioned across a narrowed coronary artery and inflated to crush the plaque and stretch the vessel wall. A stent is usually deployed to hold the artery open and prevent elastic recoil and restenosis.

History

Andreas Gruentzig performed the first successful balloon coronary angioplasty in Zurich in 1977. Coronary stents were introduced in the late 1980s.

Why It Is Done

Ordered for patients experiencing acute myocardial infarction (STEMI/NSTEMI), unstable angina, or stable angina refractory to medical therapy.

Is This a Primary or Secondary Test or Procedure

Primary emergency treatment for acute ST-elevation myocardial infarction (STEMI).

How It Is Performed

Under local anesthesia and conscious sedation. A catheter is inserted through the radial or femoral artery and guided to the heart. A balloon is inflated at the blockage site, and a stent is deployed.

Preparation

Fast for 6 hours. Establish IV access, obtain baseline lab work, and administer dual antiplatelet therapy (aspirin + clopidogrel/ticagrelor).

Contraindications and Precautions

None in acute life-threatening MI. Relative contraindications include severe renal failure (due to contrast) or active bleeding.

Risks

Coronary artery dissection or rupture, acute MI, contrast-induced nephropathy, bleeding at the access site, or stent thrombosis.

Aftercare and Recovery

Monitor vital signs and the puncture site for bleeding. Radially accessed patients can sit up immediately. Strictly adhere to dual antiplatelet therapy.

Outcome and Follow-up

Successful angioplasty restores normal coronary blood flow (grade 3 TIMI flow) and resolves chest pain/ECG changes.

Expected Results

TIMI Grade 3 flow in the target artery; stent fully expanded; stenosis reduced to <10

Follow-up

Cardiology clinic follow-up, daily dual antiplatelet therapy for 6-12 months, and cardiac rehabilitation.

References

1. MedlinePlus. Coronary Angioplasty. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 47

Cricothyrotomy

Overview

Cricothyrotomy is an emergency surgical procedure that opens the cricothyroid membrane to establish a definitive airway when endotracheal intubation is impossible or contraindicated. It provides direct access to the trachea for ventilation and oxygenation.

Alternative Names

Cricothyroidotomy, needle cricothyrotomy, surgical airway, emergency tracheostomy (less precisely)

Principle

The cricothyroid membrane lies between the thyroid and cricoid cartilages, beneath the skin of the anterior neck, and is relatively avascular and superficial. Incising or puncturing this membrane allows a tube or catheter to be placed into the subglottic trachea, bypassing the larynx and upper airway obstruction.

History

Emergency airway access through the cricothyroid membrane has been performed for over a century and became standardized in trauma and airway management algorithms. It remains the definitive rescue airway in the "cannot intubate, cannot oxygenate" scenario.

Why It Is Done

Performed in life-threatening airway obstruction or failure when mask ventilation and endotracheal intubation have failed or are impossible, including severe facial trauma, laryngeal obstruction, anaphylaxis, foreign body aspiration, or massive upper airway hemorrhage.

Is This a Primary or Secondary Test or Procedure

A secondary rescue procedure reserved for emergency situations when standard airway management fails, not a primary elective airway.

How It Is Performed

The patient is positioned with the neck extended, the laryngeal prominence is identified, and the cricothyroid membrane is palpated. In the open technique, a small vertical skin incision is made, the membrane is incised horizontally, and a small endotracheal or tracheostomy tube is guided into the trachea and secured. In the needle technique, a large-bore catheter is inserted through the membrane and connected to an oxygen source for jet ventilation.

Preparation

No preparation is possible in the emergency setting; positioning, sterilization, and immediate availability of equipment are the priorities. Consent is implied by the life-threatening nature of the situation.

Contraindications and Precautions

Relative contraindications include age under approximately 12 years (where needle cricothyrotomy may be preferred), laryngeal trauma or fracture that obliterates landmarks, and bleeding diathesis. Precautions include avoiding injury to the cricoid or thyroid cartilages and confirming intratracheal placement before ventilation.

Risks

Bleeding and hematoma, damage to the larynx or trachea, subglottic stenosis, infection, subcutaneous emphysema, and esophageal or vascular injury from misplacement.

Aftercare and Recovery

Ventilation is confirmed by chest rise and capnography, and the patient is converted to a formal tracheostomy or endotracheal airway as soon as conditions allow. The patient is monitored in an intensive care setting for bleeding, infection, and airway complications.

Outcome and Follow-up

Success is confirmed by visible chest rise, exhaled carbon dioxide, and oxygen saturation improvement after connection to a ventilation circuit. Failure to ventilate indicates esophageal or tissue placement requiring immediate repositioning.

Expected Results

A patent airway with adequate oxygenation and ventilation, stable tube position, and a subsequent planned transition to a definitive airway.

Follow-up

Formal tracheostomy or airway reassessment, chest radiography, and clinical evaluation for laryngeal or tracheal injury. Early conversion to a definitive airway and careful decannulation planning are indicated.

References

1. MedlinePlus. Cricothyrotomy. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 48

Cryotherapy

Overview

Cryotherapy uses extreme cold (typically liquid nitrogen at -196 C) to destroy abnormal or diseased tissue, such as warts, skin tags, or precancerous skin lesions.

Alternative Names

Cryosurgery, Skin freezing

Principle

Extreme cold produced by liquid nitrogen, cryoprobes, or other cryogens is applied to abnormal tissue, causing intracellular ice formation, cell membrane damage, and osmotic injury. These effects kill the target cells while surrounding tissue heals by secondary intention.

History

Dr. James Arnott first used cold mixtures to treat tumors in 1845. Liquid nitrogen became the standard cryogen in the 1940s-1950s.

Why It Is Done

Ordered to treat common warts, seborrheic keratoses, skin tags, actinic keratoses (precancerous lesions), or superficial basal cell carcinomas.

Is This a Primary or Secondary Test or Procedure

Primary treatment for common viral warts and actinic keratoses.

How It Is Performed

Liquid nitrogen is applied to the target skin lesion using a cotton-tipped applicator or a pressurized spray gun. The lesion is frozen for several seconds, allowed to thaw, and sometimes frozen again.

Preparation

No special preparation. The skin is cleaned before the procedure.

Contraindications and Precautions

Lesions suspected of melanoma (biopsy must be done instead), or patients with cold urticaria or cryoglobulinemia.

Risks

Local pain, blistering, hypopigmentation (white mark, common), hyperpigmentation, or scar formation.

Aftercare and Recovery

A blister will form at the site, which will crust over and peel off in 1-2 weeks. Keep the area clean and do not pop the blister.

Outcome and Follow-up

Successful treatment results in the sloughing off of the abnormal tissue and replacement with normal skin.

Expected Results

Destruction of the target lesion; healing of the skin with minimal scarring or discoloration.

Follow-up

Re-evaluation in 2-3 weeks; repeat freezing session if the lesion is not fully resolved.

References

1. MedlinePlus. Cryotherapy. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 49

Cystoscopy

Overview

Cystoscopy is an endoscopy of the urinary bladder via the urethra. It uses a rigid or flexible scope to inspect the bladder lining, urethra, and ureteral orifices.

Alternative Names

Cystourethroscopy, Bladder scope

Principle

A thin telescope with a lens and light source is passed through the urethra into the bladder, which is distended with sterile fluid. The bladder and urethral lining can be inspected directly, and instruments can be passed to biopsy lesions, remove stones, or treat abnormalities.

History

Philipp Bozzini invented the first endoscope ('Lichtleiter') in 1806. Max Nitze built the first modern electric cystoscope in 1877.

Why It Is Done

Ordered to investigate hematuria (blood in urine), recurrent urinary tract infections, urinary incontinence, dysuria, or suspected bladder tumors.

Is This a Primary or Secondary Test or Procedure

Primary diagnostic test for evaluating bladder pathology, urethral strictures, and hematuria.

How It Is Performed

The urethra is cleaned, and a local anesthetic gel is instilled. The cystoscope is inserted through the urethra into the bladder. Sterile water is used to fill and distend the bladder.

Preparation

Empty your bladder before the test. You may be asked to take a prophylactic antibiotic.

Contraindications and Precautions

Acute urinary tract infection (UTI) or acute prostatitis.

Risks

Urinary tract infection, hematuria, temporary dysuria (burning during urination), or urinary retention due to urethral swelling.

Aftercare and Recovery

Drink plenty of water. You may feel a burning sensation during urination for 24-48 hours. Warm baths can help soothe discomfort.

Outcome and Follow-up

Enables the urologist to detect stones, strictures, bladder cancer, or prostate enlargement.

Expected Results

Smooth, normal urethra; normal prostate size; pink, normal bladder wall with no masses, stones, or inflammation.

Follow-up

Bladder biopsy, urine cytology, or surgical intervention if abnormalities are found.

References

1. MedlinePlus. Cystoscopy. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 50

Defibrillation

Overview

Defibrillation delivers a therapeutic dose of electrical energy to the heart with a device called a defibrillator. This depolarizes the heart muscle, terminating dysrhythmias and allowing the SA node to re-establish a normal rhythm.

Alternative Names

Electrical shock therapy, Cardioversion

Principle

A powerful, unsynchronized electrical shock depolarizes the entire myocardium at once, terminating the disorganized activity of ventricular fibrillation or pulseless ventricular tachycardia. The resulting pause allows the heart's normal pacemaker to re-establish a coordinated rhythm.

History

Carl Ludwig and Florian Prevost demonstrated electrical termination of VF in dogs in 1899. Claude Beck performed the first successful human defibrillation during surgery in 1947.

Why It Is Done

Ordered immediately for pulseless Ventricular Fibrillation (VF) or pulseless Ventricular Tachycardia (VT) during cardiac arrest.

Is This a Primary or Secondary Test or Procedure

Primary curative treatment for shockable cardiac arrest rhythms (VF/pulseless VT).

How It Is Performed

Defibrillator pads are placed on the patient's chest (anterior-lateral or anterior-posterior). The machine is charged, a 'clear' command is given, and an electrical shock is delivered.

Preparation

Apply conductive gel or adhesive pads to the chest. Ensure the patient is not in standing water and no one is touching the patient.

Contraindications and Precautions

Asystole and Pulseless Electrical Activity (PEA) are non-shockable rhythms; defibrillation is contraindicated and CPR should continue.

Risks

Skin burns at the pad sites, skeletal muscle injury, or accidental shock to healthcare providers.

Aftercare and Recovery

Immediately resume chest compressions for 2 minutes before re-analyzing the heart rhythm.

Outcome and Follow-up

The rhythm is analyzed after 2 minutes of CPR to check if the shockable rhythm was terminated and replaced by an organized rhythm.

Expected Results

Termination of VF/VT; restoration of a perfusing sinus rhythm.

Follow-up

Antiarrhythmic medications (amiodarone), post-arrest care, or implantable cardioverter-defibrillator (ICD) placement.

References

1. MedlinePlus. Defibrillation. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 51

Dental Cleaning

Overview

Dental cleaning removes plaque, tartar (calculus), and stains from the teeth to prevent periodontal disease, gingivitis, and dental cavities.

Alternative Names

Prophylaxis, Dental scaling

Principle

Scaling instruments and ultrasonic devices remove plaque and calculus above and below the gumline, and polishing removes surface stains. Eliminating the bacterial biofilm and irritants prevents gingivitis, periodontitis, and caries.

History

Dental hygiene emerged as a formal profession in 1913 when Alfred Fones opened the first school of dental hygiene in Connecticut.

Why It Is Done

Ordered as a routine preventive procedure every 6 months to maintain oral health and screen for oral cancers.

Is This a Primary or Secondary Test or Procedure

Primary preventive treatment for gingivitis and periodontal disease.

How It Is Performed

A dental hygienist uses ultrasonic instruments and hand scalers to scrape tartar off teeth, polishes the teeth with a gritty paste, and performs flossing.

Preparation

Inform the hygienist if you require antibiotic prophylaxis before dental work (e.g., for specific heart valve conditions).

Contraindications and Precautions

None. Delay if the patient has severe acute oral infections or uncorrected coagulopathy.

Risks

Minor gum bleeding, mild tooth sensitivity, or rare transient bacteremia.

Aftercare and Recovery

Resume normal brushing and flossing. Avoid eating or drinking for 30 minutes if a fluoride treatment was applied.

Outcome and Follow-up

A dentist examines your clean teeth and X-rays for cavities, bone loss, or oral lesions.

Expected Results

Removal of all supragingival and subgingival plaque and calculus; pink, healthy gingiva; no active decay.

Follow-up

Routine dental cleaning in 6 months; restorative work if cavities were identified.

References

1. MedlinePlus. Dental Cleaning. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 52

Dental Crowns

Overview

A dental crown is a custom-made cap placed over a damaged or weakened tooth to restore its shape, strength, function, and appearance.

Why It Is Done

Crowns may be used for extensive decay, a fractured tooth, large fillings, root-canal-treated teeth, worn teeth, or to restore an implant.

How It Is Performed

The tooth is examined and prepared, an impression or digital scan is taken, and a temporary crown may be placed. The permanent crown is later cemented or bonded after fit and bite are checked.

Risks and Follow-up

Sensitivity, bite problems, loosening, fracture, recurrent decay, gum irritation, and need for replacement are possible. Brush, floss, and attend routine dental visits.

References

1. MedlinePlus. Dental Crowns. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 53

Dental Fillings

Overview

Dental fillings restore a tooth damaged by decay by removing the cavity and filling it with a durable material.

Alternative Names

Dental restoration; tooth filling; amalgam or composite filling

Principle

Carious tissue is removed, the cavity is shaped, and it is filled with amalgam, composite resin, or other material to restore function and prevent further decay.

History

Tooth filling has been practiced for centuries; amalgam appeared in the 19th century and modern tooth-colored composites in the late 20th century.

Why It Is Done

Performed for dental caries, broken or worn teeth, and to restore tooth structure.

Is This a Primary or Secondary Test or Procedure

Primary restorative treatment for dental decay.

How It Is Performed

Performed under local anesthesia. The decay is drilled away, the cavity is cleaned and shaped, and the filling material is placed, shaped, and cured.

Preparation

Clinical examination and radiographs to assess decay extent.

Contraindications and Precautions

Very large cavities may require crowns or root canal treatment.

Risks

Recurrent decay, filling failure or loss, sensitivity, and allergic reactions (rare).

Aftercare and Recovery

The patient avoids chewing on the filling briefly; sensitivity settles.

Outcome and Follow-up

Success is judged by a comfortable bite and no recurrent decay at review.

Expected Results

A well-sealed, functional restoration.

Follow-up

Routine examinations and radiographs.

References

1. MedlinePlus. Dental filling. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 54

Dental Implants

Overview

A dental implant is a biocompatible post placed in the jawbone to support a crown, bridge, or denture after tooth loss.

Why It Is Done

Implants restore chewing function, support dental prostheses, and help preserve jawbone structure when a patient has suitable bone and oral health.

How It Is Performed

After imaging and treatment planning, the implant is placed in the jaw and allowed to integrate with bone. An abutment and final crown or prosthesis are attached after healing.

Risks and Recovery

Pain, infection, nerve or sinus injury, failure of bone integration, peri-implant disease, and prosthesis complications are possible. Careful oral hygiene and regular dental follow-up are essential.

References

1. MedlinePlus. Dental Implants. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 55

Dentures

Overview

Dentures are removable dental prostheses that replace some or all missing teeth and surrounding oral structures.

Why It Is Done

They restore chewing, speech, and facial support when teeth are missing and fixed bridges or implants are unsuitable or not selected.

How It Is Performed

The dentist evaluates the mouth, takes impressions or scans, and provides trial and final prostheses. Complete dentures replace all teeth in an arch; partial dentures replace selected teeth and use supporting clasps or attachments.

Risks and Follow-up

Soreness, pressure areas, difficulty speaking or chewing, gagging, looseness, and gum irritation can occur. Remove and clean dentures as instructed and attend adjustments and regular oral examinations.

References

1. MedlinePlus. Dentures. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 56

Dermabrasion

Overview

Dermabrasion is a skin-resurfacing procedure in which the outer layers of damaged skin are mechanically abraded to improve its appearance and texture. It is used to treat scars, including acne scars, and to reduce the visibility of wrinkles, sun damage, and tattoos.

Alternative Names

Surgical skin planing; cosmetic skin resurfacing; microdermabrasion (superficial variant)

Principle

A rapidly rotating abrasive instrument removes the epidermis and superficial dermis, stimulating new collagen formation and re-epithelialization from adnexal structures in the deeper skin. Controlled depth of abrasion determines how aggressively the skin remodels while protecting deeper healing layers.

History

Dermabrasion was developed in the early 20th century for scar revision and gained popularity after World War II for treating acne scarring. Modern variants include microdermabrasion and laser resurfacing.

Why It Is Done

Ordered for acne scars, surgical and traumatic scars, facial wrinkles, actinic (sun) damage, and removal of superficial tattoos, usually when less invasive treatments have failed.

Is This a Primary or Secondary Test or Procedure

Primary therapeutic procedure for selected skin textural and scarring disorders.

How It Is Performed

The skin is cleaned, and a topical or injected anesthetic is applied, often with skin chilling or freezing. A handheld motorized abrader, such as a diamond burr or wire brush, is passed over the treatment area to a controlled depth. The wound is covered with a dressing or treated with an open-wound healing protocol.

Preparation

Avoidance of isotretinoin for at least six months before the procedure because of abnormal wound healing, and pretreatment with antiviral medication when treating the perioral region to prevent herpes simplex flares.

Contraindications and Precautions

Contraindications include active acne, skin infections, herpes simplex outbreak, and a history of keloid or hypertrophic scarring. Darker skin types have a higher risk of post-inflammatory pigmentary change and require extra caution.

Risks

Pain, swelling, redness, infection, scarring, hyperpigmentation or hypopigmentation, milia, and prolonged healing with deeper abrasion.

Aftercare and Recovery

The treated area crusts and heals over one to two weeks, with redness persisting for several weeks. Sun protection and gentle skin care are required during healing.

Outcome and Follow-up

Improvement in scar depth, skin texture, and wrinkle appearance is gradual over weeks to months as collagen remodeling continues. Persistent or severe textural defects may require repeat treatment.

Expected Results

Evenly resurfaced skin with a softer scar appearance and no significant pigmentary abnormality or scarring.

Follow-up

Patients are followed for wound healing and pigmentary change, and repeat sessions may be scheduled for optimal results. Procedures for further resurfacing include microdermabrasion and laser therapy.

References

1. MedlinePlus. Dermabrasion. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 57

Dialysis

Overview

Dialysis is the process of removing excess water, solutes, and toxins from the blood in patients whose kidneys can no longer perform these functions naturally (kidney failure).

Alternative Names

Renal replacement therapy, Hemodialysis, Peritoneal dialysis

Principle

Blood is cleansed across a semipermeable membrane: by diffusion against dialysis fluid in hemodialysis, or across the patient's own peritoneal membrane in peritoneal dialysis. Waste products and excess fluid move out of the blood and electrolytes are rebalanced.

History

Willem Kolff constructed the first working dialyzer in 1943 during the German occupation of the Netherlands, saving the first patient's life in 1945.

Why It Is Done

Ordered for acute kidney injury or end-stage renal disease (ESRD) to manage severe uremia, hyperkalemia, volume overload, or metabolic acidosis.

Is This a Primary or Secondary Test or Procedure

Primary treatment for life-threatening uremia, severe fluid overload, or drug toxicities.

How It Is Performed

Hemodialysis pumps blood through an external filter (dialyzer) and returns it. Peritoneal dialysis uses the lining of the abdomen as a filter, inserting dialysate fluid via a catheter.

Preparation

A vascular access (AV fistula, graft, or temporary catheter) must be surgically created prior to starting hemodialysis.

Contraindications and Precautions

No absolute contraindications for life-saving dialysis. Peritoneal dialysis is contraindicated in patients with severe abdominal adhesions.

Risks

Hypotension, muscle cramps, infection of the access site, bleeding, electrolyte imbalances, or dialysis disequilibrium syndrome.

Aftercare and Recovery

Monitor blood pressure, weight (to assess fluid removal), and post-dialysis laboratory values (BUN, creatinine, electrolytes).

Outcome and Follow-up

Efficacy is measured by Kt/V (a measure of urea clearance). A Kt/V of at least 1.2 is target for hemodialysis.

Expected Results

Adequate reduction in blood urea nitrogen (BUN) and creatinine; correction of electrolyte abnormalities (potassium, bicarbonate); normal volume status.

Follow-up

Regular nephrology clinic visits, access site monitoring, and dietary/fluid management.

References

1. MedlinePlus. Dialysis. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 58

Dilation and Curettage (D&C)

Overview

Dilation and curettage (D&C) involves dilating the cervix and scraping the uterine lining (endometrium) with a curette to remove tissue.

Alternative Names

D&C, Uterine scrapings

Principle

The cervix is dilated and a curette is used to scrape the uterine cavity, removing the endometrial tissue for diagnosis or to treat retained products of conception, abnormal bleeding, or other conditions.

History

Developed in the 19th century as a diagnostic and therapeutic tool for uterine bleeding and managing incomplete miscarriages.

Why It Is Done

Ordered to diagnose endometrial hyperplasia or cancer, treat heavy uterine bleeding, or clear retained tissue after a miscarriage or delivery.

Is This a Primary or Secondary Test or Procedure

Primary therapeutic method for heavy uterine bleeding refractory to medical therapy and for retained products of conception.

How It Is Performed

Performed under sedation or regional anesthesia. The cervix is dilated using metal dilators, and a curette (or suction device) is inserted into the uterus to scrape or suction the endometrial lining.

Preparation

Fast for 8 hours if sedation is planned. Have a negative pregnancy test (unless clearing retained products).

Contraindications and Precautions

Active pelvic inflammatory disease (PID), or suspected viable pregnancy (unless therapeutic abortion is intended).

Risks

Uterine perforation, cervical damage, post-procedure pelvic infection, or Asherman's syndrome (intrauterine adhesions).

Aftercare and Recovery

Mild cramping and spotting are common for a few days. Avoid tampons and sexual intercourse for 2 weeks to prevent infection.

Outcome and Follow-up

Tissue removed is sent to pathology to inspect for abnormal endometrial cells, polyps, or retained products of conception.

Expected Results

Successful evacuation of the uterine cavity; pathology shows benign secretory/proliferative endometrium or expected gestational tissue.

Follow-up

Pathology review; clinical follow-up in 2 weeks.

References

1. MedlinePlus. Dilation and Curettage (D&C). U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 59

Echocardiogram

Overview

An echocardiogram uses ultrasound waves to create moving images of the heart, valves, chambers, and blood flow.

Why It Is Done

It evaluates murmurs, heart failure, valve disease, cardiomyopathy, congenital heart disease, pericardial fluid, and suspected structural or functional abnormalities.

How It Is Performed

For a transthoracic study, a probe is moved over gelled areas of the chest. A transesophageal study passes a specialized probe into the esophagus after sedation to obtain closer images when needed.

Risks and Results

Transthoracic echocardiography is noninvasive. Transesophageal echocardiography carries sedation and esophageal-injury risks. Results describe chamber size, pumping function, valve structure, blood-flow gradients, and pressure estimates.

References

1. MedlinePlus. Echocardiogram. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 60

Electrocardiogram

Overview

An electrocardiogram (ECG or EKG) records the heart's electrical activity through electrodes placed on the skin.

Why It Is Done

It evaluates chest pain, palpitations, fainting, shortness of breath, arrhythmias, heart attack, conduction problems, and effects of medicines or electrolyte abnormalities.

How It Is Performed

The patient lies still while adhesive electrodes are placed on the chest and limbs. A resting tracing usually takes only a few minutes; ambulatory and exercise ECGs use related monitoring methods.

Risks and Results

A resting ECG is noninvasive and has no significant medical risk, although removing electrodes may irritate the skin. A clinician interprets rhythm, rate, intervals, conduction, and signs of ischemia or prior injury.

References

1. MedlinePlus. Electrocardiogram. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 61

Electroconvulsive Therapy

Overview

Electroconvulsive therapy (ECT) is a treatment that delivers a controlled electrical stimulus to the brain to induce a brief, generalized seizure, used for severe psychiatric disorders.

Alternative Names

ECT; shock therapy; electroshock treatment

Principle

Under anesthesia, a small electrical current is applied through scalp electrodes to provoke a therapeutic seizure. Repeated treatments produce changes in brain circuits and neurochemistry that relieve severe depression and other conditions.

History

ECT was introduced by Cerletti and Bini in 1938 and, despite stigma, remains among the most effective treatments for severe depression.

Why It Is Done

Performed for severe, psychotic, or catatonic depression, acute mania, and suicidality, especially when medications fail or rapid response is needed.

Is This a Primary or Secondary Test or Procedure

Primary treatment for severe, treatment-resistant psychiatric conditions.

How It Is Performed

Performed under general anesthesia with muscle relaxants and EEG and ECG monitoring. A brief electrical pulse induces a seizure lasting under a minute. A course typically includes 6-12 sessions.

Preparation

Pre-anesthesia evaluation, discontinuation of interacting medications where appropriate, and fasting before each session.

Contraindications and Precautions

Recent myocardial infarction, stroke, or intracranial mass with increased pressure are relative contraindications. Precaution: anesthesia and monitoring are mandatory.

Risks

Transient memory loss and confusion, headache, nausea, and cardiovascular changes.

Aftercare and Recovery

The patient recovers from anesthesia and is monitored; memory side effects usually resolve over days to weeks.

Outcome and Follow-up

Response is judged by improvement in mood, psychomotor status, and safety over the treatment course.

Expected Results

Substantial improvement in depressive or psychotic symptoms during the treatment course.

Follow-up

Maintenance ECT or medication adjustments prevent relapse.

References

1. MedlinePlus. Electroconvulsive therapy. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 62

Electrodesiccation and Curettage

Overview

Electrodesiccation and curettage (ED&C) is a dermatologic procedure in which a skin lesion is scraped away with a curette and the base is treated with a low-voltage electric current to destroy remaining tissue and control bleeding. It is used mainly to treat basal cell carcinoma and superficial skin lesions.

Alternative Names

ED&C; curettage and electrodesiccation; curettage and cautery; scrape and burn

Principle

The curette removes the soft, friable tumor tissue, which differs in consistency from normal skin, and electrodesiccation destroys residual malignant cells along the margins and base. Repeated cycles extend treatment depth and help achieve tumor clearance.

History

Curettage with electrosurgical destruction was developed in the early 20th century as a simple method for treating skin cancers and remains widely used for low-risk basal cell carcinoma.

Why It Is Done

Ordered for small, low-risk basal cell carcinomas, superficial squamous cell carcinoma in situ, actinic keratoses, seborrheic keratoses, and other benign or premalignant skin lesions when excision or Mohs surgery is not required.

Is This a Primary or Secondary Test or Procedure

Primary therapeutic procedure for selected low-risk skin lesions.

How It Is Performed

The area is cleaned and anesthetized with local anesthetic. The lesion is scraped with a dermal curette, the base is treated with an electrosurgical unit to desiccate the tissue, and the cycle is repeated two to three times. The wound is dressed and heals by secondary intention.

Preparation

Informed consent and review of the patient's bleeding risk and any implanted electronic devices, since electrosurgery may interfere with pacemakers or other implants.

Contraindications and Precautions

Precautions include lesions near the eye or nose (risk of poor cosmetic and functional results), large or deeply invasive tumors, and conditions with high recurrence risk where Mohs surgery or excision is preferred.

Risks

Pain, bleeding, infection, scarring, hypopigmentation, and recurrence of the treated lesion, particularly with larger or more aggressive tumors.

Aftercare and Recovery

The wound is kept clean and dressed until healing occurs over several weeks. The patient is advised on signs of infection, and the treated area heals with a flat scar.

Outcome and Follow-up

Successful treatment is indicated by complete removal of the lesion with healing and no local recurrence at follow-up. Recurrence may require a different treatment modality.

Expected Results

Complete elimination of the lesion with a cosmetically acceptable scar and no clinical evidence of recurrence.

Follow-up

Patients are reassessed for healing and recurrence, and periodic full skin examinations are recommended for those with a history of skin cancer. Mohs surgery or excision is considered for recurrent lesions.

References

1. MedlinePlus. Skin cancer - treatments. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 63

Electromyography and Nerve-Conduction Studies

Overview

Electromyography (EMG) and nerve-conduction studies assess the electrical function of muscles and peripheral nerves.

Why It Is Done

They help evaluate weakness, numbness, neuropathy, radiculopathy, nerve entrapment, muscle disease, and neuromuscular-junction disorders.

How It Is Performed

Surface electrodes measure nerve responses after small electrical stimuli. A fine needle electrode records electrical activity in selected muscles at rest and during contraction.

Risks and Preparation

Temporary discomfort, bruising, and rarely infection or bleeding can occur. Tell the examiner about anticoagulants, pacemakers, implanted stimulators, and bleeding disorders; avoid lotions on the test day.

References

1. MedlinePlus. Electromyography. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 64

Embolectomy and Thrombectomy

Overview

Embolectomy and thrombectomy are procedures that remove a blood clot from a blocked artery or vein, restoring blood flow to the affected organ or limb.

Alternative Names

Clot removal; mechanical thrombectomy; pulmonary embolectomy; surgical embolectomy

Principle

A catheter-based device or surgical approach captures or fragments the clot. Removing the obstruction restores perfusion and prevents irreversible tissue damage.

History

Surgical embolectomy was pioneered by Trendelenburg in 1908, and modern catheter-based thrombectomy devices have expanded treatment of stroke and venous and arterial occlusion.

Why It Is Done

Performed for acute arterial occlusion of a limb or viscera, large-vessel occlusion stroke, and massive or selected submassive pulmonary embolism.

Is This a Primary or Secondary Test or Procedure

Primary treatment for acute large-vessel occlusion.

How It Is Performed

Performed under imaging guidance (fluoroscopy or angiography). A catheter enters the vessel, and aspiration, stent-retrieval, or fragmentation removes the clot; open surgical embolectomy may be used for acute limb ischemia.

Preparation

Imaging (CT angiography or angiography), assessment of symptom onset time, and correction of coagulopathy.

Contraindications and Precautions

Hemorrhagic stroke (for stroke thrombectomy) and severe coagulopathy are contraindications.

Risks

Reperfusion injury, bleeding, vessel injury or dissection, distal embolization, and contrast nephropathy.

Aftercare and Recovery

Perfusion is monitored; anticoagulation is often continued and the cause of embolism is investigated.

Outcome and Follow-up

Restored flow is confirmed by angiography; clinical recovery depends on the duration of ischemia.

Expected Results

Complete restoration of vessel patency and tissue perfusion.

Follow-up

Imaging surveillance, anticoagulation, and treatment of the embolic source.

References

1. MedlinePlus. Embolectomy. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 65

Endometrial Ablation

Overview

Endometrial ablation is a minimally invasive procedure that destroys the lining of the uterus (endometrium) to treat abnormal uterine bleeding in women who do not wish to preserve fertility.

Alternative Names

Ablation of endometrium; endometrial resection; global endometrial ablation; NovaSure ablation

Principle

Thermal energy (radiofrequency, microwave, balloon heat, or cryotherapy) is applied to the endometrial surface, destroying the lining to a controlled depth. Scarring of the cavity reduces or eliminates menstrual bleeding.

History

Endometrial ablation was introduced with laser and resection techniques in the 1980s. Global ablation devices were approved in the late 1990s and early 2000s, simplifying the procedure.

Why It Is Done

Performed for heavy menstrual bleeding (menorrhagia) that has failed medical therapy and is confirmed benign, in women who have completed childbearing.

Is This a Primary or Secondary Test or Procedure

Primary alternative to hysterectomy for benign heavy menstrual bleeding.

How It Is Performed

Performed as an outpatient procedure, often under local anesthesia or sedation. The cervix is dilated, the cavity is measured and inspected, and the ablation device is inserted and activated for several minutes, delivering energy that destroys the endometrium.

Preparation

Preprocedure endometrial sampling to exclude hyperplasia or cancer, exclusion of pregnancy, and confirmation that childbearing is complete. Endometrial thinning agents may be used.

Contraindications and Precautions

Pregnancy, uterine infection, known or suspected uterine cancer, recent pregnancy, and desire for future fertility are contraindications. Precaution: uterine anomalies or a very thin myometrium increase perforation risk.

Risks

Uterine perforation, bleeding, infection, thermal injury to adjacent organs, hematometra, and postablation syndrome with cyclic pain.

Aftercare and Recovery

Mild cramping and watery discharge for a few weeks. Birth control is still recommended if there is a risk of pregnancy because conception can be dangerous.

Outcome and Follow-up

The procedure eliminates or markedly reduces menstrual bleeding in most women; a minority become amenorrheic. Bleeding may take several months to settle.

Expected Results

A significant reduction in menstrual flow with no complications, adequate for most women to avoid hysterectomy.

Follow-up

Review at 3-6 months for bleeding outcomes; persistent bleeding requires further evaluation for endometrial pathology.

References

1. MedlinePlus. Endometrial ablation. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 66

Endometrial Biopsy

Overview

An endometrial biopsy removes a small tissue sample from the lining of the uterus (endometrium) for histological examination.

Alternative Names

Uterine biopsy, Pipelle biopsy, Endometrial sampling

Principle

A thin sampling device or small curette is passed through the cervix into the uterine cavity and a strip of the endometrial lining is obtained. The tissue is examined microscopically for abnormal cells and to assess the hormonal phase of the lining.

History

Historically performed via dilation and curettage (D&C) in the OR. The introduction of the flexible suction curette (Pipelle) in the 1980s made it a safe office procedure.

Why It Is Done

Ordered for abnormal uterine bleeding, postmenopausal bleeding, or to screen for endometrial hyperplasia or cancer.

Is This a Primary or Secondary Test or Procedure

Primary diagnostic test for abnormal uterine bleeding in women over 45 or with risk factors.

How It Is Performed

In an office setting without anesthesia. A speculum is inserted, the cervix is cleaned, and a thin flexible tube (Pipelle) is inserted into the uterus to collect a tissue sample via suction.

Preparation

Mandatory negative pregnancy test; taking ibuprofen 1 hour prior can reduce cramping.

Contraindications and Precautions

Pregnancy, active pelvic infection, or severe cervical stenosis.

Risks

Uterine cramping, light vaginal bleeding/spotting, vasovagal reaction, or pelvic infection (rare).

Aftercare and Recovery

Expect cramping and light spotting for 1-2 days; avoid tampons and sexual intercourse for 2-3 days.

Outcome and Follow-up

Histology results describe the endometrium (e.g., proliferative, secretory, hyperplasia, or carcinoma).

Expected Results

Secretory or proliferative endometrium matching menstrual cycle phase; no hyperplasia or malignancy.

Follow-up

Review of pathology report; further treatment (hormones or surgery) if hyperplasia or cancer is found.

References

1. MedlinePlus. Endometrial Biopsy. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 67

Endoscopic Mucosal Resection (EMR)

Overview

Endoscopic mucosal resection (EMR) is an endoscopic technique used to remove superficial lesions of the gastrointestinal tract, such as polyps, adenomas, or early cancers confined to the mucosa and superficial submucosa. It is both a diagnostic and therapeutic procedure because the resected specimen is examined to determine the depth of invasion and completeness of removal.

Alternative Names

EMR, endoscopic polypectomy, wide-field endoscopic mucosal resection, submucosal injection polypectomy, cap-assisted mucosal resection

Principle

EMR uses a submucosal injection of saline or another solution to lift the lesion away from the underlying muscularis propria, creating a safety cushion. The elevated tissue is then snared and resected with electrocautery, allowing en bloc or piecemeal removal of the lesion. Histologic evaluation of the specimen determines whether the lesion is confined to the mucosa or has invaded the deeper layers. The technique relies on the principle that superficial lesions without deep invasion can be cured with endoscopic resection alone.

History

EMR was developed in Japan in the 1980s and 1990s for the treatment of early gastric cancer and later applied to the esophagus, colon, and duodenum. It represents a cornerstone of endoscopic management of early gastrointestinal neoplasia and avoids surgery for many patients.

Why It Is Done

EMR is ordered when endoscopy identifies a polyp, adenoma, or early cancer that is suspected to be confined to the superficial layers of the bowel wall. It is used to treat and stage lesions in the esophagus (including Barrett-associated dysplasia), stomach, duodenum, and colorectum. The goal is curative resection of the lesion while avoiding major surgery and its associated morbidity.

Is This a Primary or Secondary Test or Procedure

It is a primary therapeutic procedure for lesions deemed endoscopically resectable by careful inspection and optical assessment. It may be secondary when a previously biopsied or incompletely removed lesion requires definitive removal or staging.

How It Is Performed

A submucosal injection of normal saline with or without epinephrine and methylene blue is used to lift the lesion and assess for a non-lifting sign. A snare is placed around the lifted tissue, and electrocautery is applied to resect the lesion either en bloc or piecemeal. The specimen is retrieved, oriented, and submitted for histopathologic examination, and resection site clips may be placed to prevent bleeding.

Preparation

Colonoscopic EMR requires a full bowel preparation, whereas upper endoscopic EMR requires fasting. Anticoagulant and antiplatelet medications are typically held according to the risk of bleeding, and informed consent is obtained. Imaging and prior histology are reviewed to confirm that the lesion is amenable to endoscopic resection.

Contraindications and Precautions

Lesions with deep submucosal invasion, poor lifting, ulceration, or signs of invasive cancer on optical assessment are relative contraindications because of high risk of incomplete resection. Coagulopathy and inability to tolerate sedation or endoscopy preclude the procedure. Large, flat, or circumferential lesions carry higher risks of perforation, bleeding, and recurrence.

Risks

Major risks include bleeding, perforation, and postpolypectomy syndrome with pain and fever from transmural thermal injury. Incomplete resection, stricture formation, and local recurrence are additional risks, especially with piecemeal removal of large lesions.

Aftercare and Recovery

Patients are monitored for bleeding or perforation and observed for a period appropriate to the site and size of resection. Most resume a normal diet the same day or within a day, with activity restrictions as directed. A repeat endoscopy at 3–6 months is scheduled to assess the resection site for residual or recurrent tissue.

Outcome and Follow-up

The pathology report indicates whether the lesion was completely removed, the depth of invasion, and any adverse features such as poor differentiation or lymphovascular invasion. Complete removal of a superficial lesion is considered curative, whereas deep invasion or positive margins requires further endoscopic, surgical, or oncologic management.

Expected Results

There is no normal result because EMR treats a lesion; the desired outcome is complete histologic resection of the neoplastic tissue. Success is defined by negative resection margins and absence of deep invasion.

Follow-up

Surveillance endoscopy is performed at 3–6 months and then at interval guidelines to detect recurrence at the resection site and metachronous lesions. If histology shows deep invasion or high-risk features, surgical resection with lymphadenectomy may be recommended.

References

1. MedlinePlus. Endoscopic Mucosal Resection. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 68

Endoscopic Retrograde Cholangiopancreatography

Overview

Endoscopic retrograde cholangiopancreatography (ERCP) is an endoscopic procedure that examines and treats the bile ducts, gallbladder, and pancreatic ducts.

Alternative Names

ERCP; endoscopic cholangiopancreatography; biliary endoscopy

Principle

An endoscope is passed to the duodenum, and the ampulla of Vater is cannulated. Contrast is injected to image the biliary and pancreatic ducts, and instruments can open, dilate, or stent these ducts and retrieve stones.

History

ERCP was developed in the late 1960s and 1970s, initially as a diagnostic tool, and evolved into a primarily therapeutic procedure after endoscopic sphincterotomy was introduced in 1974.

Why It Is Done

Performed for obstructive jaundice, choledocholithiasis, cholangitis, bile leaks, and pancreatic disease such as strictures, pseudocysts, and some tumors, and for palliation of malignant obstruction with stents.

Is This a Primary or Secondary Test or Procedure

Primary therapeutic procedure for biliary and pancreatic obstruction.

How It Is Performed

Performed under conscious sedation or general anesthesia. The endoscope reaches the duodenum, the ampulla is cannulated, contrast outlines the ducts under fluoroscopy, and interventions such as sphincterotomy, stone extraction, dilation, or stenting are performed.

Preparation

NPO before the procedure, IV fluids, and pre-procedure labs including liver tests and coagulation studies. Antibiotics are given in selected cases.

Contraindications and Precautions

Severe acute pancreatitis without obstruction, coagulopathy, and inability to tolerate sedation are relative contraindications. Precaution: post-ERCP pancreatitis is the most common complication.

Risks

Post-ERCP pancreatitis, bleeding, perforation, cholangitis, sedation-related complications, and adverse reactions to contrast.

Aftercare and Recovery

The patient is observed for signs of pancreatitis or bleeding; blood tests and imaging may be checked the next day.

Outcome and Follow-up

Findings may show stones, strictures, leaks, or malignancy. Successful sphincterotomy and drainage are confirmed by contrast and symptom improvement.

Expected Results

A patent, non-obstructed biliary and pancreatic ductal system with no stones, stricture, or leak.

Follow-up

Liver function tests and imaging follow-up; repeat ERCP may be needed for stent exchange or recurrent stones.

References

1. MedlinePlus. Endoscopic retrograde cholangiopancreatography (ERCP). U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 69

Endoscopic Sleeve Gastroplasty

Overview

Endoscopic sleeve gastroplasty (ESG) is a minimally invasive procedure in which the stomach is reduced in size with internal sutures placed through an endoscope, restricting food intake and promoting weight loss in patients with obesity.

Alternative Names

ESG; endoscopic suturing of the stomach; sleeve gastroplasty without surgery; gastric sleeve endoscopy

Principle

An endoscopic suturing device places a series of full-thickness stitches in the gastric wall, creating a narrow tubular channel along the lesser curvature and markedly reducing the stomach's capacity, which promotes early satiety and reduced intake.

History

Endoscopic sleeve gastroplasty was developed in the 2010s as a less invasive alternative to laparoscopic sleeve gastrectomy and has gained acceptance as a primary or adjunctive weight-loss procedure.

Why It Is Done

Ordered for patients with class I or II obesity or with obesity-related complications who have failed lifestyle measures, and who prefer or qualify for an endoscopic rather than a surgical weight-loss procedure.

Is This a Primary or Secondary Test or Procedure

A primary therapeutic procedure for weight management in selected patients.

How It Is Performed

Under general anesthesia, an endoscope with a suturing device is passed into the stomach. Sutures are placed along the gastric wall to fashion the sleeve, after which the scope is removed. The procedure usually lasts under an hour.

Preparation

Preoperative evaluation including nutritional assessment and exclusion of gastric pathology, review of medications, and a clear liquid diet before the procedure.

Contraindications and Precautions

Contraindications include severe gastroparesis, gastric ulceration, prior gastric surgery, uncontrolled psychiatric disease, and bleeding disorders. Precautions include pregnancy and large hiatus hernias.

Risks

Nausea and vomiting after the procedure, pain, and rarely bleeding, perforation, suture disruption, or gastroesophageal reflux. Weight regain may occur over time.

Aftercare and Recovery

The patient follows a staged diet, progressing from liquids to soft foods over several weeks, and is monitored with regular follow-up. Nausea is managed with antiemetics.

Outcome and Follow-up

Weight loss typically begins in the first weeks and continues over months. The degree of loss is assessed against the patient's goals and the expected range for the procedure.

Expected Results

Significant and sustained weight loss with reduction of obesity-related conditions and no serious complications.

Follow-up

Ongoing nutritional monitoring and lifestyle support are required, and repeated or additional procedures may be considered if weight loss is insufficient or weight is regained.

References

1. MedlinePlus. Weight-loss procedures. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 70

Endoscopic Ultrasound (EUS)

Overview

Endoscopic ultrasound (EUS) combines endoscopy and ultrasound to obtain detailed images and information about the digestive tract and the surrounding tissue, organs, and lymph nodes.

Alternative Names

EUS, Echo-endoscopy

Principle

An endoscope with a small ultrasound transducer at its tip is passed into the esophagus, stomach, or duodenum, placing the probe close to the organs. Sound waves create detailed images of the gut wall and adjacent structures such as the pancreas, bile ducts, and lymph nodes, and fine-needle aspiration can sample suspicious lesions.

History

Developed in the early 1980s, EUS integrated high-frequency ultrasound transducers onto flexible endoscope tips, enabling high-resolution mediastinal and abdominal imaging.

Why It Is Done

Ordered to evaluate pancreatic cysts, tumors, gallstones in the bile duct, determine the depth of gastrointestinal tract cancer invasion, or obtain biopsies of deep tissues.

Is This a Primary or Secondary Test or Procedure

Primary diagnostic and staging tool for pancreatico-biliary cancers and submucosal lesions.

How It Is Performed

Performed under deep sedation. A specialized endoscope with an ultrasound probe at the tip is passed down your esophagus or into your rectum to scan adjacent structures.

Preparation

Fast for 6 to 8 hours. Arrange for transportation home due to sedation.

Contraindications and Precautions

Gastrointestinal perforation, severe coagulation disorders (if fine needle biopsy is planned).

Risks

Perforation, bleeding, pancreatitis (especially if biopsy of the pancreas is performed), or sedation complications.

Aftercare and Recovery

Monitor in recovery. Avoid driving for 24 hours. Sore throat or mild abdominal discomfort may occur.

Outcome and Follow-up

Provides highly detailed ultrasound images of the walls of the esophagus, stomach, duodenum, pancreas, and bile ducts. Helps stage tumors accurately.

Expected Results

Normal wall layers of the GI tract; normal pancreas, bile ducts, and adjacent lymph nodes; no masses or cysts.

Follow-up

Biopsy review, oncology staging discussion, or surgical consult.

References

1. MedlinePlus. Endoscopic Ultrasound (EUS). U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 71

Endoscopy (Upper GI)

Overview

An upper GI endoscopy uses a flexible scope with a light and camera (endoscope) to visualize the mucosal lining of the esophagus, stomach, and duodenum.

Alternative Names

Esophagogastroduodenoscopy, EGD, Upper endoscopy

Principle

A flexible endoscope is passed through the mouth into the esophagus, stomach, and duodenum. Air distends the lumen and the camera gives direct views of the lining, while instruments passed through the scope allow biopsy and treatment.

History

Adolf Kussmaul performed the first gastroscopy in 1868 using a rigid tube. Basil Hirschowitz developed the first fully flexible fiberoptic endoscope in 1957.

Why It Is Done

Ordered to investigate symptoms of persistent heartburn, nausea, vomiting, difficulty swallowing, abdominal pain, or upper gastrointestinal bleeding.

Is This a Primary or Secondary Test or Procedure

Primary diagnostic and therapeutic modality for upper gastrointestinal diseases.

How It Is Performed

You lie on your left side. Sedative medication is given IV, and a local anesthetic is sprayed in your throat. The endoscope is gently passed down your esophagus.

Preparation

Fast (no food or liquids) for 6 to 8 hours before the procedure. Arrange for a ride home due to sedation.

Contraindications and Precautions

Suspected visceral perforation, severe shock, or unstable cervical spine.

Risks

Bleeding, infection, or perforation of the GI tract lining (<1 in 1,000 cases). Risks associated with sedation (respiratory depression).

Aftercare and Recovery

Rest in recovery until sedation wears off. Do not drive or sign legal documents for 24 hours. Your throat may feel sore for a day.

Outcome and Follow-up

The physician inspects the lining for inflammation, ulcers, tumors, or bleeding. Biopsies may be taken for pathological evaluation.

Expected Results

Healthy, pink, smooth mucosal lining in the esophagus, stomach, and duodenum; no ulcers, bleeding, or masses.

Follow-up

H. pylori testing of biopsy, PPI therapy, or repeat endoscopy.

References

1. MedlinePlus. Endoscopy (Upper GI). U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 72

Endotracheal Intubation

Overview

Endotracheal intubation involves inserting a flexible plastic tube (endotracheal tube) through the mouth into the trachea to maintain a patent airway and facilitate mechanical ventilation.

Alternative Names

Intubation, Airway placement

Principle

A laryngoscope blade lifts the tongue and epiglottis to expose the glottis, and an endotracheal tube is advanced through the vocal cords into the trachea. Inflating the cuff seals the airway, permitting positive-pressure ventilation and protecting the lungs from aspiration.

History

William Macewen performed the first tactile oral intubation in 1878. Chevalier Jackson developed the laryngoscope in 1907, enabling direct visual intubation.

Why It Is Done

Ordered for patients with respiratory failure, airway obstruction, severe coma (Glasgow Coma Scale <= 8), or during general anesthesia.

Is This a Primary or Secondary Test or Procedure

Primary definitive method for airway management and mechanical ventilation.

How It Is Performed

A laryngoscope is used to visualize the vocal cords. The endotracheal tube is passed through the vocal cords into the mid-trachea. The cuff at the end of the tube is inflated.

Preparation

Pre-oxygenate the patient. Prepare suction, bag-valve mask, laryngoscope, tube, stylet, and medications (induction agent and paralytic).

Contraindications and Precautions

None in emergency airway compromise. Relative contraindications include severe cervical spine injury or facial trauma (which may require a surgical airway).

Risks

Esophageal intubation, hypoxia, dental injury, vocal cord injury, aspiration of gastric contents, or mainstem bronchus intubation.

Aftercare and Recovery

Confirm tube placement using end-tidal CO2 (capnography) and bilateral breath sounds. Secure the tube and obtain a chest X-ray to confirm tip position (mid-trachea).

Outcome and Follow-up

Positive capnography waveform and equal breath sounds verify correct placement. Chest X-ray should show the tube tip 3-5 cm above the carina.

Expected Results

Tube tip located in the mid-trachea; symmetric chest rise; bilateral breath sounds; end-tidal CO2 detection.

Follow-up

Mechanical ventilation management, sedation titration, and daily evaluation for weaning and extubation.

References

1. MedlinePlus. Endotracheal Intubation. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 73

Endovenous Ablation

Overview

Endovenous ablation closes a faulty superficial leg vein (most often the great saphenous vein) using heat or adhesive to treat varicose veins.

Alternative Names

Endovenous laser ablation; radiofrequency ablation of veins; vein closure; venaseal

Principle

A catheter inserted into the vein delivers laser, radiofrequency energy, or glue, damaging the vein wall so the vein closes and is reabsorbed, redirecting blood to healthy veins.

History

Endovenous ablation was introduced in the late 1990s and has largely replaced surgical vein stripping for truncal incompetence.

Why It Is Done

Performed for symptomatic varicose veins causing aching, swelling, or skin changes due to saphenous reflux.

Is This a Primary or Secondary Test or Procedure

Primary treatment for truncal varicose vein disease.

How It Is Performed

Performed under local anesthesia with ultrasound guidance. The catheter is advanced to the vein, and energy or glue is delivered as it is withdrawn.

Preparation

Duplex ultrasound confirms reflux and vein anatomy.

Contraindications and Precautions

Deep vein thrombosis, arterial disease, and pregnancy are relative contraindications.

Risks

Bruising, pain, nerve paresthesia, thrombophlebitis, and deep vein thrombosis.

Aftercare and Recovery

Compression stockings and walking; the treated vein fades over weeks.

Outcome and Follow-up

Ultrasound confirms closure of the treated vein.

Expected Results

A closed treated vein with symptom relief.

Follow-up

Ultrasound follow-up; sclerotherapy for residual branches.

References

1. MedlinePlus. Endovenous laser treatment for varicose veins. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 74

Enteral Feeding Tube Placement

Overview

Enteral feeding tube placement is the insertion of a tube into the gastrointestinal tract to deliver nutrition, fluids, and medications directly to the stomach or small intestine. Access may be short-term (nasogastric or nasoenteric tubes) or long-term (gastrostomy or jejunostomy).

Alternative Names

Nasogastric (NG) tube placement, Nasojejunal (NJ) tube placement, Gastrostomy tube (G-tube), Percutaneous endoscopic gastrostomy (PEG), Jejunostomy tube

Principle

Enteral feeding delivers formula directly into the gastrointestinal tract, preserving gut mucosal integrity and using the body's normal digestive and absorptive pathways. Nasoenteric tubes are passed through the nose into the stomach or small bowel, with position confirmed by pH of aspirate or radiography. Long-term access is created surgically or endoscopically by placing a catheter directly through the abdominal wall into the stomach or jejunum.

History

Nasogastric feeding was described in the 19th century. Percutaneous endoscopic gastrostomy, introduced in 1980, became the standard approach for long-term enteral access.

Why It Is Done

Enteral feeding is ordered when a patient cannot eat enough by mouth but has a functioning gastrointestinal tract, as in dysphagia, stroke, major trauma, burns, severe malnutrition, or prolonged critical illness. It provides nutrition, hydration, and medications and may also be used for gastric decompression.

Is This a Primary or Secondary Test or Procedure

Primary therapeutic procedure for delivering enteral nutrition when oral intake is inadequate or unsafe.

How It Is Performed

For nasoenteric tubes, the tube is lubricated and passed through the nose into the stomach, with advancement into the small bowel assisted by technique or image guidance. For gastrostomy, an endoscope identifies a safe site and the tube is placed percutaneously through the abdominal wall. Position is confirmed by aspirate pH or imaging before feeding begins.

Preparation

Informed consent is obtained; nasoenteric placement usually requires no fasting, whereas gastrostomy requires fasting and often prophylactic antibiotics. Coagulation status is checked before percutaneous placement, and cervical spine clearance is needed before nasal passage in trauma patients.

Contraindications and Precautions

Enteral feeding is avoided in bowel obstruction, severe ileus, or peritonitis. Nasoenteric tubes are avoided with facial trauma, skull base fractures, or esophageal strictures, and percutaneous placement is avoided with coagulopathy, ascites, or anatomic barriers. Use caution in patients at high risk for aspiration.

Risks

Misplacement into the airway, aspiration pneumonia, epistaxis, and tube dislodgment or clogging. Percutaneous placement carries risks of bleeding, bowel perforation, peritonitis, and site infection.

Aftercare and Recovery

Tube position is verified before each use, and feedings are started slowly and advanced as tolerated. The site is cleaned daily, and the tube is flushed after each use to prevent clogging and infection.

Outcome and Follow-up

Outcome is determined by tolerance of the feeding regimen and the absence of complications rather than by a numeric value. Success is judged by weight stabilization or gain and achievement of metabolic targets.

Expected Results

There is no quantitative normal value; the goal is a well-tolerated feeding schedule, stable weight, and no evidence of aspiration or tube-related complications.

Follow-up

Nutritional assessment and weight monitoring continue, with imaging if tube displacement is suspected. Long-term plans include transitioning back to oral feeding when safe or exchanging to a low-profile device.

References

1. MedlinePlus. Enteral feeding tube placement. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 75

Epidural Blood Patch

Overview

An epidural blood patch is the injection of the patient's own blood into the epidural space to seal a cerebrospinal fluid (CSF) leak. It is the definitive treatment for post-dural-puncture headache.

Alternative Names

Blood patch; autologous epidural blood patch

Principle

Autologous blood injected into the epidural space forms a clot that patches the dural puncture site, stopping the continued leak of CSF. Restoring CSF volume and pressure relieves the traction headache that results from intracranial hypotension.

History

The epidural blood patch was introduced by Charles B. Gormley in 1960 and rapidly became the standard treatment for post-dural-puncture headache after the report of larger series in the 1970s.

Why It Is Done

Ordered for post-dural-puncture headache that is severe, persists despite conservative therapy such as bed rest, hydration, and caffeine, or interferes with daily activity, and that follows lumbar puncture, spinal anesthesia, or epidural catheter placement.

Is This a Primary or Secondary Test or Procedure

Primary therapeutic procedure for post-dural-puncture headache.

How It Is Performed

Blood is drawn from a peripheral vein under sterile conditions. The patient is positioned prone or lateral, and a needle is placed into the epidural space at the level of or below the original dural puncture, often with fluoroscopic guidance. Fifteen to thirty milliliters of blood are injected slowly and the needle is withdrawn.

Preparation

Informed consent and review of bleeding risk and any active infection. The patient should be volume-replete, and blood is collected with aseptic technique before injection.

Contraindications and Precautions

Contraindications include local infection at the puncture site, coagulopathy or anticoagulation, patient refusal, and fever or suspected bacteremia. Precaution: placement above the original puncture site may be less effective.

Risks

Immediate backache and leg pain at injection, headache recurrence if the patch fails, and rarely infection, bleeding, or nerve injury. Repeat patches may be needed in a minority of cases.

Aftercare and Recovery

The patient lies flat for one to two hours and is advised to avoid heavy lifting and straining for several days. Headache typically resolves within hours to a day.

Outcome and Follow-up

Resolution or marked improvement of the headache and ability to resume upright activity indicate success. Persistent headache suggests an ongoing or additional leak, prompting imaging or a repeat patch.

Expected Results

Complete resolution of postural headache with return to normal upright posture without symptoms.

Follow-up

Patients with recurrent symptoms may undergo MRI to assess for persistent CSF leak and may require a second blood patch or further investigation of intracranial hypotension.

References

1. MedlinePlus. Epidural blood patch. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 76

Epidural Steroid Injection

Overview

An epidural steroid injection delivers corticosteroid into the epidural space around the spinal nerve roots to treat radicular pain from disc herniation, stenosis, or inflammation.

Alternative Names

Epidural injection; transforaminal epidural block; interlaminar epidural steroid injection; ESI

Principle

Corticosteroid is injected into the epidural space, where it reduces inflammation and swelling around irritated nerve roots, relieving radicular pain. Local anesthetic may be added for immediate relief and diagnosis.

History

Epidural steroid injections for sciatica were introduced in the 1950s and became one of the most commonly performed interventional pain procedures, refined with fluoroscopic guidance in recent decades.

Why It Is Done

Performed for radicular back or neck pain with nerve root irritation from disc herniation, spinal stenosis, or postoperative scarring that has not responded to conservative therapy.

Is This a Primary or Secondary Test or Procedure

Secondary treatment for radicular pain; primary interventional therapy when conservative care fails.

How It Is Performed

Performed in an outpatient setting with fluoroscopic guidance. The patient lies prone, the level is identified, and a needle is advanced into the epidural space using a transforaminal, interlaminar, or caudal approach. Contrast injection confirms placement, and corticosteroid is injected.

Preparation

Anticoagulants are held per guidelines, informed consent is obtained, and patients are observed briefly after the procedure. Fasting is not typically required.

Contraindications and Precautions

Active infection, bleeding disorder, anticoagulation, allergy to injectate, and local malignancy are contraindications. Precaution: transforaminal cervical and high lumbar injections carry a risk of spinal cord injury.

Risks

Transient pain flare, bleeding, infection, dural puncture with headache, nerve injury, increased blood glucose, and very rarely spinal cord injury or infarction.

Aftercare and Recovery

Patients may experience immediate relief from local anesthetic and delayed steroid effect over 3-7 days. Activity is limited briefly, and a series of injections may be planned.

Outcome and Follow-up

Relief of radicular pain for weeks to months confirms an inflammatory nerve root component; the response guides repeat treatment or surgical referral.

Expected Results

Significant reduction in radicular pain and improved function lasting weeks to months without complication.

Follow-up

Clinical reassessment after each injection; imaging or surgical evaluation if no durable benefit is obtained after a limited series.

References

1. MedlinePlus. Epidural steroid injection. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 77

Episiotomy

Overview

An episiotomy is a surgical incision in the perineum made during vaginal delivery to enlarge the vaginal opening.

Why It Is Done

It is not routine and may be considered when rapid delivery is needed, an instrument-assisted delivery requires more room, or severe perineal injury appears likely without an incision.

How It Is Performed

After local anesthetic when time permits, an incision is made in the perineum during the pushing stage. After delivery and examination for additional injury, the incision is repaired with absorbable sutures.

Risks and Recovery

Pain, bleeding, infection, wound separation, hematoma, painful intercourse, and extension into a deeper tear are possible. Keep the area clean, use comfort measures as directed, and attend follow-up if healing is abnormal.

References

1. MedlinePlus. Episiotomy. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 78

Esophageal Manometry

Overview

Esophageal manometry measures the strength and coordination of muscle contractions in the esophagus and the function of its upper and lower sphincters.

Why It Is Done

It is used to evaluate difficulty swallowing, chest pain not caused by the heart, regurgitation, or suspected motility disorders such as achalasia. It may also be performed before selected anti-reflux procedures.

How It Is Performed

After the nose and throat are numbed, a thin pressure-sensing catheter is passed through the nose into the stomach. The patient swallows small amounts of water while the catheter records pressure and muscle coordination. The catheter is then removed.

Preparation

The patient usually fasts for several hours. Some medicines that affect esophageal movement may need to be paused under medical guidance. Tell the clinician about nasal obstruction, bleeding problems, or medicines that increase bleeding risk.

Risks and Results

Temporary nasal or throat discomfort, gagging, watering of the eyes, and minor nosebleeding can occur. Results show the timing and strength of contractions and sphincter relaxation. Abnormal patterns may support diagnoses such as achalasia or esophageal spasm.

References

1. MedlinePlus. Esophageal manometry. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 79

Esophageal pH Monitoring

Overview

Esophageal pH monitoring measures how often and how long stomach acid or reflux reaches the esophagus.

Why It Is Done

It helps confirm or characterize GERD when symptoms are unclear, endoscopy is normal, symptoms persist despite treatment, or anti-reflux surgery is being considered.

How It Is Performed

A thin catheter passed through the nose or a wireless capsule attached during endoscopy records acid exposure for approximately 24 to 96 hours. The patient logs meals, symptoms, sleep, and medicines.

Risks and Results

Temporary nose or throat discomfort, gagging, and minor bleeding can occur. Results include acid-exposure time and association between reflux episodes and symptoms; interpretation depends on whether acid-suppressing medicine is continued or stopped.

References

1. MedlinePlus. Esophageal pH Monitoring. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 80

Extracorporeal Membrane Oxygenation (ECMO)

Overview

Extracorporeal membrane oxygenation (ECMO) is a form of mechanical life support that temporarily replaces the function of the heart or lungs using an external pump and oxygenator. Blood is drained from the patient, oxygenated outside the body, and returned to the circulation.

Alternative Names

ECMO, extracorporeal life support (ECLS), venovenous ECMO (VV-ECMO), venoarterial ECMO (VA-ECMO), heart-lung bypass

Principle

A membrane oxygenator adds oxygen and removes carbon dioxide from blood while a mechanical pump maintains flow, allowing the native heart and lungs time to recover. Venovenous (VV) ECMO provides isolated respiratory support, while venoarterial (VA) ECMO supports both cardiac and pulmonary function. Cannula placement and circuit configuration determine the mode of support.

History

The first successful clinical use of ECMO occurred in 1972, with early trials in neonatal respiratory failure showing benefit. Its use expanded substantially following the 2009 H1N1 influenza pandemic and the COVID-19 pandemic.

Why It Is Done

Initiated for severe, potentially reversible respiratory or cardiac failure refractory to maximal conventional therapy, including hypoxemic or hypercapnic respiratory failure, cardiogenic shock, and cardiac arrest. It also supports patients as a bridge to transplant, to ventricular assist device placement, or to recovery.

Is This a Primary or Secondary Test or Procedure

A rescue therapy used when primary treatments fail; it is supportive, not curative, and always temporary.

How It Is Performed

Cannulas are inserted percutaneously or surgically into large veins (VV-ECMO) or a vein and artery (VA-ECMO). Blood flows through the circuit to the pump and oxygenator and returns warmed to the patient, while anticoagulation is infused to prevent thrombosis. A perfusionist and critical care team manage the circuit continuously.

Preparation

Echocardiography or bedside ultrasound to guide cannulation and rule out right heart thrombus. Confirmation of type and crossmatch compatibility for circuit priming, and placement of invasive monitoring lines.

Contraindications and Precautions

Contraindications include irreversible end-organ failure, advanced age with prohibitive comorbidity, severe bleeding, and contraindication to anticoagulation. Precautions include careful patient selection and neuroprognostication, as outcomes depend heavily on candidacy.

Risks

Major bleeding, circuit thrombosis, hemolysis, infection, renal failure, limb ischemia from arterial cannulation, and neurologic complications including stroke. Mechanical or pump failure is rare but catastrophic.

Aftercare and Recovery

Patients are managed in an intensive care unit with daily circuit and end-organ assessments while underlying disease is treated. Support is weaned as native function recovers, and decannulation occurs once the patient is stable on conventional support.

Outcome and Follow-up

Improvement is tracked by arterial blood gases, hemodynamics, and echocardiographic function of the native heart and lungs. Persistent organ failure or failure to wean portends poor outcome and prompts reassessment of goals of care.

Expected Results

Stable oxygenation and ventilation on reduced circuit flows, adequate perfusion and mixed venous saturation, and recovery of native cardiac and respiratory function permitting decannulation.

Follow-up

After decannulation, continued rehabilitation, imaging for sequelae of critical illness, and long-term neurocognitive and pulmonary follow-up are required. Survivors often need physical and occupational therapy.

References

1. MedlinePlus. Extracorporeal membrane oxygenation (ECMO). U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 81

Facet Joint Injection

Overview

A facet joint injection delivers corticosteroid and anesthetic into the small joints of the spine to diagnose and treat facet-mediated back pain.

Alternative Names

Facet block; medial branch block; zygapophyseal joint injection

Principle

Under imaging guidance, medication is injected into the facet joint or its nerve supply. Symptom relief confirms the joint as the pain source and may provide therapeutic benefit.

History

Facet joint injections were introduced in the 1970s as a diagnostic and therapeutic tool for spinal pain.

Why It Is Done

Performed for axial neck or low back pain with a clinical pattern of facet joint origin, and to predict the response to radiofrequency denervation.

Is This a Primary or Secondary Test or Procedure

Primary diagnostic and therapeutic procedure for facet pain.

How It Is Performed

Performed under fluoroscopy or CT guidance. A needle enters the facet joint or reaches the medial branch nerves, and anesthetic with steroid is injected.

Preparation

Imaging and discussion of expected response; anticoagulants are adjusted.

Contraindications and Precautions

Active infection, bleeding risk, and severe allergy to contrast are contraindications.

Risks

Bleeding, infection, nerve injury, transient numbness, and steroid side effects.

Aftercare and Recovery

The patient records pain relief in a diary over the following days.

Outcome and Follow-up

Significant temporary relief supports a facet diagnosis and radiofrequency denervation.

Expected Results

Meaningful pain relief without complications.

Follow-up

Repeat injection or radiofrequency ablation depending on response.

References

1. MedlinePlus. Facet joint injection. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 82

Fine Needle Aspiration (FNA)

Overview

Fine needle aspiration (FNA) is a minimally invasive biopsy procedure in which a very thin needle is inserted into a lump or nodule to collect cells for microscopic examination, usually to determine whether a mass is benign or malignant.

Alternative Names

FNA, fine-needle aspiration biopsy, FNA biopsy, aspiration cytology

Principle

FNA uses a thin needle (22-27 gauge) to aspirate cells from a mass. Negative pressure is applied with a syringe. The aspirated material is expelled onto slides, smeared, and fixed (air-dried for Diff-Quik, ethanol-fixed for Papanicolaou). A portion may be placed in liquid medium for cell block preparation.

History

Fine needle aspiration was developed in the early 20th century at Memorial Sloan Kettering. The technique was refined in Scandinavia in the 1960s and 1970s, becoming a standard minimally invasive diagnostic method.

Why It Is Done

Your doctor orders an FNA when you have a lump or nodule that can be felt or seen on imaging, and they need to know whether it is cancerous. FNA is commonly used for thyroid nodules, lymph nodes, breast lumps, salivary gland lumps, and lung nodules seen on CT scan. It answers the question: does this lump contain cancer cells?

Is This a Primary or Secondary Test or Procedure

Primary for many lumps and nodules. FNA is less invasive than a surgical biopsy and provides rapid answers.

How It Is Performed

A very thin needle (25 to 27 gauge — thinner than the needles used for blood draws) is inserted into the lump, often guided by ultrasound. The needle is moved back and forth within the lump to collect cells. The cells are then placed on a glass slide or in a liquid preservative and sent to the laboratory for examination. The test takes about fifteen minutes. Local anesthetic may be used, but many patients do not need it.

Preparation

No special preparation is needed for most FNA procedures. If you take blood thinners, inform your doctor, though most FNAs can be performed safely.

Contraindications and Precautions

This procedure has the following contraindications and precautions:

- **Absolute contraindications**: Very few. Do not aspirate lesions suspected to be a pheochromocytoma or carotid body tumor without special precautions.
- **Relative contraindications**: Uncorrectable bleeding diathesis, severe thrombocytopenia, or therapeutic anticoagulation (hold anticoagulants when clinically safe); uncooperative patient; lesions deep or adjacent to major vascular structures without image guidance.
- **Precautions**: Use ultrasound or CT guidance for deep or small lesions; obtain informed consent about the small risk of a non-diagnostic sample; apply firm pressure after the procedure to minimize hematoma.
- **Special populations**: In pregnancy, avoid ionizing radiation by using ultrasound guidance; children may require general anesthesia or sedation for cooperation.

Risks

This procedure carries the following risks:

- **Bleeding or bruising**: Minor bleeding or bruising at the puncture site is common and usually self-limited.

- **Infection**: Very rare (<0.1%) at the puncture site. Proper skin antisepsis minimizes this risk.
- **Pain or discomfort**: Brief pain during needle insertion. Some patients experience a vasovagal reaction (fainting, dizziness, nausea).
- **Hematoma**: Collection of blood under the skin if pressure is not held adequately after the procedure.
- **Pneumothorax**: Occurs when the needle enters the chest cavity (e.g., lung or hilar lymph node FNA); may require observation or chest tube drainage.
- **Tumor seeding**: Extremely rare (<0.01%) spread of malignant cells along the needle tract.
- **Non-diagnostic sample**: Insufficient cellular material occurs in a minority of cases and may require a repeat FNA or a surgical biopsy.

Aftercare and Recovery

You may have mild bruising at the site, which resolves within a few days. Results are typically available within one to three days.

Outcome and Follow-up

FNA results are reported using established classification systems (e.g., the Bethesda System for thyroid nodules). A benign or negative result indicates no malignant cells, while atypical, suspicious, or malignant categories warrant further evaluation. Because FNA samples only a small amount of tissue, a negative result does not completely exclude cancer, and false negatives occur in a small percentage of cases.

Expected Results

A normal (negative) result shows no malignant cells. May be reported as benign, atypical, suspicious, or malignant using the appropriate classification system.

Population	Reference Range
Adult Male	No malignant cells
Adult Female	No malignant cells
Children	Same as adult
Elderly	Same as adult
Pregnancy	Same as adult

Follow-up

Core needle biopsy, open surgical biopsy, or ultrasound surveillance.

References

1. MedlinePlus. Fine Needle Aspiration (FNA). U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 83

Fine-Needle Aspiration Biopsy of the Thyroid

Overview

Thyroid fine-needle aspiration (FNA) biopsy uses a thin needle to collect cells from a thyroid nodule for cytologic examination.

Why It Is Done

It helps determine whether a thyroid nodule is benign, malignant, or indeterminate and guides decisions about surveillance, molecular testing, or surgery.

How It Is Performed

The neck is cleaned and ultrasound is used to guide one or more needle passes into the nodule. Local anesthetic may be used. The samples are sent to a pathology laboratory.

Risks and Results

Temporary discomfort, bruising, bleeding, and rarely infection or injury to nearby structures can occur. Results may be benign, malignant, nondiagnostic, or indeterminate and must be interpreted with ultrasound and clinical findings.

References

1. MedlinePlus. Thyroid Fine-Needle Aspiration Biopsy. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 84

Foley Catheter Insertion

Overview

A Foley catheter is a flexible tube inserted through the urethra into the bladder to continuously drain urine. It is held in place by an inflated balloon at the tip.

Alternative Names

Urinary catheterization, Indwelling catheter insertion

Principle

A sterile catheter with a deflated balloon at its tip is passed through the urethra into the bladder. The balloon is then inflated with sterile water to hold the catheter in place, providing a closed drainage pathway for urine.

History

Catheters have been used since antiquity. Frederic Foley designed the modern balloon retention catheter in 1929, which was commercialized in the late 1930s.

Why It Is Done

Ordered for acute urinary retention, monitoring urine output in critically ill patients, during surgical procedures, or for bladder decompression.

Is This a Primary or Secondary Test or Procedure

Primary treatment for acute urinary retention and for precise urine output monitoring.

How It Is Performed

Under sterile conditions. The periurethral area is cleaned, a sterile lubricant is applied, and the catheter is gently inserted through the urethra into the bladder. The balloon is inflated with sterile water.

Preparation

Ensure sterile gloves, catheter kit, and antiseptic prep are available. Position the patient supine (female in dorsal recumbent).

Contraindications and Precautions

Suspected urethral disruption (typically in pelvic trauma; look for blood at the meatus or high-riding prostate).

Risks

Urinary tract infection (CAUTI), urethral trauma/bleeding, false passage creation, or bladder spasms.

Aftercare and Recovery

Maintain a closed drainage system, secure the catheter to the patient's thigh, and keep the drainage bag below the level of the bladder.

Outcome and Follow-up

Immediate flow of clear urine confirms correct placement. Monitor for patency and urine volume.

Expected Results

Continuous drainage of clear yellow urine; no leakage around the catheter; no patient distress.

Follow-up

Regular catheter hygiene, monitoring for UTI symptoms, and prompt removal when no longer indicated.

References

1. MedlinePlus. Foley Catheter Insertion. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 85

Fracture Reduction

Overview

Fracture reduction aligns displaced bone fragments to their normal anatomical position, promoting proper healing and restoring limb function.

Alternative Names

Bone setting, Closed reduction, Open Reduction Internal Fixation (ORIF)

Principle

Displaced bone fragments are realigned manually, by traction, or surgically so the fracture surfaces return to anatomic or near-anatomic apposition. Immobilization then holds the fragments in alignment while healing by callus formation proceeds.

History

Hippocrates described traction and splinting techniques. Wilhelm Roentgen's discovery of X-rays in 1895 allowed visualization of alignment. Lister introduced aseptic open reduction.

Why It Is Done

Ordered for displaced bone fractures to prevent deformity, relieve pain, and avoid nerve or blood vessel compression.

Is This a Primary or Secondary Test or Procedure

Primary definitive treatment for displaced limb fractures.

How It Is Performed

Closed reduction involves manipulating the bones externally under sedation or local block. Open reduction (ORIF) is a surgical procedure where bones are aligned directly and secured with plates, screws, or rods.

Preparation

Perform neurovascular assessment of the distal limb. Obtain pre-reduction X-rays. Fast if sedation or general anesthesia is required.

Contraindications and Precautions

None when a fracture is present. Active soft-tissue infection is a relative contraindication for immediate internal fixation.

Risks

Nerve or vascular injury during manipulation, compartment syndrome, infection (in open reduction), or nonunion/malunion.

Aftercare and Recovery

Immobilize the limb immediately with a splint or cast. Obtain post-reduction X-rays to verify alignment. Re-assess distal pulses and sensation.

Outcome and Follow-up

Post-reduction X-rays must show acceptable bone alignment and length restoration.

Expected Results

Anatomical or acceptable functional alignment of bone fragments; intact distal neurovascular status.

Follow-up

Follow-up X-rays in 1, 2, and 6 weeks to monitor healing; physical therapy.

References

1. MedlinePlus. Fracture Reduction. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 86

Gallbladder Scan (HIDA Scan)

Overview

A hepatobiliary iminodiacetic acid (HIDA) scan, or cholescintigraphy, uses a small radioactive tracer and a gamma camera to evaluate bile production and flow through the liver, gallbladder, and bile ducts.

Why It Is Done

It helps evaluate suspected acute cholecystitis, bile-duct obstruction, bile leaks, impaired gallbladder emptying, or selected postoperative complications.

How It Is Performed

The tracer is injected into a vein and sequential images are obtained as it passes through the hepatobiliary system. A medicine may be given to stimulate gallbladder contraction.

Preparation, Risks, and Results

Fasting and medicine instructions must be followed. Mild injection discomfort and rare tracer reactions are possible; radiation exposure is low. Results assess tracer filling, drainage, and gallbladder ejection when measured.

References

1. MedlinePlus. Hepatobiliary Scan. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 87

Gastric Balloon Insertion

Overview

Gastric balloon insertion places a temporary, soft silicone balloon inside the stomach to help with weight loss. The balloon is positioned through the mouth using an endoscope and is then filled with saline or air. It remains in place for 6 to 12 months and is later removed endoscopically.

Alternative Names

Intragastric balloon, gastric balloon therapy, endoscopic balloon weight loss device

Principle

The inflated balloon takes up space in the stomach, reducing its capacity for food and increasing the sensation of fullness after small meals. This early satiety helps the person eat less and adopt more sustainable eating habits. Because the balloon is temporary, it is intended to support lifestyle change rather than to act as a permanent solution.

History

The concept of an intragastric balloon dates to the early 1980s, with modern inflatable devices approved in the 2010s. Balloon designs and placement techniques have evolved to improve safety, tolerability, and weight loss results. Today it is a recognized nonsurgical option in the management of obesity.

Why It Is Done

It is offered to adults with a body mass index (BMI) in the moderate obesity range who have not achieved lasting weight loss through diet and exercise alone. It may be used when the person does not qualify for or does not desire bariatric surgery. Weight loss from the balloon can also improve conditions linked to obesity, such as type 2 diabetes and high blood pressure.

Is This a Primary or Secondary Test or Procedure

It is a secondary procedure, used when lifestyle measures alone have been insufficient. It may also serve as a bridge to more definitive bariatric surgery by helping people lose weight and reduce surgical risk beforehand.

How It Is Performed

Under sedation, an endoscope is passed through the mouth and esophagus into the stomach, and the deflated balloon is placed and then inflated with saline or air. The procedure typically takes 20 to 30 minutes and usually does not require an overnight stay. The balloon is removed with the endoscope after several months.

Preparation

You must fast for several hours before the procedure. Medications that increase bleeding risk may need to be paused, and your doctor will review any prior stomach surgery or conditions. A blood test and discussion of your medical history are usually completed in advance.

Contraindications and Precautions

- Prior stomach or esophageal surgery, active peptic ulcer disease, or gastrointestinal bleeding.
- Pregnancy or breastfeeding.
- Severe reflux disease, large hiatal hernia, or an inflamed esophagus that has not been evaluated.

Risks

- Nausea, vomiting, and abdominal cramping, which are common in the first days after placement.
- Stomach ulceration, irritation, or bleeding from the balloon surface.
- Balloon deflation, which risks blockage of the intestine (obstruction) and requires prompt removal.

Aftercare and Recovery

Nausea and discomfort are expected for the first few days and are managed with medication. A structured diet program and exercise plan are essential to maximize and maintain weight loss. The balloon is removed at the planned time, usually 6 to 12 months after insertion.

Outcome and Follow-up

Weight loss is gradual and varies with diet, activity, and adherence to the program. Many people lose a meaningful portion of their excess weight over the life of the balloon. Results depend on sustaining healthy habits after removal, since the balloon alone is temporary.

Expected Results

A successful outcome is safe placement and removal of the balloon with significant, sustained weight loss. Improvement in obesity-related conditions, such as blood sugar and blood pressure, is also a marker of success.

Follow-up

Regular visits with the weight management team monitor weight, nutrition, and tolerance of the balloon. Bariatric surgery may be discussed after removal if further weight loss is needed. Routine follow-up continues for at least a year to support long-term weight maintenance.

References

1. MedlinePlus. Gastric Balloon. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 88

Gastric Emptying Study

Overview

A gastric emptying study measures how quickly food leaves the stomach using a small amount of radiolabeled material and serial imaging.

Why It Is Done

It evaluates suspected gastroparesis, rapid gastric emptying, unexplained nausea and vomiting, early fullness, bloating, or upper-abdominal symptoms.

How It Is Performed

After an overnight fast, the patient eats a standardized meal containing a tiny tracer. A gamma camera obtains images over several hours while the patient remains otherwise relaxed.

Preparation, Risks, and Results

Follow fasting and medicine instructions; diabetes medicines may require special planning. Radiation exposure is low, and allergic reactions are not expected from the tracer. Results report the percentage of meal remaining at specified times.

References

1. MedlinePlus. Gastric Emptying Study. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 89

Gastric Lavage

Overview

Gastric lavage is the washing out of the stomach using a large-bore tube with instillation and aspiration of fluid. It is used to remove ingested toxins or to manage gastrointestinal bleeding.

Alternative Names

Stomach pumping, Gastric irrigation

Principle

A wide-bore orogastric tube is passed into the stomach, and small aliquots of fluid, typically warm saline, are instilled and aspirated in repeated cycles. This mechanically removes gastric contents and unabsorbed toxin. Removal depends on mechanical clearance rather than neutralization or absorption of the substance.

History

Gastric lavage has been performed for more than a century as a method of gastric decontamination after poisoning.

Why It Is Done

Gastric lavage is ordered for potentially life-threatening ingestions when performed soon after the event, ideally within one hour, and when the substance is not contraindicated. It may also be used therapeutically for gastrointestinal bleeding or to obtain gastric contents for analysis.

Is This a Primary or Secondary Test or Procedure

Secondary procedure of declining use; it is reserved for selected toxic ingestions rather than routine poisoning management.

How It Is Performed

The patient is positioned head down on the left side, and a lubricated large-bore orogastric tube is passed orally into the stomach. After confirming tube position, 200–300 mL aliquots of warm saline are instilled and aspirated until the returns are clear. The tube is then removed.

Preparation

Airway protection is essential; patients with impaired consciousness or a depressed gag reflex are intubated first. The substance, time of ingestion, and vital signs are documented, and informed consent is obtained when possible.

Contraindications and Precautions

Avoid lavage after ingestion of hydrocarbons, corrosives, or substances with high aspiration risk, and avoid sharp objects. It is contraindicated in esophageal varices, recent esophageal surgery, and severe bleeding diatheses.

Risks

Aspiration pneumonia, laryngospasm, and esophageal or gastric perforation. Epistaxis and electrolyte disturbances from fluid shifts may also occur.

Aftercare and Recovery

The patient is observed for complications such as aspiration, and supportive care continues. Poisoning management proceeds with activated charcoal, antidotes, and laboratory monitoring as indicated.

Outcome and Follow-up

Lavage serves decontamination rather than diagnosis; effectiveness depends on timing, quantity, and type of toxin. Clinical assessment and toxicologic monitoring determine the outcome.

Expected Results

Clear return of lavage fluid with no further recovery of gastric contents; no quantitative diagnostic value is expected.

Follow-up

Observation for delayed toxicity with supportive laboratory monitoring and serial clinical assessment. Referral to toxicology services and psychiatric evaluation is appropriate after intentional ingestion.

References

1. MedlinePlus. Gastric lavage. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 90

Hand Arthroplasty

Overview

Hand arthroplasty is joint replacement surgery of the hand, most commonly involving a finger or thumb joint damaged by arthritis, trauma, or another destructive condition.

Alternative Names

Finger joint replacement, thumb joint replacement, small-joint arthroplasty

Principle

Damaged joint surfaces are removed or reshaped and replaced with an implant to reduce pain and preserve useful motion. Fusion or another reconstruction may be preferable when stability is more important than motion.

Why It Is Done

It may be considered for persistent pain, deformity, or loss of function that has not responded to medication, splinting, injections, therapy, or activity modification.

How It Is Performed

With regional or general anesthesia, the surgeon exposes the joint, removes damaged surfaces, prepares the bone, and inserts the selected implant. Tendons and soft tissues are repaired, and the hand is splinted.

Preparation, Risks, and Recovery

Imaging and hand-function assessment guide implant selection. Risks include infection, stiffness, nerve or tendon injury, implant loosening or wear, instability, fracture, persistent pain, and revision surgery. Therapy restores motion while protecting the repair.

References

1. MedlinePlus. Joint Replacement—Hand. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 91

Hemorrhoid Banding

Overview

Hemorrhoid banding, or rubber band ligation, is an outpatient procedure that treats internal hemorrhoids by placing a small elastic band around their base to cut off their blood supply.

Alternative Names

Rubber band ligation; banding of hemorrhoids; hemorrhoidal ligation

Principle

A tight rubber band is applied to the base of an internal hemorrhoid. The banded tissue becomes ischaemic, necroses, and sloughs off within days, reducing the hemorrhoid.

History

Rubber band ligation of hemorrhoids was popularized by Barron in the 1960s and remains the most common outpatient treatment for symptomatic internal hemorrhoids.

Why It Is Done

Performed for bleeding or prolapsing internal hemorrhoids (grades 1-3) that fail dietary and stool-softening measures.

Is This a Primary or Secondary Test or Procedure

Primary treatment for symptomatic internal hemorrhoids.

How It Is Performed

Performed in the office with an anoscope. A ligator applies a band around the hemorrhoid base above the dentate line, where the tissue is pain-insensitive.

Preparation

No sedation is required. A bowel preparation is generally unnecessary; the patient's bowel may be emptied.

Contraindications and Precautions

External hemorrhoids, bleeding disorders, and anticoagulant use are relative contraindications. Precaution: banding below the dentate line causes severe pain.

Risks

Pelvic pain or tenesmus, bleeding, vasovagal reaction, and rarely pelvic infection or septicaemia.

Aftercare and Recovery

The patient may feel pressure for a day; mild bleeding is expected when the band sloughs. Stool softeners prevent straining.

Outcome and Follow-up

Improvement is judged by resolution of bleeding and prolapse; several sessions may be needed for multiple hemorrhoids.

Expected Results

Symptom relief with no significant bleeding, pain, or infection.

Follow-up

Follow-up examination; repeat banding for residual hemorrhoids.

References

1. MedlinePlus. Hemorrhoid banding. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 92

Hemorrhoidectomy

Overview

Hemorrhoidectomy is surgical removal of enlarged hemorrhoidal tissue, usually for severe or recurrent hemorrhoids.

Why It Is Done

It may be considered for large prolapsing hemorrhoids, persistent bleeding or pain, strangulation, or symptoms that do not improve with dietary measures and office treatments.

How It Is Performed

Under anesthesia, the surgeon excises selected hemorrhoidal columns and controls the blood supply. Open, closed, or stapled techniques may be used depending on the anatomy.

Risks and Recovery

Pain, bleeding, urinary retention, infection, anal stenosis, recurrence, and temporary bowel-control problems are possible. Stool-softening measures, fluids, local care, and activity guidance are important during recovery.

References

1. MedlinePlus. Hemorrhoidectomy. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 93

Hyperbaric Oxygen Therapy

Overview

Hyperbaric oxygen therapy (HBOT) is a treatment in which the patient breathes 100% oxygen while inside a pressurized chamber at greater than atmospheric pressure. It is used to treat conditions involving tissue hypoxia, gas bubbles, or infection.

Alternative Names

Hyperbaric oxygen therapy (HBOT); hyperbaric medicine; high-pressure oxygen therapy; HBO

Principle

Breathing 100% oxygen under increased ambient pressure markedly raises the partial pressure of oxygen dissolved in plasma and delivered to tissues, independent of hemoglobin. This corrects tissue hypoxia, promotes angiogenesis and wound healing, exerts antimicrobial effects, and shrinks gas bubbles in decompression illness by creating a pressure gradient for gas reabsorption.

History

Hyperbaric oxygen use dates to the 19th century, and after the success of compressed-air chambers in diving medicine, its modern clinical applications were developed in the mid-20th century.

Why It Is Done

Ordered for decompression sickness, arterial gas embolism, carbon monoxide poisoning (especially with neurologic signs or myocardial injury), and radiation injury to bone and soft tissue. It is also used for chronic nonhealing diabetic wounds, necrotizing soft-tissue infections, crush injuries, compromised skin grafts, and certain refractory osteomyelitis.

Is This a Primary or Secondary Test or Procedure

A primary treatment for decompression sickness, arterial gas embolism, and severe carbon monoxide poisoning, and an adjunctive therapy for most other indications.

How It Is Performed

The patient enters a monoplace chamber or a multiplace chamber where a technician controls pressure. The chamber is pressurized to approximately 2 to 3 atmospheres absolute while the patient breathes 100% oxygen, typically for sessions of 60 to 120 minutes. Multiple daily sessions are usually required over several weeks.

Preparation

The patient should have no recent alcohol use and should avoid smoking during treatment. A pretreatment assessment includes a chest radiograph or pulmonary evaluation and oximetry, and the patient wears only cotton clothing to reduce the fire risk of oxygen-rich environments.

Contraindications and Precautions

The only absolute contraindication is untreated pneumothorax, because trapped gas expands on ascent. Relative contraindications include COPD with bullae, active infections, severe cardiac disease, pregnancy, claustrophobia, and current use of drugs such as bleomycin or doxorubicin that potentiate oxygen toxicity.

Risks

Common adverse effects are reversible middle-ear or sinus barotrauma during compression and temporary myopia. Serious but rare complications include oxygen seizures, pulmonary oxygen toxicity, and pneumothorax, and a brief risk of fire exists within the oxygen-rich chamber.

Aftercare and Recovery

The patient is observed briefly after each session for ear discomfort or dizziness and instructed on pressure-equalization techniques. The course continues with serial assessment of the wound or underlying condition to determine the number of sessions needed.

Outcome and Follow-up

Response is assessed by wound healing, resolution of neurologic signs, or improvement in imaging and clinical status. The number of sessions is adjusted based on these endpoints.

Expected Results

Healing of the target wound, resolution of gas bubble or carbon monoxide effects, and recovery of neurologic or tissue function within the expected treatment course.

Follow-up

Wound care continues with regular measurement and photography, and serial examinations document neurologic recovery. When the course is complete, the patient is discharged with instructions to avoid flying or high-altitude exposure until cleared, and long-term follow-up addresses any residual deficits.

References

1. MedlinePlus. Hyperbaric oxygen therapy. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 94

Hysteroscopy

Overview

Hysteroscopy is the inspection of the uterine cavity by endoscopy. A hysteroscope is inserted through the cervix to diagnose and treat intrauterine pathology.

Alternative Names

Uterine endoscopy

Principle

A thin telescope with a light and camera is passed through the cervix into the uterine cavity, which is distended with fluid or gas. The endometrial cavity is inspected directly, and instruments allow biopsy, removal of polyps or fibroids, and treatment of adhesions or septa.

History

Pantaleoni performed the first hysteroscopy in 1869 using a modified cystoscope to identify a uterine polyp.

Why It Is Done

Ordered for abnormal uterine bleeding, repeated miscarriages, infertility, checking for uterine fibroids or polyps, or locating a displaced IUD.

Is This a Primary or Secondary Test or Procedure

Primary diagnostic and therapeutic tool for endometrial and intrauterine pathology.

How It Is Performed

Performed under local, regional, or general anesthesia. The hysteroscope is passed through the vagina and cervix into the uterus. Liquid or gas is used to expand the cavity.

Preparation

The procedure is usually scheduled when you are not menstruating. Fasting is required if regional or general anesthesia is planned.

Contraindications and Precautions

Active pelvic infection, pregnancy, or known cervical or uterine cancer.

Risks

Uterine perforation (<1

Aftercare and Recovery

You may experience mild cramping and light vaginal bleeding or spotting for a few days. Avoid sexual intercourse until bleeding stops.

Outcome and Follow-up

Enables direct visualization of the endometrium, fallopian tube openings, and structural anomalies. Biopsies or polypectomy can be performed.

Expected Results

Normal cervix and uterine cavity; smooth endometrial lining; patent tubal ostia; no polyps or fibroids.

Follow-up

Endometrial biopsy review, hormonal therapy, or surgical scheduling.

References

1. MedlinePlus. Hysteroscopy. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 95

Immunotherapy

Overview

Immunotherapy stimulates or trains the body's own immune system to recognize and destroy cancer cells. This includes immune checkpoint inhibitors, monoclonal antibodies, and T-cell therapies.

Alternative Names

Biological therapy, Immuno-oncology

Principle

Treatments are used to stimulate, suppress, or modulate the immune system: allergen immunotherapy induces tolerance, checkpoint inhibitors release antitumor T-cell responses, and biologic agents target inflammatory pathways. The immune response against the disease is thereby altered.

History

William Coley used bacterial toxins ('Coley's toxins') in the 1890s to stimulate the immune system. James Allison and Tasuku Honjo won the 2018 Nobel Prize for discovering immune checkpoint inhibition.

Why It Is Done

Ordered to treat various advanced cancers, including melanoma, lung cancer, kidney cancer, and lymphoma, often when standard therapies have failed.

Is This a Primary or Secondary Test or Procedure

Primary systemic treatment for advanced melanoma and specific non-small cell lung cancers.

How It Is Performed

Usually administered intravenously in an outpatient clinic. Infusions occur every few weeks in cycles, similar to chemotherapy.

Preparation

Complete blood count, metabolic panel, and thyroid function tests are performed prior to each infusion to screen for immune-related adverse events.

Contraindications and Precautions

Active severe autoimmune diseases (e.g., lupus, Crohn's) are relative contraindications due to the risk of exacerbating the disease.

Risks

Immune-related adverse events (irAEs) including colitis, pneumonitis, hepatitis, thyroiditis, hypophysitis, or severe skin rashes.

Aftercare and Recovery

Immediately report any new symptoms of shortness of breath, severe diarrhea, abdominal pain, or extreme fatigue to your oncologist.

Outcome and Follow-up

Assessed using imaging scans. Response may take longer than chemotherapy, and tumors may temporarily appear to grow before shrinking (pseudoprogression).

Expected Results

Tumor response or disease stability; absence of severe immune-mediated organ toxicity.

Follow-up

Frequent monitoring of thyroid function, liver enzymes, and clinical symptoms.

References

1. MedlinePlus. Immunotherapy. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 96

Implantable Venous Access Port

Overview

An implantable venous access port (Port-a-Cath) is a small reservoir and catheter placed under the skin that provides durable central venous access for repeated therapies such as chemotherapy, fluids, or blood sampling.

Alternative Names

Port-a-Cath; subcutaneous port; implanted port; central venous port

Principle

A catheter is inserted into a central vein, typically the internal jugular or subclavian, and tunneled to a subcutaneous reservoir port placed on the chest wall. The port is accessed through the skin with a non-coring needle for infusion or aspiration.

History

Implantable ports were developed in the 1980s following the growth of long-term chemotherapy, building on earlier central venous catheter and subcutaneous reservoir technology.

Why It Is Done

Performed for patients needing frequent or long-term intravenous therapy, most commonly chemotherapy, total parenteral nutrition, antibiotics, or blood products, when peripheral access is inadequate or undesirable.

Is This a Primary or Secondary Test or Procedure

Primary durable vascular access; secondary to peripheral or external central access for long-term use.

How It Is Performed

Performed under local anesthesia with or without sedation, usually with ultrasound and fluoroscopy. The vein is accessed percutaneously, the catheter is positioned with its tip at the cavoatrial junction, and the port is placed in a subcutaneous pocket on the chest and connected to the catheter. The incisions are closed.

Preparation

Preoperative assessment of coagulation, site examination, and NPO if sedation is planned. Aseptic technique is essential.

Contraindications and Precautions

Active bacteremia or infection at the site, and severe coagulopathy are relative contraindications. Precaution: careful technique avoids pneumothorax and arterial puncture.

Risks

Pneumothorax, bleeding, infection, catheter thrombosis, pinch-off syndrome, catheter migration, and wound complications.

Aftercare and Recovery

The port is flushed and used for therapy. Patients may shower after a few days; the port remains accessible for years with regular flushing.

Outcome and Follow-up

Fluoroscopy confirms correct tip position. Patency and absence of complications are verified before use.

Expected Results

A functional port that delivers and aspirates blood easily, with no infection, thrombosis, or mechanical complication.

Follow-up

Flushing every 4-6 weeks when unused, and prompt evaluation of any swelling, pain, or fever for infection or thrombosis.

References

1. MedlinePlus. Port-a-Cath. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 97

In Vitro Fertilization

Overview

In vitro fertilization (IVF) is an assisted reproductive technology in which eggs are retrieved from the ovaries, fertilized with sperm in the laboratory, and resulting embryos transferred to the uterus. It is the most effective form of assisted reproduction for many causes of infertility.

Alternative Names

IVF, assisted reproductive technology, embryo transfer

Principle

Controlled ovarian hyperstimulation with gonadotropins matures multiple follicles, which are aspirated under ultrasound guidance. Retrieved eggs are fertilized with sperm in the laboratory, and fertilized embryos are cultured for several days. One or more embryos are transferred into the uterine cavity, and any viable embryos not transferred may be frozen for later use.

History

The first live birth from IVF, Louise Brown, occurred in 1978 in the United Kingdom. Subsequent advances included intracytoplasmic sperm injection (ICSI) in 1992 and the routine use of preimplantation genetic testing and embryo cryopreservation.

Why It Is Done

IVF is ordered for tubal factor infertility, severe male factor infertility, advanced maternal age, endometriosis, and unexplained infertility that has failed other treatments. It is also used for preimplantation genetic testing and for fertility preservation with planned egg or embryo freezing.

Is This a Primary or Secondary Test or Procedure

It is a primary treatment for many severe causes of infertility. It is secondary when used after failed intrauterine insemination or other fertility treatments.

How It Is Performed

After ovulation suppression and 8-14 days of gonadotropin injections monitored by ultrasound and estradiol levels, eggs are retrieved transvaginally under sedation. Eggs are fertilized with sperm (conventional or ICSI), and embryos are cultured for 3-6 days. One or more embryos are transferred through a thin catheter, with remaining embryos often frozen.

Preparation

Patients take daily gonadotropin injections for about 2 weeks with frequent ultrasound and blood monitoring. A full bladder is required before egg retrieval, and fasting is needed if sedation is used. Tobacco, alcohol, and some medications are discontinued.

Contraindications and Precautions

Contraindications include active pelvic infection, severe uterine abnormality, and medical conditions making pregnancy unsafe. Precautions include screening for HIV and hepatitis, counseling on the risk of multiple gestation, and monitoring for ovarian hyperstimulation syndrome.

Risks

Risks include ovarian hyperstimulation syndrome, multiple pregnancy, bleeding or infection from egg retrieval, and a small increase in preterm birth. The emotional and financial burden of treatment is significant.

Aftercare and Recovery

The patient rests briefly after embryo transfer and receives progesterone support for the luteal phase. A pregnancy test is performed about 10-14 days after transfer.

Outcome and Follow-up

A positive hCG test about 2 weeks after transfer indicates implantation, confirmed later by ultrasound. A negative result ends the cycle, and the frozen embryo bank or a new stimulation cycle may be discussed with the fertility specialist.

Expected Results

Successful retrieval of mature eggs, normal fertilization, development of good-quality embryos, and a viable intrauterine pregnancy on follow-up ultrasound.

Follow-up

Serial beta-hCG levels and an early ultrasound confirm a viable pregnancy, followed by routine obstetric care. Failed cycles may be followed by frozen embryo transfer or a subsequent stimulation cycle.

References

1. MedlinePlus. In Vitro Fertilization (IVF). U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 98

Incision and Drainage

Overview

Incision and drainage (I&D) is a minor surgical procedure to release pus from an abscess, boil, or infected cyst. It relieves pressure, reduces pain, and allows the infection to heal.

Alternative Names

I&D, Abscess drainage

Principle

The skin and subcutaneous tissue over an abscess are incised to open the cavity, pus is drained, and loculations are broken down. The wound is often left open or packed so infection can drain while the cavity granulates and heals from the base outward.

History

The surgical maxim 'Ubi pus, ibi evacua' (Where there is pus, evacuate it) has been followed since the Roman physician Celsus in the 1st century AD.

Why It Is Done

Ordered for fluctuant skin abscesses or localized purulent infections that do not resolve with antibiotics alone.

Is This a Primary or Secondary Test or Procedure

Primary definitive treatment for localized, fluctuant soft-tissue abscesses.

How It Is Performed

The area is cleaned, and local anesthetic is injected. An incision is made over the center of the abscess. The pus is drained, loculations are broken, and the cavity may be packed with gauze.

Preparation

Irrigate the site. Obtain scalpel, packing gauze, forceps, and local anesthetic.

Contraindications and Precautions

Facial abscesses in the 'danger triangle' (risk of cavernous sinus thrombosis) or abscesses close to major blood vessels may require surgical referral.

Risks

Bleeding, pain, recurrence, scar formation, or spread of infection.

Aftercare and Recovery

Keep the wound clean and covered. If packed, the packing is removed or changed in 24-48 hours. Warm compresses are encouraged.

Outcome and Follow-up

Complete drainage leads to immediate pain relief, decreased swelling, and resolution of local infection.

Expected Results

Evacuation of purulent fluid; collapse of the abscess cavity; immediate relief of pressure pain.

Follow-up

Wound check and packing removal in 24-48 hours; monitoring for systemic signs of infection (fever).

References

1. MedlinePlus. Incision and Drainage. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 99

Inferior Vena Cava Filter Placement

Overview

Inferior vena cava (IVC) filter placement is an interventional procedure in which a metal filter is deployed in the inferior vena cava to trap emboli traveling from the lower extremities. It is a mechanical means of preventing pulmonary embolism.

Alternative Names

IVC filter, vena cava filter, Greenfield filter, retrievable vena cava filter

Principle

The filter is positioned in the infrarenal IVC, where it spans the lumen and captures thrombus migrating from pelvic or lower extremity veins while allowing normal blood flow. Trapped clot is subsequently broken down by the body's endogenous fibrinolytic system, and the filter is removed once the risk of embolism resolves.

History

The concept of caval interruption dates to the 1960s, and the first percutaneously placed IVC filter was introduced in the 1970s. Retrievable filters developed in the 1990s allow removal once protection is no longer needed.

Why It Is Done

Placed in patients with proven venous thromboembolism and a contraindication to anticoagulation, failure or complications of anticoagulation, or recurrent pulmonary embolism despite adequate therapy. It is also considered for prophylaxis in select high-risk trauma patients or those undergoing major surgery with a high bleeding risk.

Is This a Primary or Secondary Test or Procedure

A secondary procedure to prevent pulmonary embolism when pharmacologic anticoagulation is contraindicated or has failed, rather than a primary treatment of the thrombus itself.

How It Is Performed

Venous access is obtained, usually through the femoral or internal jugular vein, and a venogram or ultrasound confirms IVC anatomy and the level of the renal veins. A sheath containing the collapsed filter is advanced to the infrarenal IVC, the filter is deployed, and a completion venogram confirms position and patency.

Preparation

Informed consent, imaging to assess IVC size, diameter, and anatomy, and assessment of renal function before iodinated contrast venography. Anticoagulation is held or reversed as appropriate for the bleeding risk.

Contraindications and Precautions

Contraindications include an IVC diameter exceeding the filter's capacity, severe thrombus in the IVC, and inability to access the venous system; caution is needed in pregnancy. Precautions include documenting the filter in registries, planning timely retrieval, and avoiding prolonged placement because of the risk of filter thrombosis and fracture.

Risks

Filter thrombosis or IVC occlusion, filter migration or tilting, strut fracture, vein injury or bleeding at the access site, and complications of retrieval including perforation.

Aftercare and Recovery

The patient resumes anticoagulation once safe and undergoes regular follow-up imaging to assess filter position. Retrieval is attempted when the indication for protection has resolved, ideally within weeks to months.

Outcome and Follow-up

A completion venogram showing the filter centered in the infrarenal IVC below the renal veins with brisk flow indicates successful placement. Imaging at follow-up showing stable filter position, patent IVC, and no new pulmonary emboli reflects a good outcome.

Expected Results

A well-positioned filter with preserved caval flow, no filter migration or fracture, and no pulmonary embolism during the period of protection.

Follow-up

Periodic radiography or CT to check filter position and integrity, and plans for elective retrieval. Patients are evaluated for resuming or continuing anticoagulation and for management of the underlying thrombus.

References

1. MedlinePlus. Vena cava filter. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 100

Injections

Overview

An injection is the administration of a medication or fluid into the body using a needle and syringe. The most common routes are intramuscular (IM) and subcutaneous (SC) injections, which deliver drugs for local or systemic effect.

Alternative Names

IM injection; SC injection; subcutaneous injection; intramuscular injection; parenteral drug administration

Principle

A drug is deposited into skeletal muscle or subcutaneous fat, where it is absorbed into the circulation. Intramuscular injection provides faster absorption than subcutaneous delivery because of the rich muscle blood supply, while subcutaneous injection is used for drugs requiring slow, sustained absorption.

History

The hypodermic syringe was developed in the 19th century, and by the 20th century injection techniques became standard for vaccination, insulin, analgesics, and antibiotics, replacing many older oral and topical routes.

Why It Is Done

Ordered to administer vaccines, insulin, anticoagulants, antibiotics, analgesics, hormones, biologics, and other medications when the oral route is unavailable, unreliable, or undesirable, or when rapid or controlled absorption is required.

Is This a Primary or Secondary Test or Procedure

A primary therapeutic procedure when the sole purpose is drug administration; it is a delivery technique rather than a diagnostic or curative intervention.

How It Is Performed

The site is selected and cleaned. For IM injection the needle is inserted at 90 degrees into a large muscle such as the deltoid, vastus lateralis, or ventrogluteal; for SC injection the needle enters the fatty layer at 45-90 degrees after a skin pinch. After aspirating to confirm the needle is not in a vessel, the medication is injected slowly and the needle is withdrawn with pressure applied.

Preparation

Informed consent, verification of the drug, dose, and route, and selection of the appropriate needle length and gauge. The site is assessed for suitability, and the patient's allergy and medication history is reviewed.

Contraindications and Precautions

Avoid injecting into an inflamed, infected, bruised, or previously irradiated area, and avoid sites near major nerves or vessels. Use caution in thin or emaciated patients, and aspirate before injection to prevent intravascular delivery.

Risks

Pain, bleeding, bruising, infection, abscess, nerve injury, and unintended intravascular injection. Systemic risks include anaphylaxis, adverse drug reactions, and, with insulin, local lipohypertrophy or lipoatrophy.

Aftercare and Recovery

The site is observed briefly for immediate reactions, and the patient is monitored for the expected drug effect and any delayed side effects. Injection sites are rotated for repeated administration.

Outcome and Follow-up

The response is evaluated by the clinical effect of the medication, such as analgesia, glucose control, or immune response. Failure of effect may reflect incorrect site, dose, or technique.

Expected Results

Correct drug delivery with the expected therapeutic response and no significant local or systemic adverse reaction.

Follow-up

Patients on repeated injections are taught proper technique and site rotation. Drug levels may be monitored for medications with narrow therapeutic windows, and injection sites are inspected for complications.

References

1. MedlinePlus. Giving an IM (intramuscular) injection. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 101

Intra-articular Steroid Injection

Overview

Intra-articular steroid injection is the delivery of a corticosteroid directly into a joint space to reduce local inflammation and pain. It is commonly used for inflammatory and degenerative joint diseases.

Alternative Names

Joint injection; corticosteroid joint injection; intra-articular corticosteroid; IA steroid injection

Principle

Corticosteroids suppress local inflammation by inhibiting leukocyte migration, cytokine release, and prostaglandin synthesis within the joint. Delivering the drug directly into the joint achieves a high local concentration with limited systemic exposure, and the effect typically lasts weeks to months.

History

Intra-articular corticosteroid injection was introduced by Hollander and colleagues in the early 1950s and remains a cornerstone of local therapy for inflammatory arthritis.

Why It Is Done

Ordered for symptomatic relief of osteoarthritis, rheumatoid arthritis flare, gout, pseudogout, and other inflammatory arthropathies, and to bridge care while systemic therapy takes effect.

Is This a Primary or Secondary Test or Procedure

A secondary, adjunctive treatment for joint inflammation; it provides symptom relief rather than disease modification.

How It Is Performed

The joint is identified by palpation or ultrasound guidance, and the skin is cleaned with antiseptic. A needle is inserted through the skin into the joint space, synovial fluid may be aspirated for analysis, and a corticosteroid, sometimes with local anesthetic, is injected before the needle is withdrawn.

Preparation

Informed consent, review of anticoagulation and bleeding risk, and confirmation that the joint is not septic are needed. The patient should report signs of active infection, and steroid should generally be held for one to two weeks before elective surgery on a joint.

Contraindications and Precautions

Septic arthritis, bacteremia, skin infection over the joint, an unstable or prosthetic joint, and allergy to the injected agents are contraindications. Precautions include limiting injection frequency, usually no more than every three months and no more than three to four times per joint per year, and exercising caution in diabetic patients.

Risks

Risks include post-injection flare, local infection, bleeding or bruising, tendon rupture, skin or soft tissue atrophy, transient glucose elevation, and rarely intra-articular crystal deposition with cartilage damage from repeated injections.

Aftercare and Recovery

The joint is rested for 24 to 48 hours and iced, and the patient is advised that a brief post-injection flare may occur. Relief typically develops within a few days and may last weeks to several months.

Outcome and Follow-up

Durability of pain relief and improved joint mobility indicate a response, while early recurrence suggests more severe disease and the need for systemic or surgical therapy. A hot, red, or swollen joint after injection warrants evaluation for infection.

Expected Results

Marked reduction in joint pain, swelling, and stiffness with improved range of motion for weeks to months after injection indicates a good result.

Follow-up

Physical therapy and strengthening are continued, and systemic disease-modifying therapy is optimized for inflammatory arthritis. Repeat injection or arthroplasty is considered when relief is short-lived or disease progresses.

References

1. MedlinePlus. Steroid injection (intra-articular). U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 102

Intracranial Pressure Monitoring (Ventriculostomy)

Overview

Intracranial pressure (ICP) monitoring is the continuous measurement of pressure within the cranial vault using a catheter placed in the lateral ventricle or a sensor placed in the brain parenchyma or subdural space. An intraventricular catheter (ventriculostomy) also permits cerebrospinal fluid (CSF) drainage to reduce ICP.

Alternative Names

ICP monitoring; intraventricular catheter; external ventricular drain; EVD; ventriculostomy; ICP bolt

Principle

The intracranial contents (brain, blood, and CSF) are contained within the rigid skull, so pressure rises as volume increases. A catheter or sensor transmits pressure to an external transducer, and an intraventricular drain additionally allows CSF to be diverted to lower the pressure.

History

Continuous ICP monitoring was pioneered by Nils Lundberg in the 1960s, and external ventricular drains became standard for monitoring and treating raised ICP in severe traumatic brain injury and other conditions.

Why It Is Done

Ordered in patients with severe traumatic brain injury (typically Glasgow Coma Scale score of 8 or less with abnormal imaging), hydrocephalus, intracranial hemorrhage, cerebral edema, or after neurosurgery to guide management and maintain cerebral perfusion pressure.

Is This a Primary or Secondary Test or Procedure

A primary monitoring and therapeutic procedure in neurocritical care for patients with actual or suspected raised intracranial pressure.

How It Is Performed

Under sterile technique, usually at the bedside with or without a burr hole, a catheter is inserted into the frontal horn of the lateral ventricle or a sensor into the brain parenchyma. The system is connected to a pressure transducer zeroed at the level of the external auditory canal, and a CSF collection chamber is set at the desired drainage height.

Preparation

Informed consent from the patient or surrogate, review of the coagulation profile with correction of coagulopathy, and confirmation that imaging shows no contraindication to the planned insertion site are required.

Contraindications and Precautions

Coagulopathy, sepsis, and scalp infection at the insertion site are contraindications, while a small or shifted ventricle makes placement more difficult. Precautions include strict sterile technique, confirming catheter position with imaging, and placing the drain so that overdrainage does not occur.

Risks

Risks include intracranial hemorrhage along the catheter tract, catheter-related ventriculitis or meningitis, infection, overdrainage or obstruction of the catheter, and neurological injury.

Aftercare and Recovery

ICP and cerebral perfusion pressure are monitored continuously, and CSF may be drained against a pressure threshold. Sedation, hyperosmolar therapy, or surgery may be added when pressures remain elevated.

Outcome and Follow-up

Normal adult ICP is below 10 to 15 mmHg, and sustained elevation above 20 mmHg is generally treated. Low pressure may indicate overdrainage or a leak, while failure of treatment to control ICP is a poor prognostic sign.

Expected Results

Intracranial pressure below 10 to 15 mmHg with a normal waveform in an adult, or age-appropriate values in children, indicates satisfactory intracranial compliance.

Follow-up

The drain is weaned or removed once ICP is controlled, and serial imaging and neurological examination assess for hydrocephalus, hemorrhage, or the need for a permanent shunt. CSF samples may be sent for culture if infection is suspected.

References

1. MedlinePlus. Intracranial pressure (ICP) monitoring. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 103

Intraosseous Access

Overview

Intraosseous (IO) access is the placement of a needle directly into the medullary cavity of a bone to provide rapid vascular access in emergencies. It allows infusion of fluids and medications when intravenous cannulation fails or is impractical.

Alternative Names

IO line, intraosseous infusion, intraosseous cannulation, bone marrow needle placement

Principle

The highly vascular bone marrow provides a non-collapsible venous plexus that communicates directly with the systemic circulation. Infused fluids and drugs reach the central circulation reliably even in severe hypovolemia or cardiac arrest, when peripheral veins are collapsed.

History

Intraosseous infusion was described as early as the 1920s and was used widely during World War II and in pediatric medicine. Its use has expanded to adult emergency care, trauma, and prehospital settings with modern powered insertion devices.

Why It Is Done

Performed when rapid vascular access is needed and peripheral or central intravenous access fails, such as in cardiac arrest, severe shock, trauma, burns, status epilepticus, or sepsis in infants and adults.

Is This a Primary or Secondary Test or Procedure

A secondary rescue procedure used when standard intravenous access is unavailable, but a primary approach in pediatric resuscitation protocols.

How It Is Performed

The site, usually the proximal tibia, proximal humerus, or distal femur, is identified and prepared with antiseptic. A manual or powered needle is inserted through the cortex into the marrow cavity with a popping sensation, the stylet is removed, and aspiration of blood or marrow confirms placement. The line is secured and flushed, then connected to fluids and medications.

Preparation

Informed consent when feasible, and preparation of the insertion site with sterile antiseptic. No fasting or laboratory testing is required in the emergency setting.

Contraindications and Precautions

Contraindications include fracture of the target bone, recent IO placement in the same bone, infection or burns at the site, and severe osteoporosis in some cases. Precautions include avoiding extravasation of medications, confirming placement by aspiration, and removing the line within 24 hours or once venous access is established.

Risks

Extravasation and compartment syndrome, osteomyelitis, fracture, fat embolism, and rarely injury to neurovascular structures.

Aftercare and Recovery

The IO line is used for resuscitation while standard intravenous or central venous access is obtained. It is removed as soon as alternative access is available, and the site is covered with a sterile dressing and observed for infection.

Outcome and Follow-up

The procedure is successful when aspiration or flushing confirms placement and fluids and medications infuse freely without significant resistance or extravasation. Failure to infuse suggests malposition, occlusion, or dislodgement requiring reinsertion.

Expected Results

Free aspiration of blood or marrow, a stable insertion site without swelling, and successful delivery of fluids and medications constitute a satisfactory result.

Follow-up

The patient is transitioned to intravenous access and monitored for local complications such as extravasation, infection, or compartment syndrome. Radiographic or laboratory confirmation of position may be obtained if malposition is suspected.

References

1. MedlinePlus. Intraosseous infusions. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 104

Intrathecal Baclofen Pump Therapy

Overview

Intrathecal baclofen therapy delivers the muscle relaxant baclofen directly into cerebrospinal fluid through an implanted pump and catheter.

Why It Is Done

It is used for severe spasticity that is inadequately controlled by oral medicines or causes functional disability, painful spasms, or difficult caregiving.

How It Is Performed

After a screening dose or trial, a programmable pump is implanted under the abdominal skin and connected to a catheter entering the spinal canal. The dose is adjusted electronically and the pump is refilled periodically.

Risks and Follow-up

Infection, catheter malfunction, overdose, withdrawal, respiratory depression, and need for revision are possible. Abrupt interruption can cause life-threatening withdrawal, so refills and emergency planning are essential.

References

1. MedlinePlus. Intrathecal Baclofen. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 105

Intrauterine Device (IUD) Insertion

Overview

An IUD is a small, T-shaped contraceptive device inserted into the uterus. It prevents pregnancy by releasing copper (non-hormonal) or levonorgestrel (hormonal), altering the uterine environment.

Alternative Names

IUD placement, Coil insertion

Principle

A small T-shaped device is loaded into an insertion tube, passed through the cervix into the uterine cavity, and released so its arms open and it seats at the fundus. It prevents pregnancy through local inflammation, thickened cervical mucus, and, for hormonal devices, progestin release.

History

Early intrauterine devices were developed in the early 20th century. Plastic, T-shaped copper IUDs were developed in the 1960s, and hormonal IUDs in the 1990s.

Why It Is Done

Ordered by patients seeking highly effective, long-acting reversible contraception (LARC) or management of heavy menstrual bleeding (hormonal IUD).

Is This a Primary or Secondary Test or Procedure

Primary, highly effective method of long-acting reversible contraception.

How It Is Performed

The clinician performs a bimanual exam, inserts a speculum, stabilizes the cervix with a tenaculum, measures uterine depth, and inserts the IUD through the cervix using a sterile applicator tube.

Preparation

A negative pregnancy test is mandatory. Taking an NSAID (ibuprofen) 1 hour before can help reduce insertion cramping.

Contraindications and Precautions

Pregnancy, active pelvic infection (PID, mucopurulent cervicitis), unexplained vaginal bleeding, or uterine anomalies.

Risks

Severe uterine cramping during insertion, vasovagal reaction, expulsion (<5

Aftercare and Recovery

Cramping and irregular spotting are common for several weeks. Check the IUD strings monthly. Avoid tampon use for the first 24-48 hours.

Outcome and Follow-up

Proper placement is confirmed if the strings are visible at the cervical os and the device is confirmed intrauterine on ultrasound (if checked).

Expected Results

IUD positioned in the fundus of the uterus; strings visible at the external cervical os; no signs of displacement.

Follow-up

String check in 4-6 weeks.

References

1. MedlinePlus. Intrauterine Device (IUD) Insertion. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 106

Intrauterine Insemination

Overview

Intrauterine insemination (IUI) is a fertility treatment that places concentrated, washed sperm directly into the uterine cavity around the time of ovulation. It brings sperm closer to the egg to increase the chance of fertilization.

Alternative Names

IUI, artificial insemination, donor insemination, sperm insemination

Principle

A semen sample is washed and concentrated to remove seminal fluid and select motile sperm. The prepared sample is injected through a thin catheter across the cervix into the uterus near ovulation. This shortens the distance sperm must travel and overcomes mild cervical or semen factors impairing natural conception.

History

Early attempts at artificial insemination date to the 18th century. Modern intrauterine insemination with washed sperm became established in the 1980s and remains a first-line fertility treatment.

Why It Is Done

It is ordered for unexplained infertility, mild male factor infertility, cervical mucus abnormalities, or ejaculatory dysfunction. It is also used with donor sperm for single women and same-sex female couples, and may be combined with ovulation induction in women with anovulation.

Is This a Primary or Secondary Test or Procedure

It is a primary, low-complexity treatment for mild fertility problems. It may be secondary when used after ovulation induction or as an alternative to more invasive treatments such as in vitro fertilization.

How It Is Performed

Ovulation timing is tracked with ultrasound and sometimes urine luteinizing hormone tests, with or without ovulation induction medication. The male partner provides a semen sample by masturbation, which is washed and concentrated in the laboratory. The prepared sperm is injected through a thin, soft catheter into the uterus; the procedure takes minutes and requires no anesthesia.

Preparation

Ovulation is monitored with serial ultrasounds, and the cycle may be coordinated with ovulation induction medication. The partner abstains from ejaculation for 2-5 days before providing the semen sample. A full bladder is sometimes requested before the procedure.

Contraindications and Precautions

Contraindications include tubal obstruction, active pelvic infection, and severe male factor infertility. Precautions include screening for sexually transmitted infections in both partners and counseling about the risk of multiple gestation when ovulation induction is used.

Risks

Risks are minimal and include mild cramping, spotting, and a small risk of infection. Ovarian hyperstimulation syndrome and multiple pregnancy may occur when ovulation induction medication is combined with IUI.

Aftercare and Recovery

The patient rests for 10-15 minutes and may resume normal activities, avoiding strenuous exercise for the day. A pregnancy test is performed about 2 weeks later.

Outcome and Follow-up

A positive pregnancy test approximately 2 weeks after insemination indicates conception. If the cycle is unsuccessful, the fertility specialist reviews the response and may adjust medications or consider additional cycles or more advanced treatment.

Expected Results

A properly timed insemination with an adequate number of motile sperm delivered into the uterine cavity and a subsequent positive pregnancy test.

Follow-up

A serum hCG pregnancy test is done about 2 weeks after the procedure, and an early ultrasound confirms viability if positive. Unsuccessful cycles may be followed by repeat IUI or progression to in vitro fertilization.

References

1. MedlinePlus. Intrauterine Insemination (IUI). U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 107

Intravenous (IV) Therapy

Overview

IV therapy delivers fluids, nutrients, or medications directly into a person's vein, allowing immediate absorption and rapid distribution throughout the circulatory system.

Alternative Names

IV fluid administration, Intravenous infusion

Principle

A cannula placed in a vein delivers fluids, electrolytes, medications, blood products, or nutrition directly into the bloodstream. Absorption is immediate and bypasses the digestive tract, allowing rapid effect and continuous administration.

History

Sir Christopher Wren performed the first IV injection in 1656 using a quill and bladder. The development of plastic cannulas in the 1950s made IV therapy safe and routine.

Why It Is Done

Ordered to treat dehydration, administer antibiotics, chemotherapy, blood products, or emergency medications that cannot be given orally.

Is This a Primary or Secondary Test or Procedure

Primary method for acute fluid resuscitation, electrolyte correction, and urgent drug delivery.

How It Is Performed

A nurse inserts a small plastic catheter (IV line) into a vein, typically in your hand or arm, using a needle which is then removed. Fluids or medications are connected to the catheter.

Preparation

No special preparation. Tell the nurse if you are allergic to latex, adhesives, or antiseptics like chlorhexidine.

Contraindications and Precautions

Avoid IV insertion in a limb with an active infection, lymphedema, or an AV fistula.

Risks

Infiltration (fluid leaking into surrounding tissue), phlebitis (vein inflammation), infection at the insertion site, or fluid overload.

Aftercare and Recovery

The catheter is removed, and pressure is applied to the site to prevent bleeding. Monitor for redness or swelling.

Outcome and Follow-up

Successful therapy is indicated by the resolution of symptoms (e.g., rehydration, pain control, or target drug levels).

Expected Results

Successful catheter insertion with patent flow; no infiltration, pain, swelling, or redness at the site.

Follow-up

Monitoring vital signs, fluid balance, and clinical improvement.

References

1. MedlinePlus. Intravenous (IV) Therapy. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 108

Intravenous Cannulation

Overview

Intravenous (IV) cannulation is the insertion of a plastic cannula into a peripheral vein to provide access to the bloodstream. It allows the administration of fluids, medications, blood products, and contrast media, and is a fundamental skill in acute and hospital care.

Alternative Names

Peripheral intravenous line insertion; peripheral IV access; PIVC insertion; IV start

Principle

A hollow needle with an overlying plastic catheter is advanced into a distended vein under tourniquet control. Once a flashback of blood confirms entry into the lumen, the needle is withdrawn and the catheter is advanced fully into the vein, where it is secured and connected to infusion tubing.

History

The practice of intravenous infusion began with early experimental injections in the 17th century and became a routine therapy after the introduction of the indwelling intravenous cannula and improved equipment in the mid-20th century.

Why It Is Done

Ordered to deliver intravenous fluids and electrolytes, administer medications and antibiotics, give blood products, provide parenteral nutrition, administer contrast for imaging, and establish emergency vascular access when oral or other routes are unsuitable.

Is This a Primary or Secondary Test or Procedure

A primary supportive procedure that establishes a route for therapy; it is an access procedure rather than a diagnostic or curative intervention.

How It Is Performed

A tourniquet is applied and a vein on the dorsal hand, forearm, or antecubital fossa is selected. The site is disinfected and the skin is punctured with the needle-catheter assembly bevel up at a shallow angle. On observing flashback, the needle is removed, the catheter is threaded into the vein, the tourniquet is released, and the catheter is flushed, capped, and secured with a transparent dressing.

Preparation

Informed consent and verification of the planned therapy. The operator selects the smallest adequate gauge and warms the limb or uses a tourniquet to improve vein distension when needed.

Contraindications and Precautions

Avoid cannulating a limb with an arteriovenous fistula, lymphedema, thrombophlebitis, cellulitis, or previous mastectomy on that side. Use caution in fragile or rolling veins, in anticoagulated patients, and in children, and do not persist with repeated failed attempts.

Risks

Pain, bleeding, hematoma, phlebitis, extravasation of fluid or medication, infection, thrombophlebitis, and rarely nerve or arterial injury. Cannula occlusion, dislodgement, and infiltration are common complications during therapy.

Aftercare and Recovery

The line is labelled with the insertion date and the site is inspected regularly for signs of phlebitis or infection. Peripheral cannulas are typically replaced or removed after 72-96 hours or when no longer needed.

Outcome and Follow-up

Cannulation success is confirmed by blood flashback, free flush, and secure placement; continued patency is indicated by good flow and the absence of pain or swelling during infusion.

Expected Results

A patent cannula that flushes freely without pain, swelling, or leakage, and an insertion site that remains clean with no erythema, tenderness, or discharge.

Follow-up

The site is monitored for complications, and cannula removal with site care is performed when therapy is complete. Persistent or difficult access may require ultrasound guidance or central venous catheter placement.

References

1. MedlinePlus. Intravenous line. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 109

Intravitreal Injection

Overview

An intravitreal injection delivers medication directly into the vitreous cavity of the eye to treat retinal diseases, most commonly anti-vascular endothelial growth factor agents for macular degeneration and diabetic eye disease.

Alternative Names

Intravitreal injection; anti-VEGF injection; eye injection

Principle

A fine needle is passed through the sclera into the vitreous cavity, depositing medication adjacent to the retina. This achieves high intraocular drug concentrations with minimal systemic exposure.

History

Intravitreal injections were used for antibiotics and gas in the 1970s. The advent of anti-VEGF therapy in the 2000s made injection the leading treatment for neovascular age-related macular degeneration, diabetic retinopathy, and retinal vein occlusion.

Why It Is Done

Performed for neovascular age-related macular degeneration, diabetic macular edema, retinal vein occlusion, uveitis, endophthalmitis, and other retinal diseases, and for corticosteroid delivery.

Is This a Primary or Secondary Test or Procedure

Primary first-line therapy for many sight-threatening retinal diseases.

How It Is Performed

Performed as an office procedure. The eye is anesthetized with topical or subconjunctival anesthetic, the surface is sterilized with povidone-iodine, a speculum is placed, and a small-gauge needle is introduced through the pars plana to deliver the drug. The eye is then examined and monitored.

Preparation

Usually none beyond stopping topical contact lenses. A discussion of injection frequency and follow-up is essential because treatment is often repeated monthly or every few months.

Contraindications and Precautions

Active ocular infection or inflammation is a contraindication. Precaution: aseptic technique is critical because endophthalmitis, while rare, is devastating.

Risks

Transient discomfort, subconjunctival hemorrhage, elevated intraocular pressure, floaters, cataract, retinal detachment, and the rare but serious risks of endophthalmitis and retinal tears.

Aftercare and Recovery

The patient is monitored briefly, and visual acuity and pressure may be checked. Most people resume normal activity immediately.

Outcome and Follow-up

Vision improvement or stabilization over weeks confirms response. Failure to improve may indicate non-responder status, and the regimen is adjusted.

Expected Results

Stabilization or improvement of vision with reduction in retinal fluid on optical coherence tomography.

Follow-up

Frequent injections and monitoring with visual acuity testing and OCT; long-term surveillance for disease recurrence.

References

1. MedlinePlus. Intravitreal injection. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 110

Kidney Biopsy

Overview

A kidney biopsy is a procedure that obtains a small sample of kidney tissue, usually through a needle, to diagnose and guide treatment of medical kidney disease.

Alternative Names

Renal biopsy; percutaneous kidney biopsy; renal biopsy specimen

Principle

A biopsy needle is advanced into the lower pole of the kidney, typically under ultrasound or CT guidance. Obtained cores are processed for light microscopy, immunofluorescence, and electron microscopy to characterize glomerular, tubular, and interstitial disease.

History

Needle biopsy of the kidney was introduced by Iversen and Brun in 1951 and refined with real-time ultrasound guidance in the 1980s, becoming the cornerstone of nephrology diagnosis.

Why It Is Done

Performed for nephrotic syndrome, unexplained glomerulonephritis, unexplained acute kidney injury, suspected hereditary kidney disease, and to guide therapy and assess disease activity.

Is This a Primary or Secondary Test or Procedure

Primary diagnostic procedure for medical kidney disease.

How It Is Performed

Performed in an outpatient or inpatient setting. The patient lies prone, the lower pole is localized by ultrasound, the site is anesthetized, and a spring-loaded biopsy needle obtains two to three cores under real-time guidance. Pressure is held after removal.

Preparation

Coagulation studies, blood pressure control, and discontinuation of anticoagulants and aspirin. NPO is usually not required. Blood type and screen are obtained in case of bleeding.

Contraindications and Precautions

Uncontrolled hypertension, severe coagulopathy, a solitary kidney, or a bleeding disorder are relative contraindications. Precaution: a small, atrophic or cystic kidney may be unsuitable for biopsy.

Risks

Bleeding with macroscopic hematuria, perinephric hematoma, arteriovenous fistula, infection, and rarely the need for embolization or nephrectomy.

Aftercare and Recovery

The patient is observed for 6-24 hours with bed rest and serial urinalysis and hematocrit monitoring before discharge.

Outcome and Follow-up

Histology classifies glomerular disease (e.g., minimal change, membranous nephropathy, IgA nephropathy, lupus nephritis) and guides immunosuppression.

Expected Results

Normal glomeruli, tubules, interstitium, and vessels with no immune deposits.

Follow-up

Repeat biopsies may assess treatment response or disease progression; the plan follows the histologic diagnosis.

References

1. MedlinePlus. Kidney biopsy. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 111

Kyphoplasty

Overview

Kyphoplasty is a variation of vertebroplasty where a balloon is inflated inside the fractured vertebra to restore height before injecting bone cement.

Alternative Names

Balloon kyphoplasty, BKP, Spinal balloon treatment

Principle

A balloon is inserted through a needle into a compressed vertebral body and inflated to restore height and create a cavity; bone cement is then injected into the cavity. The cement hardens to stabilize the fracture and relieve pain.

History

Developed in the late 1990s and FDA-approved in 1998 to reduce the risk of cement leakage and correct spinal deformity (kyphosis).

Why It Is Done

Ordered for acute or subacute painful osteoporotic vertebral compression fractures, especially when there is significant height loss or kyphotic deformity.

Is This a Primary or Secondary Test or Procedure

Primary interventional standard for acute painful vertebral compression fractures with height loss.

How It Is Performed

Under local anesthesia and sedation. Under fluoroscopy, a balloon catheter is placed in the vertebra, inflated to create a cavity and restore height, then deflated. The cavity is then filled with PMMA.

Preparation

MRI of the spine to verify fracture age and edema; check coagulation profile; NPO if sedation is used.

Contraindications and Precautions

Severe vertebrae destruction preventing instrument placement; active spinal infection; asymptomatic fractures.

Risks

Balloon rupture, cement leakage, nerve injury, infection, or adjacent segment fractures.

Aftercare and Recovery

Lie flat for 1 hour; monitor access sites; most patients experience rapid pain relief and go home same day.

Outcome and Follow-up

X-rays show restoration of vertebral body height and stable cement placement.

Expected Results

Restored vertebral height; stable cement deposition; no leakage; resolution of focal back pain.

Follow-up

Follow-up visit in 2 weeks; treat underlying osteoporosis.

References

1. MedlinePlus. Kyphoplasty. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 112

Laparoscopy

Overview

Laparoscopy is a minimally invasive procedure in which a camera and surgical instruments are passed through small incisions, called ports, in the abdomen. It allows the surgeon to see inside the abdomen and pelvis without a large incision. It can be used for diagnosis, treatment, or both.

Alternative Names

Keyhole surgery, minimally invasive abdominal surgery, diagnostic laparoscopy, therapeutic laparoscopy

Principle

A narrow telescope with a light and camera is inserted through a small incision. Carbon dioxide gas is used to inflate the abdomen so that organs can be seen clearly. Instruments are passed through additional ports to inspect and treat structures.

History

The first laparoscopic procedures were performed in the early 1900s, and technical advances expanded the technique through the 1970s. The first laparoscopic cholecystectomy in 1985 transformed abdominal surgery. Laparoscopy is now the standard approach for many operations.

Why It Is Done

Diagnostic laparoscopy is used to inspect the pelvic and abdominal organs when imaging is unclear. Therapeutic laparoscopy treats conditions such as gallbladder disease, hernias, endometriosis, and ovarian cysts. It is often preferred because recovery is faster than with open surgery.

Is This a Primary or Secondary Test or Procedure

It is a primary procedure for many elective operations and a secondary tool when noninvasive tests are inconclusive. It may be diagnostic, therapeutic, or both in the same operation.

How It Is Performed

The patient is placed under general anesthesia, and the abdomen is inflated with carbon dioxide. A port is created at the navel for the camera, and additional small ports are placed for instruments. The surgeon inspects or treats the organs and then removes the ports and closes the small wounds.

Preparation

Fasting for six to eight hours before the procedure and emptying the bladder and bowels as directed. Blood-thinning medications are stopped beforehand. A pregnancy test is done when appropriate.

Contraindications and Precautions

- Severe cardiopulmonary disease that cannot tolerate gas inflation
- Extensive prior abdominal surgery with adhesions
- Uncorrected coagulopathy
- Late pregnancy

Risks

Bleeding, infection, and injury to abdominal organs or blood vessels are possible. Port-site hernias can develop at the incision sites. Gas-related shoulder and abdominal discomfort is common and temporary.

Aftercare and Recovery

Most patients go home the same day or stay overnight. Mild pain, bloating, and shoulder ache from the gas are expected. Activity can usually be resumed within a few days to a week.

Outcome and Follow-up

A diagnostic laparoscopy confirms the presence or absence of disease seen at surgery. A therapeutic laparoscopy is successful when the condition is corrected and the patient recovers without complications.

Expected Results

Normal-appearing abdominal and pelvic organs with no findings requiring treatment. For therapeutic procedures, successful completion without conversion to open surgery or major complications.

Follow-up

A follow-up visit is scheduled to review the surgical findings and check the wounds. Heavy lifting is restricted for several weeks to prevent port-site hernias. Additional testing is guided by the findings and any tissue sent to the laboratory.

References

1. MedlinePlus. Laparoscopy. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 113

Laparoscopy and Dye Test

Overview

Laparoscopy and dye test is a diagnostic fertility procedure in women that examines the uterus and fallopian tubes. A colored dye is passed through the uterus into the fallopian tubes while the organs are viewed with a laparoscope. It checks whether the tubes are open and whether sperm and egg can travel freely.

Alternative Names

Dye test, chromoperturbation, hydrotubation with laparoscopy, tubal patency test with laparoscopy

Principle

A thin telescope is inserted through a small incision in the navel to view the pelvis. Dye is then injected through the cervix into the uterus and fallopian tubes. If the dye flows out of the tube ends, the tubes are patent (open).

History

Injection of dye to test tubal patency was described in the mid-1900s. Combining the test with laparoscopy allowed direct visualization and improved accuracy. It remains a standard part of the fertility workup.

Why It Is Done

It is ordered as part of an infertility evaluation to determine whether the fallopian tubes are blocked. It is also used to assess the uterus, ovaries, and pelvic cavity. It may be recommended after pelvic infection, surgery, or endometriosis.

Is This a Primary or Secondary Test or Procedure

It is a primary test for tubal patency in women who have not conceived. It is secondary when used to confirm findings seen on ultrasound or hysterosalpingography.

How It Is Performed

The procedure is done under general anesthesia. After inflating the abdomen with gas, a laparoscope is inserted through the navel, and dye is injected through the cervix. The surgeon observes whether dye passes through each tube and also inspects the ovaries and pelvis.

Preparation

The test is timed for the first half of the menstrual cycle, before ovulation. Fasting for several hours is required because general anesthesia is used. Pregnancy is excluded beforehand.

Contraindications and Precautions

- Confirmed or suspected pregnancy
- Active pelvic infection
- Allergy to the dye used
- Heavy bleeding during the current menstrual period

Risks

Bleeding, infection, and injury to pelvic organs are possible. Shoulder and abdominal pain may occur from the gas used to inflate the abdomen. Minor adhesions may be treated at the same time.

Aftercare and Recovery

Most women go home the same day after recovery from anesthesia. Mild cramping, bloating, and a little vaginal spotting are common. Strenuous activity is usually avoided for a few days.

Outcome and Follow-up

If dye flows freely from both tubes, the tubes are patent and no tubal cause for infertility is found. If dye does not pass, the tube is blocked, and the level and site of blockage guide treatment.

Expected Results

Free flow of dye through both fallopian tubes, confirming patency. The uterus, ovaries, and pelvic lining appear normal.

Follow-up

Results are discussed with a fertility specialist who plans the next steps, such as timed intercourse or assisted reproduction. If blockages are found, surgery or in vitro fertilization may be considered. Repeat testing is only done if symptoms suggest new problems.

References

1. MedlinePlus. Laparoscopy and Dye Test. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 114

Laryngoscopy

Overview

Laryngoscopy is an examination of the back of the throat, including the larynx (voice box) and vocal cords, using a mirror (indirect) or a flexible endoscope (direct).

Alternative Names

Vocal cord exam, Laryngeal endoscopy

Principle

A rigid or flexible scope visualizes the larynx and vocal cords: flexible scopes pass through the nose under local anesthesia, while rigid scopes are used in the operating room. The direct view allows diagnosis, biopsy, and treatment of laryngeal lesions.

History

Manuel Garcia, a singing teacher, invented the first laryngoscope in 1854 using sunlight and dental mirrors. Flexible scopes were introduced in the late 20th century.

Why It Is Done

Ordered for persistent hoarseness, chronic throat pain, difficulty swallowing, sensation of a lump in the throat (globus), or chronic cough.

Is This a Primary or Secondary Test or Procedure

Primary diagnostic test for voice disorders, vocal cord nodules, and laryngeal tumors.

How It Is Performed

A local anesthetic spray is applied to numb your throat. A thin, flexible scope is passed through your nose and down into your throat while you make vocal sounds.

Preparation

Avoid eating or drinking for 2 hours before the test to prevent gagging and aspiration if a numbing spray is used.

Contraindications and Precautions

Acute epiglottitis (risk of airway occlusion, especially in children).

Risks

Minor throat irritation, gagging, or rare nosebleeds from the scope passage.

Aftercare and Recovery

Do not eat or drink until your throat numbness wears off (about 1 hour) to avoid choking. You can then resume normal activities.

Outcome and Follow-up

Visualizes vocal cord movement, inflammation, polyps, nodules, or tumors.

Expected Results

Normal mucosal lining; symmetrical vocal cord movement; no nodules, polyps, or masses.

Follow-up

Vocal therapy, biopsy of lesions, or CT scan of the neck.

References

1. MedlinePlus. Laryngoscopy. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 115

Laser Photocoagulation of the Retina

Overview

Laser photocoagulation of the retina is an ophthalmic procedure in which a laser is used to seal leaking blood vessels, treat new abnormal vessels, or create burns in the retina to prevent vision loss in conditions such as diabetic retinopathy and retinal tears.

Alternative Names

Retinal laser photocoagulation; laser therapy for the retina; panretinal photocoagulation; focal laser treatment

Principle

Laser energy is absorbed by retinal tissue and converted to heat, coagulating tissue and blood vessels. This seals leaking vessels, closes abnormal new vessels, and creates chorioretinal adhesions around tears that prevent retinal detachment.

History

Retinal laser photocoagulation was developed by Franciscus Donders' successors and others from the 1960s, and large trials in the 1970s established it as standard treatment for proliferative diabetic retinopathy and macular edema.

Why It Is Done

Ordered for proliferative diabetic retinopathy, clinically significant diabetic macular edema, retinal tears, central serous retinopathy, and certain vascular retinal diseases to prevent or limit vision loss.

Is This a Primary or Secondary Test or Procedure

A primary therapeutic procedure for many retinal diseases, used alone or with other treatments such as anti-VEGF injections.

How It Is Performed

After the pupil is dilated and eye drops applied, a contact lens is placed on the eye and laser pulses are delivered through a slit lamp to the target areas of the retina. The number and pattern of burns depend on the condition being treated.

Preparation

Dilation of the pupil with eye drops and review of the patient's medications, including blood thinners, which may be continued in most cases.

Contraindications and Precautions

Precautions include media opacity such as cataract or vitreous hemorrhage that blocks the laser beam, and caution with perifoveal treatment, which risks central scotoma.

Risks

Temporary blurred vision, discomfort, scarring, decreased night or peripheral vision after panretinal photocoagulation, and rarely bleeding, retinal detachment, or increased eye pressure.

Aftercare and Recovery

The patient may experience transient visual blurring and uses prescribed eye drops. Vision may take weeks to stabilize, and follow-up examinations monitor the response.

Outcome and Follow-up

Control of the treated condition, with reduced leakage and regression of new vessels, indicates success. Persistent or progressive disease may require additional laser or other treatment.

Expected Results

Stabilized or improved vision with sealing of leaking vessels, regression of abnormal new vessels, and no evidence of ongoing retinal disease progression.

Follow-up

Regular dilated eye examinations are required, and further laser sessions, injections, or vitrectomy may be offered depending on the response.

References

1. MedlinePlus. Laser eye surgery for diabetic retinopathy. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 116

Laser Skin Treatment

Overview

Laser skin treatment uses focused beams of light (CO2, Erbium, or pulsed dye lasers) to remove damaged outer skin layers or target specific vascular/pigment structures to stimulate collagen growth.

Alternative Names

Laser skin resurfacing, Laser dermatology

Principle

Laser light of a specific wavelength is absorbed by targeted chromophores in the skin such as pigment, hemoglobin, or water, converting light into heat. The heat destroys or remodels the target tissue while minimizing damage to surrounding skin.

History

The ruby laser was first used in dermatology in the 1960s by Leon Goldman. Carbon dioxide (CO2) lasers were developed in the 1980s for skin resurfacing.

Why It Is Done

Ordered to treat acne scars, facial wrinkles, sun damage, birthmarks, port-wine stains, or tattoo removal.

Is This a Primary or Secondary Test or Procedure

Primary aesthetic and corrective treatment for superficial scarring, vascular lesions, and skin rejuvenation.

How It Is Performed

A topical numbing cream is applied. You wear protective goggles. The clinician passes the laser handpiece over the target skin, delivering quick pulses of light.

Preparation

Avoid sun exposure and tanning for 4 weeks before the procedure. Stop using retinoids 1 week before.

Contraindications and Precautions

Active acne or skin infection in the target area, history of keloid scarring, or active use of isotretinoin (Accutane) within 6 months.

Risks

Redness, swelling, hyperpigmentation or hypopigmentation, scarring, or reactivation of herpes simplex (cold sores).

Aftercare and Recovery

The skin will feel like a sunburn. Apply gentle moisturizers and strict sun protection. Avoid scratching peeling skin.

Outcome and Follow-up

Over weeks, the skin heals, showing smoother texture, reduced pigmentation, and increased elasticity.

Expected Results

Successful treatment of the targeted areas; peeling followed by regeneration of healthy, smooth skin.

Follow-up

Post-treatment skin check-up; repeat sessions as planned.

References

1. MedlinePlus. Laser Skin Treatment. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 117

LASIK Eye Surgery

Overview

LASIK is a refractive eye procedure that reshapes the cornea with a laser to reduce dependence on glasses or contact lenses.

Why It Is Done

It is an elective treatment for suitable adults with stable nearsightedness, farsightedness, or astigmatism after a comprehensive eye evaluation.

How It Is Performed

A thin corneal flap is created, an excimer laser reshapes the underlying cornea, and the flap is repositioned. The procedure is usually performed with topical anesthetic.

Risks and Recovery

Dry eye, glare, halos, under- or over-correction, infection, corneal ectasia, flap problems, and need for additional treatment are possible. Vision often improves quickly, but stabilization takes longer.

References

1. National Eye Institute. LASIK. 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 118

LEEP (Loop Electrosurgical Excision Procedure)

Overview

LEEP removes abnormal cervical tissue using a thin wire loop carrying an electrical current. The tissue is sent to a laboratory to diagnose and treat precancerous cervical changes.

Why It Is Done

It is commonly recommended for biopsy-confirmed high-grade cervical dysplasia or selected persistent abnormal cervical changes. It can both remove the abnormal area and provide a specimen for examination.

How It Is Performed

The patient lies in the position used for a pelvic examination. After a speculum is placed, the cervix is numbed with local anesthetic. The clinician uses the energized loop to remove transformation-zone tissue, then controls bleeding with cautery or a topical agent.

Preparation

Pregnancy should be excluded, and the clinician should be told about blood-thinning medicines or bleeding disorders. The procedure is usually scheduled when the patient is not menstruating and is generally performed in an office or outpatient setting.

Risks and Recovery

Cramping, light bleeding, and watery or dark discharge are common for a short time. Less common risks include infection, heavier bleeding, cervical narrowing, and a small increase in the risk of preterm birth in a future pregnancy. Patients are usually advised to avoid vaginal intercourse, tampons, and douching while the cervix heals.

Outcome and Follow-up

The pathology report describes the grade of cervical intraepithelial neoplasia and whether the margins are clear. Follow-up usually includes HPV testing and/or cervical cytology at an interval recommended by the clinician; a positive margin or persistent abnormality may require additional evaluation.

References

1. MedlinePlus. Loop electrosurgical excision procedure (LEEP). U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 119

Lithotripsy

Overview

ESWL uses shock waves generated outside the body to break kidney or ureteral stones into tiny fragments (sand) that can then pass naturally in the urine.

Alternative Names

Extracorporeal Shock Wave Lithotripsy, ESWL, Kidney stone blast

Principle

Shock waves are generated outside the body and focused on a kidney or ureteral stone, traveling through tissue to produce compressive and tensile forces. These forces fragment the stone into small pieces that can pass in the urine.

History

Developed in Germany by Dornier Medizintechnik in the late 1970s. The first human treatment occurred in 1980, transforming kidney stone management.

Why It Is Done

Ordered for symptomatic kidney or ureteral stones, typically between 5 mm and 20 mm in size, that fail to pass spontaneously.

Is This a Primary or Secondary Test or Procedure

Primary non-invasive treatment option for simple renal pelvic and upper ureteral stones.

How It Is Performed

Performed under sedation or general anesthesia. You lie on a table. Fluoroscopy or ultrasound is used to locate the stone. A shock-wave generator delivers thousands of targeted shock waves through a water cushion.

Preparation

Fast for 8 hours. Stop all blood thinners at least 7-10 days prior to prevent kidney hematoma.

Contraindications and Precautions

Pregnancy, active urinary tract infection, uncorrected bleeding disorders, or severe obesity preventing stone targeting.

Risks

Renal hematoma, hematuria (blood in urine, expected), renal colic (pain from passing fragments), or 'steinstrasse' (ureteral obstruction by stone fragments).

Aftercare and Recovery

Drink plenty of water to flush out stone fragments. Strain your urine to collect fragments for analysis. Take prescribed alpha-blockers.

Outcome and Follow-up

Success is determined by the fragmentation of the stone and complete passage of fragments, confirmed by follow-up X-ray or ultrasound.

Expected Results

Complete fragmentation of the target stone; passage of stone gravel; no residual stones on post-op imaging.

Follow-up

Follow-up X-ray (KUB) or ultrasound in 2 weeks; dietary counseling for stone prevention.

References

1. MedlinePlus. Lithotripsy. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 120

Liver Biopsy

Overview

A liver biopsy is a procedure that obtains a small core of liver tissue, usually percutaneously, to diagnose and stage liver disease.

Alternative Names

Percutaneous liver biopsy; liver needle biopsy; hepatic biopsy

Principle

A spring-loaded or suction biopsy needle is advanced into the liver through the intercostal space or via a transjugular route, capturing a tissue core for histologic evaluation of inflammation, fibrosis, steatosis, and other pathology.

History

Percutaneous liver biopsy was introduced by Iversen and Roholm in 1939 and became a standard diagnostic tool, increasingly guided by ultrasound in recent decades.

Why It Is Done

Performed for unexplained liver enzyme elevation, chronic hepatitis of uncertain cause, staging of fibrosis in chronic liver disease, evaluation of autoimmune hepatitis, and characterization of hepatic masses.

Is This a Primary or Secondary Test or Procedure

Primary diagnostic procedure for parenchymal liver disease; secondary when non-invasive tests are inconclusive.

How It Is Performed

Performed with the patient supine, usually under ultrasound guidance. The biopsy site is marked and anesthetized, the needle is inserted during a breath-hold, and one or two cores are obtained. Pressure is applied, and the patient is observed.

Preparation

Coagulation studies, platelet count, and blood type. Anticoagulants and antiplatelet agents are held, and informed consent is obtained. NPO is generally not required.

Contraindications and Precautions

Coagulopathy, thrombocytopenia, ascites, or an uncooperative patient are relative contraindications. Precaution: a transjugular approach may be used when bleeding risk is high.

Risks

Pain, bleeding with hemoperitoneum or subcapsular hematoma, biliary leak, and rarely puncture of other organs or death.

Aftercare and Recovery

The patient is observed for 2-4 hours with vital sign monitoring before discharge; severe pain or hypotension prompts urgent evaluation.

Outcome and Follow-up

Histology grades necroinflammation, stages fibrosis, identifies the cause (steatohepatitis, autoimmune hepatitis, viral hepatitis, drug injury), and assesses response to therapy.

Expected Results

Normal hepatocytes with no significant inflammation, fibrosis, steatosis, or other abnormality.

Follow-up

Results guide treatment; serial non-invasive fibrosis markers or repeat biopsy monitor disease progression.

References

1. MedlinePlus. Liver biopsy. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 121

Lumbar Drain Placement

Overview

A lumbar drain is a small catheter inserted into the lumbar subarachnoid space to continuously drain cerebrospinal fluid (CSF) to lower intracranial pressure or manage CSF leaks.

Alternative Names

Spinal catheter placement, Continuous CSF drainage

Principle

A catheter is inserted through a needle between the lumbar vertebrae into the subarachnoid space and connected to a closed drainage system. It allows controlled, continuous drainage of cerebrospinal fluid to lower intracranial pressure or remove blood and debris.

History

Evolved from the lumbar puncture technique first described by Heinrich Quincke in 1891. Continuous lumbar drainage was developed in the mid-20th century.

Why It Is Done

Ordered to treat postoperative CSF leaks, reduce brain tension during neurosurgery, or evaluate patients for normal pressure hydrocephalus (NPH).

Is This a Primary or Secondary Test or Procedure

Primary temporary therapeutic option for persistent cranial/spinal CSF leaks.

How It Is Performed

Under sterile conditions, a needle is inserted into the L3-L4 or L4-L5 space. A flexible catheter is threaded through the needle into the subarachnoid space and connected to a collection bag.

Preparation

Check coagulation profile (coagulopathy is a strict contraindication). Position the patient sitting or in lateral decubitus.

Contraindications and Precautions

Obstructive hydrocephalus or space-occupying lesion with mass effect (risk of brain herniation), or local skin infection.

Risks

Over-drainage (headache, subdural hematoma, herniation), meningitis, nerve root irritation (radicular pain), or catheter blockage.

Aftercare and Recovery

The drainage system must be leveled carefully (usually at the shoulder or ear level) as ordered. Monitor neurological status and drainage rate hourly.

Outcome and Follow-up

Clear CSF is collected. Symptoms of CSF leak should resolve.

Expected Results

Drainage of clear, colorless CSF at the prescribed flow rate (typically 5-10 mL/hour); patient neurologically intact.

Follow-up

Daily CSF laboratory analysis (cell count, protein, glucose) if infection is suspected; catheter removal within 5-7 days.

References

1. MedlinePlus. Lumbar Drain Placement. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 122

Lumbar Puncture

Overview

Lumbar puncture, also known as a spinal tap, is a procedure that removes cerebrospinal fluid (CSF) from the subarachnoid space of the lumbar spine for diagnosis or to deliver medication.

Alternative Names

Spinal tap; CSF collection; lumbar tap

Principle

A needle is inserted between the lumbar vertebrae, typically L3-L4 or L4-L5, into the subarachnoid space. CSF pressure is measured and fluid is collected for analysis or medication such as anesthetic, chemotherapy, or antibiotics is instilled.

History

Lumbar puncture was introduced by Heinrich Quincke in 1891 and rapidly became a cornerstone of neurologic diagnosis, especially for meningitis and subarachnoid hemorrhage.

Why It Is Done

Performed to evaluate suspected meningitis or encephalitis, subarachnoid hemorrhage, demyelinating disease, inflammatory or neoplastic processes of the central nervous system, and to administer intrathecal medication or measure CSF pressure.

Is This a Primary or Secondary Test or Procedure

Primary diagnostic procedure for central nervous system infection and other intrathecal pathology.

How It Is Performed

The patient lies on the side in the fetal position or sits leaning forward. The back is prepped and anesthetized, and a spinal needle is advanced between the spinous processes until a "pop" is felt. Opening pressure is measured, CSF is collected, and the needle is removed.

Preparation

Informed consent, review of anticoagulants and bleeding risk, and imaging before the procedure when papilledema, focal neurologic deficit, or suspected mass is present.

Contraindications and Precautions

Increased intracranial pressure from a mass lesion, bleeding diathesis or anticoagulation, and skin infection over the puncture site are contraindications. Precaution: obtain imaging first if a space-occupying lesion is suspected.

Risks

Post-dural-puncture headache, bleeding, infection, nerve root irritation, and rarely cerebral herniation when performed despite elevated intracranial pressure.

Aftercare and Recovery

The patient lies flat or semireclined and is encouraged to hydrate. The sample is sent immediately for cell count, protein, glucose, culture, and other studies.

Outcome and Follow-up

A low CSF glucose with elevated protein and neutrophils suggests bacterial meningitis; lymphocytic pleocytosis suggests viral, mycobacterial, or fungal causes; xanthochromia supports subarachnoid hemorrhage.

Expected Results

Clear, colorless CSF with normal opening pressure, low white cell count, normal protein and glucose, and no organisms.

Follow-up

Results guide antimicrobial and supportive therapy; repeat puncture may assess response or reclassify uncertain cases.

References

1. MedlinePlus. Lumbar puncture (spinal tap). U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 123

Lymph Node Biopsy

Overview

A lymph node biopsy removes all or part of an enlarged or abnormal lymph node for histologic and molecular diagnosis, most commonly to evaluate suspected lymphoma or metastatic cancer.

Alternative Names

Excisional lymph node biopsy; sentinel lymph node biopsy; lymphadenectomy; node sampling

Principle

The lymph node is excised intact (excisional biopsy) or sampled (incisional or core biopsy) and examined by histology, flow cytometry, and molecular testing. Excisional biopsy provides the architecture needed for lymphoma classification.

History

Lymph node biopsy became a cornerstone of pathology in the 19th century, and excisional biopsy remains the standard for suspected lymphoma, complemented by fine-needle and core biopsy in modern practice.

Why It Is Done

Performed for unexplained persistent lymphadenopathy, suspected lymphoma or metastatic disease, and for sentinel node evaluation in breast cancer, melanoma, and other cancers.

Is This a Primary or Secondary Test or Procedure

Primary diagnostic procedure for lymphadenopathy of unknown cause.

How It Is Performed

Performed under local or general anesthesia. The node is localized by palpation or ultrasound, an incision is made along skin creases, and the node is carefully dissected and removed intact with hemostasis. The wound is closed and the specimen is oriented and sent fresh for processing.

Preparation

Usually none beyond routine preoperative assessment. The surgeon and pathologist coordinate handling to ensure fresh tissue for flow cytometry and molecular studies.

Contraindications and Precautions

Bleeding disorders and infection over the site are relative contraindications. Precaution: identify and protect adjacent structures such as vessels and nerves, especially in the neck and axilla.

Risks

Bleeding, infection, seroma, nerve injury (e.g., spinal accessory nerve in the neck), lymphedema, and anesthetic risks.

Aftercare and Recovery

Most patients go home the same day. Mild pain is controlled with analgesics, and the wound is kept clean and dry.

Outcome and Follow-up

Histology distinguishes reactive hyperplasia from lymphoma, metastasis, granulomatous disease, or infection; flow cytometry and cytogenetics subclassify lymphomas.

Expected Results

Reactive lymphoid hyperplasia with a preserved architecture and no evidence of malignancy.

Follow-up

Results guide staging, referral, and treatment; persistent or recurring adenopathy elsewhere prompts further evaluation.

References

1. MedlinePlus. Lymph node biopsy. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 124

Mechanical Ventilation

Overview

Mechanical ventilation is a life-support treatment that helps a patient breathe when they cannot breathe adequately on their own, using a machine (ventilator).

Alternative Names

Ventilator support, Life support, Mechanical breathing support

Principle

A machine delivers pressurized gas into the lungs through an endotracheal tube or mask, generating positive pressure that inflates the lungs and performs the work of breathing. The ventilator controls tidal volume, rate, and oxygen concentration to support gas exchange.

History

The iron lung (negative pressure) was widely used during the polio epidemics. Modern positive-pressure ventilators were developed in the mid-20th century.

Why It Is Done

Ordered for acute respiratory failure, severe hypoxemia, hypercapnia, coma, or during major surgery under general anesthesia.

Is This a Primary or Secondary Test or Procedure

Primary life-support intervention for acute respiratory failure.

How It Is Performed

The ventilator is connected to the patient's airway via an endotracheal tube or tracheostomy tube. The machine delivers gas (air/oxygen mix) using positive pressure.

Preparation

Perform endotracheal intubation; secure the tube; verify placement; connect to ventilator.

Contraindications and Precautions

DNR/DNI orders (unless temporary/reversible). Relative includes untreated pneumothorax (must place chest tube first).

Risks

Ventilator-associated pneumonia (VAP), barotrauma/pneumothorax, lung injury (VALI), or hypotension.

Aftercare and Recovery

Perform daily sedation vacation and spontaneous breathing trials (SBTs) to evaluate for extubation.

Outcome and Follow-up

Monitored using arterial blood gases (ABGs), chest X-rays, and ventilator graphics (tidal volume, airway pressures).

Expected Results

Adequate gas exchange (PaO2 >60 mmHg, pH 7.35-7.45); patient synchrony with the ventilator.

Follow-up

Continuous respiratory therapy monitoring; weaning protocol.

References

1. MedlinePlus. Mechanical Ventilation. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 125

Medical Abortion

Overview

Medical abortion is the termination of a pregnancy using medications rather than surgery. It is effective for early pregnancy, generally up to about 10-11 weeks of gestation, and is performed under clinician supervision.

Alternative Names

Abortion pill; medication abortion; nonsurgical abortion; mifepristone-misoprostol abortion; RU-486

Principle

A combination of two medications is used: mifepristone blocks the action of progesterone, which is needed to maintain the pregnancy, and misoprostol, taken 24-48 hours later, causes uterine contractions that expel the pregnancy. The process produces bleeding and cramping similar to a heavy period.

History

Mifepristone (RU-486) was first approved in France in 1988 and in the United States in 2000. The mifepristone-misoprostol regimen has since become the standard medical method for early abortion worldwide.

Why It Is Done

Chosen by a person who wishes to end an early pregnancy and prefers to avoid surgery. It is also used for incomplete or missed miscarriage management. The clinician must confirm gestational age before proceeding.

Is This a Primary or Secondary Test or Procedure

A primary method of early pregnancy termination, offering a safe alternative to surgical abortion for eligible patients.

How It Is Performed

Mifepristone is taken by mouth in the clinic or at home, followed 24-48 hours later by misoprostol, which may be taken by mouth or placed in the cheek, under the tongue, or in the vagina. Bleeding and cramping typically begin within hours of the misoprostol dose, and a follow-up visit confirms that the abortion is complete.

Preparation

Pregnancy dating by ultrasound or last menstrual period, counseling about options, and a plan for follow-up. Rh-negative patients may need Rh immune globulin, and arrangements for pain control and after-hours contact should be made.

Contraindications and Precautions

Medical abortion is not used in pregnancies beyond the approved gestational age, in ectopic pregnancy, or when there is an allergy to the medications. Severe anemia, bleeding disorders, and long-term corticosteroid or anticoagulant use require special caution.

Risks

- Incomplete abortion requiring surgical completion
- Heavy or prolonged bleeding
- Infection
- Persistent pain, nausea, or diarrhea from the medications

Aftercare and Recovery

Cramping and bleeding usually resolve within 1-2 weeks, and a follow-up visit or home test confirms the pregnancy has ended. Normal activities may resume once bleeding settles, but intercourse and tampon use should be avoided until the bleeding stops.

Outcome and Follow-up

The procedure is successful when bleeding occurs and a follow-up evaluation shows no remaining pregnancy tissue. Symptoms of pregnancy, such as nausea and breast tenderness, usually resolve within days.

Expected Results

Complete expulsion of the pregnancy with return of a normal menstrual period, usually within 4-6 weeks, and no need for further intervention.

Follow-up

A follow-up visit 1-2 weeks later with ultrasound or a pregnancy test to confirm completion. Surgical evacuation is performed if the abortion is incomplete or bleeding is excessive.

References

1. MedlinePlus. Medical Abortion. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 126

Mohs Surgery

Overview

Mohs surgery is a tissue-sparing treatment for selected skin cancers. The visible tumor is removed in stages, and each layer is examined microscopically until no cancer cells remain at the edges.

Why It Is Done

It is commonly used for basal cell carcinoma and squamous cell carcinoma, especially when the cancer is on the face, hands, feet, or other areas where preserving healthy tissue is important, or when the tumor has a high risk of recurrence.

How It Is Performed

The area is numbed with local anesthetic. The surgeon removes the visible tumor and a thin surrounding layer, maps and examines the tissue, and removes additional small layers only where cancer remains. The wound is then allowed to heal or is closed with stitches or a reconstruction.

Preparation and Recovery

Tell the clinician about blood thinners, allergies, and medical conditions. Most patients return home the same day. Wound care, bleeding precautions, and follow-up depend on the size and location of the wound.

Risks

Bleeding, infection, pain, scarring, numbness, and recurrence are possible. The risks vary with the cancer and the repair required.

Outcome and Follow-up

The procedure is complete when the examined margins contain no remaining cancer cells. Continued skin examinations are important because a history of skin cancer increases the risk of developing another lesion.

References

1. National Cancer Institute. Mohs surgery. 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 127

Myringotomy and Tube Insertion

Overview

Myringotomy with tympanostomy tube placement creates a small opening in the eardrum and inserts a tiny tube to ventilate the middle ear and prevent fluid accumulation.

Alternative Names

Ear tube surgery; tympanostomy tube placement; grommet insertion; ventilation tube insertion

Principle

Incision of the tympanic membrane allows drainage of middle-ear fluid and ventilation of the middle ear, equalizing pressure. A tube keeps the opening patent, bypassing a dysfunctional eustachian tube.

History

Myringotomy has been performed since the 19th century, and tympanostomy tubes were introduced by Armstrong in 1954. They remain the most common pediatric surgical procedure.

Why It Is Done

Performed for recurrent acute otitis media, persistent middle-ear effusion (chronic otitis media with effusion) with hearing loss, and eustachian tube dysfunction that has failed medical therapy.

Is This a Primary or Secondary Test or Procedure

Primary surgical treatment for recurrent or chronic middle-ear disease; secondary to antibiotic therapy.

How It Is Performed

Performed under general anesthesia in children, or local anesthesia in adults. The ear canal is examined microscopically, a small incision is made in the tympanic membrane, fluid is aspirated, and a tube is inserted. The procedure takes only minutes.

Preparation

NPO before general anesthesia, and an examination by an ear, nose, and throat specialist. Audiometry is often performed preoperatively.

Contraindications and Precautions

Cholesteatoma or a perforation that is already draining is a contraindication. Precaution: the patient should avoid water entering the ear unless protected.

Risks

Tube obstruction or premature extrusion, persistent perforation after tube removal, scarring of the eardrum, recurrent drainage (otorrhea), and rarely cholesteatoma.

Aftercare and Recovery

Hearing usually improves promptly. The tube stays in place for 6-18 months before extruding spontaneously.

Outcome and Follow-up

Resolution of effusion and improved hearing confirm success. Drainage from the ear prompts evaluation for infection.

Expected Results

A patent tube with a dry, aerated middle ear and restored hearing.

Follow-up

Audiometry and otoscopy at intervals; tubes that fail to extrude or recurrent disease may require repeat placement or removal.

References

1. MedlinePlus. Myringotomy and tubes. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 128

Nasal Packing (Epistaxis Management)

Overview

Nasal packing is the placement of an absorbent or inflatable material into the nasal cavity to control epistaxis that does not respond to direct pressure or cautery. It is an acute management technique used by emergency and ear, nose, and throat (ENT) clinicians to achieve hemostasis.

Alternative Names

Anterior nasal packing; posterior nasal packing; epistaxis packing; nasal tamponade

Principle

Packing applies direct mechanical pressure to the bleeding nasal mucosa, promoting clot formation and tamponade of the vessel. Moisture-retaining materials hydrate the mucosa and prevent crusting, while inflatable balloons provide controllable compression. The packing is left in place for 24 to 72 hours until the underlying bleeding source has sealed.

History

Nasal packing has been used for centuries, with early practitioners employing cloth and gauze; modern absorbent sponges and inflatable balloons were developed in the 20th century.

Why It Is Done

Ordered for epistaxis that persists after initial measures such as digital pressure, vasoconstrictive sprays, and chemical or electrical cautery fail to control bleeding. It is also used to stabilize brisk posterior epistaxis, to manage bleeding in coagulopathic patients, and to protect the nasal mucosa after sinonasal surgery.

Is This a Primary or Secondary Test or Procedure

A primary therapeutic procedure for acute epistaxis when conservative measures fail.

How It Is Performed

The nose is examined with a headlight and nasal speculum, and the bleeding site is identified after suctioning. An anterior pack (absorbent sponge or ribbon gauze layered from floor to roof of the cavity) is placed along the nasal floor; for posterior bleeding, a balloon catheter or posterior pack is advanced and positioned behind the posterior choana. The pack is then secured externally with tape, and the patient is observed for rebleeding.

Preparation

The patient sits upright with the head tilted forward and applies pressure while the nose is inspected. Topical vasoconstrictor and local anesthetic are sprayed. Coagulation studies and blood pressure are reviewed, particularly in anticoagulated or hypertensive patients.

Contraindications and Precautions

Placement is avoided or performed with caution in suspected cerebrospinal fluid leak or skull base fracture, because of the risk of intracranial migration. Severe septal deviation, nasal tumors, and uncorrected coagulopathy may require specialty consultation rather than blind packing.

Risks

Rebleeding, local pain, nasal trauma, and dislodgement; posterior packing carries risks of aspiration, hypoxia, and the trigeminovagal reflex, and is associated with a rare risk of toxic shock syndrome.

Aftercare and Recovery

The pack is left in place for 24 to 72 hours and removed in the clinic once bleeding has stopped. Patients are advised to avoid nose blowing, heavy lifting, and anticoagulant use when possible, and are often started on saline spray or petroleum jelly to keep the mucosa moist.

Outcome and Follow-up

Resolution of active bleeding confirms adequate tamponade. Recurrent or continued bleeding after removal may indicate a persistent vessel, coagulopathy, or posterior source requiring further intervention.

Expected Results

No active bleeding after pack removal, with a healed nasal mucosa and no recurrence during the observation period.

Follow-up

Nasal endoscopy to identify the bleeding source, and evaluation of coagulation status or blood pressure if rebleeding occurs. Patients with recurrent epistaxis may benefit from cautery, embolization, or ligation of the responsible vessel.

References

1. MedlinePlus. Nosebleed. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 129

Nasogastric Tube Insertion

Overview

Nasogastric (NG) tube insertion passes a flexible tube through the nose, down the esophagus, and into the stomach for decompression, feeding, or medication administration.

Alternative Names

NG tube placement; nasogastric intubation; gastric tube

Principle

A lubricated tube is advanced through the nasal passage into the stomach. Correct placement is confirmed by aspiration of gastric contents, pH testing, or radiography. The tube drains the stomach, delivers nutrition, or gives medication.

History

Nasogastric tubes have been used since the 19th century, with modern flexible tubes and enteral feeding developed in the 20th century.

Why It Is Done

Performed for gastric decompression in bowel obstruction or ileus, short-term enteral feeding, medication administration, gastric lavage after overdose, and obtaining gastric contents for analysis.

Is This a Primary or Secondary Test or Procedure

Primary short-term intervention for gastric access or decompression.

How It Is Performed

The tube is measured and marked, lubricated, and advanced through the nose while the patient swallows. Placement is verified by aspirating gastric contents, pH testing, or chest radiograph. The tube is secured to the nose.

Preparation

Usually none. The patient may be given a local anesthetic spray and sits upright. Coagulation status is reviewed.

Contraindications and Precautions

Facial or skull base trauma (risk of intracranial placement), severe coagulopathy, esophageal varices, and recent esophageal surgery are relative contraindications.

Risks

Nosebleed, sinusitis, aspiration, esophageal injury, and malposition (including bronchial placement) if position is not confirmed.

Aftercare and Recovery

The tube is connected to suction or used for feeds, flushed regularly, and its position is rechecked before each use.

Outcome and Follow-up

Aspiration of acidic gastric fluid and a confirmatory radiograph ensure safe use.

Expected Results

A correctly placed tube providing decompression or nutrition without complication.

Follow-up

Daily position checks, skin care at the nares, and removal when the indication resolves.

References

1. MedlinePlus. Nasogastric tube. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 130

Nebulizer Treatment

Overview

Nebulizer treatment converts liquid medicine into a fine mist that can be inhaled through a mouthpiece or mask. It delivers medicine directly to the airways.

Why It Is Done

It is commonly used to deliver bronchodilators or other inhaled medicines during asthma attacks, COPD exacerbations, wheezing, and selected respiratory illnesses when a nebulizer is easier to use than an inhaler.

How It Is Performed

The prescribed medicine is placed in the nebulizer cup. The patient sits upright, seals the lips around the mouthpiece or wears a mask, and breathes normally until the treatment is complete. The equipment should be cleaned and dried according to instructions.

Preparation and Precautions

Use only the prescribed medicine and dose. Tell the clinician about heart rhythm problems, tremor, glaucoma, or medicine allergies. A bronchodilator may cause temporary shakiness, nervousness, palpitations, or a fast heart rate.

Results and Risks

Breathing, wheezing, oxygen level, and work of breathing are assessed. Cough, throat irritation, unpleasant taste, or temporary rapid heartbeat may occur. Severe or worsening breathing difficulty requires urgent medical attention rather than repeated home treatments alone.

References

1. MedlinePlus. Nebulizer treatment. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 131

Needle Decompression of Pneumothorax

Overview

Needle decompression of pneumothorax is an emergency procedure in which a large-bore needle or catheter is inserted into the pleural space to release trapped air. It is the immediate, temporizing treatment for tension pneumothorax.

Alternative Names

Needle thoracostomy, needle aspiration of pneumothorax, emergency pleural decompression

Principle

In tension pneumothorax, a one-way valve effect allows air to enter the pleural space during inspiration but prevents its escape, progressively compressing the lung and mediastinum. Inserting a cannula into the pleural space converts the pressure to atmospheric, decompressing the lung and restoring venous return until a chest tube can be placed.

History

Needle decompression has been a standard emergency intervention for tension pneumothorax since the mid-20th century and is taught in all advanced trauma and resuscitation courses. Its routine use outside tension physiology has been re-examined as simple aspiration for primary pneumothorax.

Why It Is Done

Performed emergently for suspected or confirmed tension pneumothorax with respiratory distress, hypotension, or cardiac arrest, and sometimes as simple aspiration for a large, symptomatic primary spontaneous pneumothorax.

Is This a Primary or Secondary Test or Procedure

A primary emergency intervention for tension pneumothorax, and a secondary temporizing measure before definitive chest tube drainage in other settings.

How It Is Performed

The insertion site, typically the second intercostal space in the midclavicular line or the fourth or fifth intercostal space in the anterior axillary line, is identified and prepared. A large-bore over-the-needle catheter is inserted perpendicular to the chest wall just above the rib, a rush of air or blood confirms entry into the pleural space, and the catheter is advanced while the needle is withdrawn. The catheter is left open to air or connected to a one-way valve while definitive chest tube placement is arranged.

Preparation

None is feasible in the emergency setting; sterile preparation of the chest wall and a readily available decompression kit are required. Chest radiography should not delay decompression in suspected tension pneumothorax.

Contraindications and Precautions

There are essentially no absolute contraindications in a dying patient with tension pneumothorax; relative precautions include chest wall infection or coagulopathy. Care must be taken to avoid injury to the intercostal neurovascular bundle and to recognize that a single decompression may fail, especially in obese or muscular patients.

Risks

Failure of decompression with recurrent tension, pneumothorax if no pneumothorax was present, bleeding, infection, lung laceration, and injury to mediastinal structures.

Aftercare and Recovery

The patient is reassessed for clinical improvement and undergoes chest radiography, followed by definitive tube thoracostomy when tension physiology is confirmed. The decompression catheter may be removed after a functioning chest tube is in place.

Outcome and Follow-up

Clinical success is marked by immediate improvement in blood pressure, oxygenation, and breath sounds with a rush of air on entry. Radiographic confirmation shows lung re-expansion and correct catheter position before definitive drainage.

Expected Results

Relief of tension physiology, re-expansion of the lung, and successful transition to chest tube drainage without recurrence.

Follow-up

Chest radiography, arterial blood gas or pulse oximetry monitoring, and definitive chest tube placement are the standard follow-up. Repeat decompression may be required if the catheter occludes or the tension reaccumulates.

References

1. MedlinePlus. Pneumothorax. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 132

Nephrostomy Placement

Overview

Nephrostomy placement inserts a tube through the skin into the kidney to drain urine when the ureter is blocked.

Alternative Names

Percutaneous nephrostomy; nephrostomy tube insertion; PCN

Principle

Under imaging guidance, a needle enters the renal pelvis and a catheter is placed to drain urine, bypassing an obstruction of the ureter.

History

Percutaneous nephrostomy was developed in the 1950s and is now a routine interventional radiology procedure.

Why It Is Done

Performed for ureteral obstruction causing hydronephrosis, infection, or kidney injury, and to access the kidney for stone treatment.

Is This a Primary or Secondary Test or Procedure

Primary drainage procedure for obstructive uropathy.

How It Is Performed

Performed under sedation with ultrasound and fluoroscopy. The collecting system is punctured, a wire is passed, and a drainage catheter is placed and secured.

Preparation

Imaging, coagulation review, and antibiotics in selected cases.

Contraindications and Precautions

Severe coagulopathy is a relative contraindication.

Risks

Bleeding, infection or sepsis, injury to adjacent organs, and tube dislodgement or blockage.

Aftercare and Recovery

The tube is monitored and flushed; urine output is recorded.

Outcome and Follow-up

Drainage and imaging confirm decompression of the kidney.

Expected Results

Free drainage of urine with resolution of hydronephrosis.

Follow-up

Tube exchange and imaging; definitive treatment of the obstruction.

References

1. MedlinePlus. Percutaneous nephrostomy. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 133

Nerve Block

Overview

A nerve block is an injection of local anesthetic, often with corticosteroid, near a nerve or nerve plexus to relieve pain, diagnose its source, or provide anesthesia.

Alternative Names

Nerve block injection; peripheral nerve block; sympathetic block; plexus block

Principle

Local anesthetic placed adjacent to a nerve temporarily interrupts nerve conduction, relieving pain and providing anesthesia. Corticosteroid reduces local inflammation. Fluoroscopy or ultrasound guidance targets the nerve precisely.

History

Nerve blocks evolved from early regional anesthesia in the late 19th century into a mainstay of chronic pain management and surgical anesthesia in the 20th century.

Why It Is Done

Performed for chronic pain conditions (e.g., facet arthropathy, trigeminal neuralgia, complex regional pain syndrome), to provide surgical or obstetric anesthesia, and as a diagnostic tool to identify pain generators.

Is This a Primary or Secondary Test or Procedure

Primary interventional treatment for selected pain syndromes; secondary to conservative management.

How It Is Performed

Performed in an outpatient or surgical setting. The target is identified by palpation, fluoroscopy, or ultrasound, the skin is prepared and anesthetized, and the needle is advanced to the nerve. Contrast and test doses may confirm placement before anesthetic injection.

Preparation

Anticoagulants are usually held, informed consent is obtained, and patients are monitored during and briefly after injection.

Contraindications and Precautions

Coagulopathy, anticoagulation, active infection, and allergy to local anesthetic are contraindications. Precaution: needle placement near major vessels, nerves, and the neuraxis requires image guidance.

Risks

Transient numbness or weakness, bleeding, infection, nerve injury, systemic local anesthetic toxicity, and vasovagal reactions.

Aftercare and Recovery

The block resolves over hours; corticosteroid effects may take days. Patients are observed and warned about temporary weakness if a motor block occurred.

Outcome and Follow-up

Temporary relief confirms the nerve as the pain source; prolonged relief indicates a therapeutic response.

Expected Results

Significant pain relief lasting weeks to months without complication.

Follow-up

Serial assessments; repeated blocks or referral to surgery may be considered based on response.

References

1. MedlinePlus. Nerve block. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 134

Newborn Hearing Screening

Overview

Newborn hearing screening is a quick, painless screening procedure performed soon after birth to identify babies who may have hearing loss.

How It Is Performed

While the baby is asleep or calm, the clinician places a small probe in the ear for an otoacoustic-emissions test or soft earphones and surface sensors for an automated auditory brainstem-response test. The equipment records the ear's or auditory pathway's response to sound.

Why It Is Done

Early identification allows confirmatory testing and timely support for speech, language, and learning. A screening result is not a diagnosis of permanent hearing loss.

Preparation and Results

No special preparation is needed. The baby should be comfortable and quiet. A result of "pass" means no further immediate screening is usually needed, while a "refer" result requires repeat screening or diagnostic evaluation. Fluid or debris in the ear and movement can affect the result.

Risks

The procedure is noninvasive and does not cause pain. The main limitation is that screening can occasionally miss hearing loss or produce a false-positive result.

References

1. National Institute on Deafness and Other Communication Disorders. Your baby's hearing and communicative development checklist. 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 135

Noninvasive Ventilation

Overview

Noninvasive ventilation (NIV) is the delivery of positive-pressure ventilatory support through a mask without an endotracheal tube. It is used to support patients with acute or chronic respiratory failure while preserving airway defenses and speech.

Alternative Names

NIV, bi-level positive airway pressure (BiPAP), continuous positive airway pressure (CPAP), mask ventilation, noninvasive positive pressure ventilation

Principle

A face or nasal mask delivers pressurized gas to the upper airway, providing inspiratory support and/or continuous positive end-expiratory pressure. This unloads the respiratory muscles, recruits collapsed alveoli, and splints the airway open, improving ventilation and oxygenation while avoiding the complications of intubation.

History

Continuous positive airway pressure for obstructive sleep apnea was introduced in 1981, and noninvasive ventilation for acute respiratory failure became established in the 1980s and 1990s. It is now a standard first-line therapy for acute hypercapnic respiratory failure and cardiogenic pulmonary edema.

Why It Is Done

Ordered for acute exacerbations of COPD with respiratory acidosis, cardiogenic pulmonary edema, and acute asthma or pneumonia when selected, as well as for hypoxemic respiratory failure and immunocompromised patients to avoid intubation. It is also used chronically for sleep apnea, obesity hypoventilation syndrome, and neuromuscular disease.

Is This a Primary or Secondary Test or Procedure

A primary treatment for selected acute and chronic respiratory failure, and an adjunct that may avoid or defer the need for invasive mechanical ventilation.

How It Is Performed

A well-fitted mask is secured over the nose or nose and mouth, and a ventilator or portable device delivers pressure with settings adjusted for inspiratory and expiratory positive airway pressure, fraction of inspired oxygen, and rate. The patient is coached on mask tolerance and synchronization, and response is monitored with pulse oximetry, blood gases, and clinical assessment.

Preparation

Patient education, careful mask fitting and securing of the airway, and baseline blood gas and oximetry measurements. The patient should be awake, cooperative, and able to protect the airway.

Contraindications and Precautions

Contraindications include respiratory or cardiac arrest, inability to protect the airway, copious secretions, vomiting, severe facial trauma or surgery, and hemodynamic instability. Precautions include gastric insufflation with aspiration risk, pressure-related skin injury, and the need for early conversion to invasive ventilation if NIV fails.

Risks

Facial pressure sores, skin breakdown, eye irritation, gastric distention and aspiration, air leak, and failure of therapy with respiratory deterioration.

Aftercare and Recovery

The patient is monitored closely, and therapy is titrated based on blood gases and respiratory effort, with early reassessment within one to two hours. NIV is weaned gradually as the patient improves and discontinued when stable on supplemental oxygen.

Outcome and Follow-up

Improvement in pH, PaCO2, oxygenation, and respiratory rate indicates a successful response, whereas worsening acidosis, inability to tolerate the mask, or hemodynamic decline signals NIV failure requiring intubation.

Expected Results

Restoration of near-normal pH, ventilation, and oxygenation with comfortable breathing and a reduced work of breathing.

Follow-up

Repeat blood gases and oximetry during titration, and a definitive plan for home NIV if chronic indications are present. Patients who fail NIV proceed to invasive mechanical ventilation and further evaluation of the cause of respiratory failure.

References

1. MedlinePlus. Noninvasive ventilation. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 136

Normal Labor and Vaginal Delivery

Overview

Normal labor and vaginal delivery is the process by which uterine contractions open the cervix and the baby is delivered through the vagina, followed by delivery of the placenta.

How Labor Progresses

Labor is commonly described in three stages: cervical dilation and effacement, birth of the baby, and delivery of the placenta. The mother and baby are monitored throughout labor, and pain relief may include breathing techniques, medicines, or neuraxial anesthesia.

Preparation and Care

The care team reviews the pregnancy history, checks the baby's position and heart rate, and monitors maternal vital signs and contractions. A birth plan may address support persons, mobility, pain relief, and newborn care, while remaining flexible for safety.

Risks and Complications

Possible complications include prolonged labor, heavy bleeding, infection, tears of the birth canal, shoulder dystocia, and changes in the baby's heart rate. Forceps, vacuum assistance, or cesarean delivery may be needed if spontaneous delivery is not safe.

Recovery

After delivery, the mother is monitored for bleeding, blood pressure, pain, and urination. The newborn receives an examination and routine preventive care. Recovery and follow-up depend on the delivery and any complications.

References

1. MedlinePlus. Labor and delivery. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 137

Occupational Therapy

Overview

Occupational therapy is a rehabilitation service that helps a person perform everyday activities safely and independently after illness, injury, disability, or developmental difficulty.

Why It Is Done

It may help with arthritis, stroke, neurological conditions, hand injuries, chronic pain, developmental conditions, or recovery after hospitalization. The therapist focuses on activities meaningful to the person's home, school, work, and community life.

How It Is Performed

The therapist evaluates movement, strength, coordination, cognition, sensation, and daily tasks. Treatment may include guided exercises, practice of self-care, energy-conservation strategies, adaptive equipment, splinting, and changes to the home or work environment.

Preparation and Safety

Wear comfortable clothing and bring information about symptoms, medicines, and activities that are difficult. Exercises should be progressed gradually. Tell the therapist about pain, dizziness, new weakness, or worsening symptoms.

Results and Follow-up

Progress is measured by improvements in function, safety, range of motion, strength, and ability to complete daily activities. The therapist may provide a home program and coordinate care with physicians, physical therapists, speech therapists, and caregivers.

References

1. MedlinePlus. Occupational therapy. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 138

Operative Vaginal Delivery

Overview

Operative vaginal delivery uses forceps or a vacuum extractor to assist delivery of a fetus through the vagina when spontaneous delivery cannot be achieved safely.

Alternative Names

Forceps delivery; vacuum-assisted delivery; instrumental delivery; ventouse delivery

Principle

Forceps cradle the fetal head with two curved blades; a vacuum extractor applies suction to the scalp. Traction during contractions assists descent of the head, shortening the second stage when indicated.

History

Obstetric forceps were developed in the 17th century and held as family secrets by the Chamberlen family. The vacuum extractor was developed in the mid-20th century as an alternative.

Why It Is Done

Performed for prolonged second stage, fetal distress, maternal exhaustion, or the need to shorten pushing for medical conditions such as cardiac or hypertensive disease.

Is This a Primary or Secondary Test or Procedure

Secondary obstetric intervention when spontaneous delivery is not progressing safely.

How It Is Performed

Performed after full cervical dilation with ruptured membranes and an engaged head. Regional anesthesia is preferred. Forceps blades are applied to the fetal head and traction is applied with contractions; a vacuum cup is placed on the occiput and gentle traction applied. Delivery is then completed.

Preparation

Assessment of fetal position, station, and presentation; bladder emptying; consent; and readiness for emergency cesarean if delivery fails.

Contraindications and Precautions

Non-engaged head, unknown fetal position, face presentation, fetal macrosomia with a small pelvis, and fetal bone disorders are contraindications. Precaution: failed instrumentation mandates cesarean delivery.

Risks

Maternal perineal or vaginal trauma, third- or fourth-degree lacerations, hemorrhage, and fetal scalp injury, cephalohematoma, or rarely intracranial hemorrhage.

Aftercare and Recovery

The perineum is examined and repaired, and the mother and baby are monitored. Pain and perineal care follow standard postpartum care.

Outcome and Follow-up

Successful vaginal delivery with minimal trauma confirms a good outcome. Neonatal scalp findings typically resolve.

Expected Results

A safe delivery of a healthy infant with controlled maternal trauma and a favorable neonatal course.

Follow-up

Postpartum perineal care, newborn examination, and follow-up for any maternal or neonatal complications.

References

1. MedlinePlus. Forceps delivery. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 139

Oxygen Therapy

Overview

Oxygen therapy provides supplemental oxygen to the lungs to treat hypoxemia (low blood oxygen levels). It increases alveolar oxygen tension and improves tissue oxygenation.

Alternative Names

Supplemental oxygen, O2 therapy

Principle

Supplemental oxygen is delivered by mask, nasal cannula, or other devices at concentrations above ambient air, increasing the fraction of inspired oxygen. This raises arterial oxygen saturation and improves oxygen delivery to tissues when normal breathing is insufficient.

History

Joseph Priestley discovered oxygen in 1774. Alvan Barach developed modern oxygen tents, masks, and nasal cannulas in the early 20th century.

Why It Is Done

Ordered for acute respiratory failure, COPD exacerbation, pneumonia, heart failure, sleep apnea, or carbon monoxide poisoning.

Is This a Primary or Secondary Test or Procedure

Primary supportive therapy for patients with acute or chronic hypoxemic respiratory failure.

How It Is Performed

Oxygen is delivered via a nasal cannula (prongs in the nose), a face mask, or high-flow nasal therapy. The flow rate is adjusted based on pulse oximetry.

Preparation

No preparation. Arterial blood gas (ABG) may be performed to establish baseline oxygenation.

Contraindications and Precautions

No absolute contraindications in hypoxemic patients. Use with caution in patients with hypercapnic respiratory failure (COPD) to avoid suppressing respiratory drive.

Risks

Oxygen toxicity, carbon dioxide retention, fire hazard, or nasal dryness and irritation.

Aftercare and Recovery

Monitor oxygen saturation and work of breathing. Adjust flow rate to maintain target saturation (typically 92-96

Outcome and Follow-up

Monitored using pulse oximetry (SpO2) and arterial blood gases. Resolution of hypoxemia is the primary goal.

Expected Results

Maintenance of target arterial oxygen saturation (SpO2 92-96

Follow-up

Pulse oximetry checks, titration of oxygen flow, or pulmonary consult.

References

1. MedlinePlus. Oxygen Therapy. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 140

Paracentesis

Overview

Paracentesis is a procedure that removes fluid from the abdominal cavity through a needle or catheter, performed for diagnosis and treatment of ascites.

Alternative Names

Abdominal paracentesis; abdominal tap; ascites drainage; therapeutic paracentesis

Principle

Fluid accumulates in the peritoneal cavity in cirrhosis, malignancy, infection, and heart failure. A needle or catheter is inserted through the abdominal wall to sample or drain this fluid for analysis or symptom relief.

History

Paracentesis has been performed since antiquity, and its role in diagnosing spontaneous bacterial peritonitis and relieving massive ascites was established in the 20th century.

Why It Is Done

Performed for new or worsening ascites to determine its cause, to evaluate suspected spontaneous bacterial peritonitis, and to relieve respiratory or abdominal discomfort from large-volume ascites.

Is This a Primary or Secondary Test or Procedure

Primary diagnostic and therapeutic procedure for ascites.

How It Is Performed

Performed at the bedside. The patient lies supine with the head elevated, the puncture site is identified (usually the left or right lower quadrant, avoiding surgical scars and the epigastric vessels), the skin is prepped and anesthetized, and a needle or catheter is advanced through the abdominal wall. Fluid is aspirated or drained into collection bags.

Preparation

Coagulation assessment and platelet count, bladder emptying, and ultrasound may be used to localize fluid in difficult cases.

Contraindications and Precautions

Relative contraindications include severe coagulopathy, bowel obstruction, and overlying skin infection. Precaution: avoid the inferior epigastric artery and surgical scars where bowel may be adherent.

Risks

Bleeding, infection, bowel perforation, and post-procedure hypotension with large-volume drainage; albumin is often given to prevent circulatory dysfunction.

Aftercare and Recovery

The patient is observed briefly and monitored for bleeding or hypotension. Fluid is sent for cell count, culture, albumin, and cytology as indicated.

Outcome and Follow-up

The serum-ascites albumin gradient (SAAG) distinguishes portal hypertension from other causes; an elevated neutrophil count indicates spontaneous bacterial peritonitis.

Expected Results

Clear fluid with a low cell count and no organisms, and relief of the symptoms of ascites.

Follow-up

Repeat paracentesis for recurrent symptoms, serial evaluation of the underlying liver or malignant disease, and monitoring for infection.

References

1. MedlinePlus. Paracentesis. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 141

Pelvic Examination

Overview

A pelvic examination assesses the external and internal reproductive organs, including the vulva, vagina, cervix, uterus, and adnexa. It may be part of routine care or an evaluation of symptoms.

Why It Is Done

It can help evaluate pelvic pain, abnormal bleeding, discharge, lesions, urinary or sexual symptoms, pregnancy-related concerns, and changes identified during screening. Samples for cervical screening or infection testing may be collected when appropriate.

How It Is Performed

After discussion and consent, the patient lies on an examination table. The clinician inspects the external area, may use a speculum to view the vaginal walls and cervix, and may perform a gentle bimanual examination to assess the uterus and adnexa. The patient may ask to stop at any time.

Preparation and Risks

No special preparation is usually needed. Emptying the bladder may improve comfort. Pressure or brief discomfort can occur, but severe pain should be reported. Minor spotting may occur after sampling.

Outcome and Follow-up

Findings may be normal or may suggest infection, inflammation, structural changes, or a need for additional testing. A pelvic examination does not replace cervical cancer screening or other tests when those are indicated.

References

1. MedlinePlus. Pelvic examination. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 142

Percutaneous Endoscopic Gastrostomy

Overview

Percutaneous endoscopic gastrostomy (PEG) is a procedure that places a feeding tube directly into the stomach through the abdominal wall using endoscopic guidance, for long-term enteral nutrition.

Alternative Names

PEG tube placement; gastrostomy tube; feeding tube; percutaneous gastrostomy

Principle

An endoscope is passed into the stomach, and the stomach wall is pushed against the abdominal wall. A needle is advanced percutaneously into the stomach, a guidewire is introduced and pulled through the mouth, and a feeding tube is drawn back through the abdominal wall into the stomach.

History

Percutaneous endoscopic gastrostomy was introduced by Gauderer, Ponsky, and Izant in 1980 and rapidly replaced surgical gastrostomy for most patients needing long-term tube feeding.

Why It Is Done

Performed for patients who cannot maintain adequate oral nutrition, such as those with dysphagia from stroke, neurologic disease, head and neck cancer, or severe malnutrition.

Is This a Primary or Secondary Test or Procedure

Primary long-term enteral access procedure.

How It Is Performed

Performed under conscious sedation or anesthesia. An endoscope confirms position and transillumination guides the puncture site. After prepping, the stomach is entered percutaneously, the tube is placed, and its position is confirmed endoscopically. The tube is secured at the skin.

Preparation

NPO before the procedure, IV antibiotics, and assessment of coagulation. The patient's head is elevated, and the puncture site is chosen below the xiphoid.

Contraindications and Precautions

Total gastrectomy, severe coagulopathy, and severe ascites are contraindications. Precaution: prior abdominal surgery with interposed bowel may preclude safe placement.

Risks

Bleeding, infection at the site, peritonitis, perforation, tube migration or blockage, aspiration, and buried bumper syndrome.

Aftercare and Recovery

Feeds begin within hours, and the site is kept clean and dry. The tract matures over 2-4 weeks, after which the tube can be replaced if needed.

Outcome and Follow-up

Successful tube placement is confirmed endoscopically; feeds are tolerated and weight stabilizes or improves.

Expected Results

A functioning tube delivering nutrition with a healed, non-infected site.

Follow-up

Ongoing nutrition monitoring, tube care, and replacement as the tube wears or if the patient regains oral intake.

References

1. MedlinePlus. PEG tube. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 143

Percutaneous Lung Biopsy

Overview

Percutaneous lung biopsy is a procedure that uses a needle through the chest wall to sample a lung nodule or mass for diagnosis.

Alternative Names

Transthoracic needle biopsy; CT-guided lung biopsy; lung needle aspiration

Principle

Under CT or ultrasound guidance, a needle is advanced into the lesion and tissue or cells are aspirated for cytology and histology.

History

Transthoracic needle biopsy developed in the mid-20th century and became a standard, image-guided method for diagnosing peripheral lung lesions.

Why It Is Done

Performed for peripheral lung nodules or masses when bronchoscopy cannot reach them and malignancy is suspected.

Is This a Primary or Secondary Test or Procedure

Primary diagnostic test for selected peripheral lung lesions.

How It Is Performed

Performed under local anesthesia with CT guidance. A needle is advanced into the lesion, samples are taken, and the patient is positioned to minimize pneumothorax risk.

Preparation

Imaging review, and holding anticoagulants.

Contraindications and Precautions

Severe emphysema, bleeding risk, and inability to tolerate pneumothorax are relative contraindications.

Risks

Pneumothorax, bleeding, hemoptysis, and infection.

Aftercare and Recovery

A chest X-ray is often taken to check for pneumothorax.

Outcome and Follow-up

Cytology and histology determine benign versus malignant disease.

Expected Results

Benign tissue with no evidence of malignancy.

Follow-up

Repeat imaging or surgical biopsy if sampling is nondiagnostic.

References

1. MedlinePlus. Lung biopsy. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 144

Percutaneous Nephrolithotomy

Overview

Percutaneous nephrolithotomy (PCNL) is a procedure that removes large or complex kidney stones through a small incision and tract made directly into the kidney.

Alternative Names

PCNL; percutaneous stone removal; nephrolithotripsy

Principle

A tract is created from the skin into the renal collecting system, a nephroscope visualizes the stone, and ultrasound, pneumatic, or laser energy fragments it so fragments can be extracted.

History

PCNL was introduced in the 1970s as an alternative to open surgery for large kidney stones and remains the standard for stones larger than 2 cm.

Why It Is Done

Performed for kidney stones larger than 2 cm, staghorn calculi, stones that fail shock wave lithotripsy, and stones in abnormal kidneys.

Is This a Primary or Secondary Test or Procedure

Primary treatment for large and complex renal stones.

How It Is Performed

Performed under general anesthesia. A needle and guidewire create the tract under fluoroscopy or ultrasound, it is dilated, the nephroscope enters the kidney, and the stone is fragmented and removed. A nephrostomy tube or stent may be placed.

Preparation

Preoperative imaging, urine culture and treatment of infection, and anticoagulation management.

Contraindications and Precautions

Active untreated urinary infection, bleeding diathesis, and pregnancy are contraindications.

Risks

Bleeding, perforation of the collecting system, infection or sepsis, injury to adjacent organs, and residual stone fragments.

Aftercare and Recovery

The nephrostomy tube is removed after drainage clears; imaging confirms residual stones.

Outcome and Follow-up

Postoperative imaging and stone analysis guide further therapy and metabolic evaluation.

Expected Results

Complete stone clearance with no significant residual fragments.

Follow-up

Metabolic stone evaluation and imaging surveillance; additional procedures may be needed for residual stones.

References

1. MedlinePlus. Percutaneous nephrolithotomy. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 145

Percutaneous Tumor Ablation

Overview

Percutaneous tumor ablation is a minimally invasive technique in which image guidance is used to place a needle or probe directly into a tumor to destroy it by extreme cold (cryoablation) or heat (radiofrequency or microwave ablation). It is performed through the skin without open surgery, typically as an outpatient or short-stay procedure.

Alternative Names

Cryoablation, cryotherapy, radiofrequency ablation (RFA), microwave ablation (MWA), thermal ablation, tumor ablation, image-guided tumor ablation

Principle

All three modalities rely on a probe positioned within the tumor under computed tomography, ultrasound, or magnetic resonance imaging guidance. Cryoablation destroys cells by rapid freeze–thaw cycles that cause ice-crystal formation, osmotic injury, and microvascular thrombosis. Radiofrequency and microwave ablation generate frictional and molecular heat, respectively, raising tissue temperature above 50–60°C to produce coagulative necrosis. The goal is complete necrosis of the target lesion while sparing adjacent structures.

History

Percutaneous ablation was developed in the 1990s and expanded rapidly as image-guidance technology improved. Cryoablation predates heat-based methods, with modern imaging-compatible systems now allowing real-time monitoring of the ice ball.

Why It Is Done

Percutaneous ablation is used to treat primary and metastatic tumors of the liver, kidney, lung, bone, and adrenal glands when surgery is not feasible because of patient comorbidities, tumor number or location, or poor surgical candidacy. It is also used for pain palliation of osseous metastases and to control tumor burden in patients awaiting transplant or other systemic therapy. In selected cases it offers local control comparable to surgery with less morbidity and shorter recovery.

Is This a Primary or Secondary Test or Procedure

It is a primary therapeutic procedure for appropriately selected small tumors. It is a secondary or alternative option when resection, transplant, or stereotactic radiation is contraindicated or declined.

How It Is Performed

The patient is positioned, and the target tumor is localized with CT, ultrasound, or MRI. After sterile preparation and local or general anesthesia, one or more ablation probes are inserted percutaneously into the lesion. Energy is delivered in cycles to create a margin of necrosis around the tumor, and for cryoablation the ice ball is monitored by real-time imaging. Multiple ablations may be performed during the same session to cover larger or irregular tumors.

Preparation

Preprocedural imaging and laboratory evaluation, including coagulation studies, are required, and anticoagulant or antiplatelet therapy is typically held before the procedure. Patients fast for several hours and are instructed about anesthesia and postprocedural observation. Written informed consent is obtained after a discussion of risks, benefits, and alternatives.

Contraindications and Precautions

Uncorrectable coagulopathy, active infection, and inability to safely reach the tumor are relative contraindications. Lesions adjacent to major vessels, bile ducts, bowel, or other critical structures carry an increased risk of collateral injury. Tumor size, location, and number must be assessed because larger or exophytic tumors have higher incomplete-ablation rates.

Risks

Risks include bleeding, infection, pneumothorax or pleural effusion for lung lesions, bile duct or bowel injury, thermal injury to adjacent organs, and tumor seeding along the needle track. Pain and a postablation syndrome of fever, malaise, and nausea may occur transiently.

Aftercare and Recovery

Patients are monitored in a recovery area for several hours, and imaging is performed to assess for complications such as bleeding or pneumothorax. Most patients return home the same day or after a short overnight stay and resume normal activity within a few days. Contrast-enhanced imaging at 1–3 months is used to document complete necrosis.

Outcome and Follow-up

A successful ablation is indicated by an avascular zone larger than the original tumor on follow-up imaging, with absence of enhancement. Persistent or rim-like enhancement at the ablation margin suggests residual viable tumor and may require repeat treatment or an alternative therapy.

Expected Results

There is no normal result because ablation is a therapeutic intervention; the expected outcome is complete necrosis of the treated lesion. Success is defined by absence of residual enhancement at the site on follow-up imaging.

Follow-up

Regular surveillance imaging (CT, MRI, or PET/CT) is performed at 1–3 months and then at intervals to detect local recurrence or new disease. Patients with metastatic disease usually continue systemic therapy under the direction of their oncology team.

References

1. MedlinePlus. Tumor Ablation. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 146

Pericardiocentesis

Overview

Pericardiocentesis is a procedure that uses a needle and catheter to remove fluid from the pericardial sac around the heart, either for diagnosis or to relieve cardiac tamponade.

Alternative Names

Pericardial tap; pericardial fluid drainage

Principle

A needle is advanced into the pericardial space, usually with echocardiographic guidance, to aspirate fluid. Drainage reduces intrapericardial pressure and restores cardiac filling in tamponade.

History

Pericardiocentesis was described in the 1840s and refined with the introduction of echocardiographic guidance and catheter drainage in the late 20th century.

Why It Is Done

Performed urgently in cardiac tamponade, and for diagnostic evaluation of pericardial effusion of uncertain cause, including suspected infection, malignancy, or inflammation.

Is This a Primary or Secondary Test or Procedure

Primary emergency procedure for tamponade and a diagnostic tool for pericardial effusion.

How It Is Performed

Performed with echocardiographic or fluoroscopic guidance, typically via a subxiphoid or apical approach. The needle enters the pericardial space, fluid is aspirated, and a drain may be left in place.

Preparation

Echocardiography to confirm the effusion and guide the approach; emergency equipment is prepared for complications.

Contraindications and Precautions

Coagulopathy and a posterior effusion are relative contraindications in the elective setting. Precaution: tamponade is an emergency in which the procedure is performed despite other risks.

Risks

Cardiac puncture, pneumothorax, bleeding, arrhythmia, infection, and reaccumulation of fluid.

Aftercare and Recovery

The patient is monitored; a drain may remain for days. Fluid is sent for cell count, cytology, culture, and chemistry.

Outcome and Follow-up

Findings may indicate infection (purulent or culture-positive), malignancy (positive cytology), uremia, or inflammation. Rapid clinical improvement follows tamponade relief.

Expected Results

A small, physiologic amount of clear fluid with no evidence of infection, malignancy, or bleeding.

Follow-up

Repeat echocardiography; the underlying cause is treated.

References

1. MedlinePlus. Pericardiocentesis. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 147

Peripheral Angioplasty and Stenting

Overview

Peripheral angioplasty and stenting is a minimally invasive procedure that opens narrowed or blocked arteries of the legs using a balloon and, when needed, a stent, to treat peripheral artery disease.

Alternative Names

Lower-extremity angioplasty and stenting; peripheral vascular intervention; angioplasty of the leg arteries

Principle

A balloon catheter is advanced over a guidewire through the arterial occlusion and inflated to dilate the vessel; a stent is placed to scaffold it open. This restores blood flow and relieves claudication or critical limb ischemia.

History

Percutaneous transluminal angioplasty was developed by Andreas Gruentzig in the 1970s and applied to peripheral arteries soon after, with stents added in the 1990s.

Why It Is Done

Performed for symptomatic peripheral artery disease (claudication, rest pain, or tissue loss) that is not controlled by medical therapy, and for acute limb ischemia from thrombosis or embolism.

Is This a Primary or Secondary Test or Procedure

Primary or secondary revascularization for peripheral artery disease.

How It Is Performed

Performed in an angiography suite under local anesthesia. Access is obtained (usually the femoral or popliteal artery), angiography maps the lesion, and a guidewire crosses it. The balloon is inflated and a stent is deployed as needed. Completion angiography confirms flow.

Preparation

Preoperative imaging with duplex ultrasound, CT, or MR angiography; antiplatelet therapy; and assessment of renal function and contrast allergy.

Contraindications and Precautions

Active infection, severe contrast allergy, and unreconstructable disease are relative contraindications. Precaution: calcified or long occlusions carry higher risk.

Risks

Vessel injury or rupture, dissection, distal embolization, thrombosis, bleeding or pseudoaneurysm at the access site, and contrast nephropathy.

Aftercare and Recovery

Observation, antiplatelet therapy, and access-site care. Most patients go home within a day.

Outcome and Follow-up

Post-procedure imaging confirms a patent vessel with improved flow; clinical improvement in walking or rest pain confirms benefit.

Expected Results

A patent, non-stenotic artery with restored distal flow and relief of symptoms.

Follow-up

Duplex surveillance, risk-factor management, and repeat intervention for restenosis or progression.

References

1. MedlinePlus. Peripheral artery angioplasty and stenting. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 148

Peripherally Inserted Central Catheter

Overview

Peripherally inserted central catheter (PICC) line placement inserts a long catheter through a vein in the arm to reach the large central veins near the heart for medication, nutrition, or blood draws.

Alternative Names

PICC line; peripherally inserted central catheter; PICC placement

Principle

A flexible catheter is inserted into a peripheral vein of the upper arm and advanced so its tip rests in the superior vena cava. The tip position provides reliable access for irritant or long-term therapies.

History

PICC lines were developed in the 1970s to provide central venous access from the arm, reducing the risks of chest placement and repeated needle sticks.

Why It Is Done

Performed for patients needing several weeks of IV therapy such as antibiotics, chemotherapy, or total parenteral nutrition, and when peripheral access is difficult.

Is This a Primary or Secondary Test or Procedure

Primary vascular access for intermediate-duration central venous therapy.

How It Is Performed

Performed at the bedside or in a procedure room with ultrasound guidance. The vein is accessed in the upper arm, the catheter is threaded to the cavoatrial junction, position is confirmed (often by ECG guidance or radiograph), and the line is secured and dressed.

Preparation

Review of bleeding risk and arm history (mastectomy, pacemaker, prior surgery), and ultrasound assessment of the veins.

Contraindications and Precautions

Local infection, severe coagulopathy, and unsuitable arm veins are relative contraindications. Precaution: use the non-dominant arm and avoid the side of a prior mastectomy.

Risks

Bleeding, infection, thrombosis, arrhythmia from tip migration, catheter malposition or breakage, and nerve injury.

Aftercare and Recovery

The line is flushed, and the site is cared for with sterile dressing changes. Weekly flushing maintains patency when not in use.

Outcome and Follow-up

A confirmatory radiograph shows the tip at the cavoatrial junction. Clinical use and site checks confirm function.

Expected Results

A patent, well-positioned PICC line with no infection, thrombosis, or mechanical complication.

Follow-up

Routine site inspection and flushing; the line is removed when therapy is complete.

References

1. MedlinePlus. Peripherally inserted central catheter (PICC) line. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 149

Peritoneal Dialysis Catheter Placement

Overview

Peritoneal dialysis catheter placement inserts a soft tube into the abdomen that allows peritoneal dialysis to filter waste using the lining of the abdominal cavity.

Alternative Names

PD catheter insertion; Tenckhoff catheter placement; peritoneal access

Principle

A catheter is placed through the abdominal wall into the peritoneal cavity. Dialysate fills and drains through it, and the peritoneum acts as the semipermeable membrane for solute and fluid exchange.

History

Peritoneal dialysis was established with the Tenckhoff catheter in the 1960s and has grown into a home-based alternative to hemodialysis.

Why It Is Done

Performed to establish access for patients who choose peritoneal dialysis, prefer home therapy, or lack good vascular access for hemodialysis.

Is This a Primary or Secondary Test or Procedure

Primary procedure for peritoneal dialysis access.

How It Is Performed

Performed surgically or percutaneously under anesthesia. The catheter is placed into the pelvic region of the peritoneal cavity, tunneled through the subcutaneous tissue, and exits at a convenient site.

Preparation

Bowel preparation, bladder emptying, and prophylaxis against infection.

Contraindications and Precautions

Extensive abdominal adhesions, hernias, and conditions limiting the peritoneal surface are relative contraindications.

Risks

Infection (peritonitis, exit-site infection), catheter migration or obstruction, leakage, and bleeding.

Aftercare and Recovery

The catheter is flushed regularly; dialysis typically starts after a break-in period to allow healing.

Outcome and Follow-up

Function is judged by dialysate flow, drainage volumes, and absence of leakage or infection.

Expected Results

Free inflow and outflow of dialysate with no infection.

Follow-up

Ongoing exit-site care and monitoring; repositioning or replacement for dysfunction.

References

1. MedlinePlus. Peritoneal dialysis. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 150

Phlebotomy Considerations in Children

Overview

Phlebotomy in children is the procedure of collecting blood samples from pediatric patients using age-appropriate techniques that minimize pain, distress, and blood volume loss.

Alternative Names

Pediatric phlebotomy, children's blood draw, child venipuncture, pediatric blood collection, infant blood sampling

Principle

Pediatric phlebotomy uses age-appropriate techniques: heel stick for neonates (warmed heel, lancet depth <2 mm), scalp vein or jugular vein for infants, and antecubital venipuncture for older children. Maximum blood draw volume is calculated as 1-3

History

Pediatric phlebotomy techniques were developed in the mid-20th century as awareness grew that children require different approaches than adults. The use of topical anesthetics and smaller-gauge needles became standard.

Why It Is Done

This section addresses the practical challenges of collecting blood from children of different ages. Understanding the best approach for each age group helps your doctor or phlebotomist obtain accurate results while minimizing pain and distress for you and your child. It answers the question: what is the safest and least stressful way to collect blood from a child of this age?

Is This a Primary or Secondary Test or Procedure

Informational guideline.

How It Is Performed

Age Group	Preferred Collection Sites
Newborns (0–1 month)	Heel prick, jugular vein, femoral vein, PICC line
Infants (1–12 months)	Jugular vein, femoral vein, saphenous vein
Toddlers (1–3 years)	Antecubital fossa (with distraction techniques)
Children (3+ years)	Antecubital fossa preferred

Key principles for pediatric blood collection:

- Minimize the volume of blood drawn — use pediatric tubes
- Use distraction techniques (toys, videos, storytelling) for young children
- Consider point-of-care testing when available to reduce the need for venous draws
- Use non-invasive alternatives when possible (transcutaneous bilirubin instead of blood bilirubin, pulse oximetry instead of arterial blood gas)
- Comfort and reassure your child throughout the process

Preparation

Psychological preparation of child and parents; application of topical anesthetic (EMLA cream) 1 hour prior.

Contraindications and Precautions

Avoid crying or hyperventilation before collection to prevent transient shifts in blood gas parameters.

Risks

Bruising, minor bleeding, or psychological distress/needle phobia.

Aftercare and Recovery

Hold pressure for 3-5 minutes; reward the child for cooperation; monitor for hematoma.

Outcome and Follow-up

Results should be interpreted using age-specific pediatric reference ranges.

Expected Results

Successful blood collection with minimal hemolysis or trauma.

Follow-up

Monitoring serial labs if necessary.

References

1. MedlinePlus. Pediatric phlebotomy. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 151

Photorefractive Keratectomy (PRK)

Overview

PRK is a laser vision-correction procedure that reshapes the cornea without creating a corneal flap.

Why It Is Done

It is an elective alternative to LASIK for suitable patients, including some with thin corneas or occupations where flap injury is a concern.

How It Is Performed

The surface corneal epithelium is removed, an excimer laser reshapes the cornea, and a bandage contact lens is placed while the surface heals.

Risks and Recovery

Pain during early healing, haze, infection, dry eye, glare, residual refractive error, and regression are possible. Visual recovery is slower than after LASIK and requires prescribed drops and follow-up.

References

1. National Eye Institute. Refractive Surgery. 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 152

Phototherapy

Overview

Phototherapy is a treatment that exposes the skin to specific wavelengths of ultraviolet or visible light to treat skin disease, hyperbilirubinemia, or certain systemic disorders. It is delivered as narrowband or broadband UVB, UVA with photosensitizing agents (PUVA), or blue light for neonatal jaundice.

Alternative Names

Light therapy; UV light therapy; ultraviolet phototherapy; PUVA; blue-light therapy; heliotherapy

Principle

Light of specific wavelengths is absorbed by target molecules in the skin and produces therapeutic effects. In psoriasis and eczema, UVB and PUVA suppress epidermal hyperproliferation and modulate local immune responses. In neonatal jaundice, blue light within the 400 to 490 nm range converts unconjugated bilirubin to water-soluble photo-isomers that are excreted without conjugation.

History

Phototherapy dates to ancient sunbathing traditions, and Niels Finsen won the Nobel Prize in 1903 for using light to treat tuberculosis of the skin. The use of blue light for neonatal jaundice was introduced in the 1950s.

Why It Is Done

Ordered for moderate to severe psoriasis, vitiligo, atopic dermatitis, cutaneous T-cell lymphoma (mycosis fungoides), and other photosensitive skin disorders when topical therapy is inadequate. Blue-light phototherapy is ordered for significant neonatal unconjugated hyperbilirubinemia to prevent kernicterus.

Is This a Primary or Secondary Test or Procedure

A primary treatment for neonatal hyperbilirubinemia and a second-line disease-modifying therapy for many dermatologic conditions.

How It Is Performed

The patient stands in a cabinet lined with fluorescent tubes or LED arrays that emit the chosen wavelength while eyes are protected, with exposure times and doses titrated to skin type and response. PUVA requires taking a psoralen drug before exposure to sensitize the skin to UVA. In neonates, the infant is undressed, eyes are covered, and lights are positioned above the incubator or bassinet.

Preparation

Skin type and minimal erythema dose are assessed, and photosensitizing drugs and natural sunlight are avoided where relevant. For PUVA, the psoralen is taken at a set time before exposure, and eye protection with UV-blocking glasses is used on treatment days. Bilirubin level is measured before starting infant phototherapy.

Contraindications and Precautions

Conditions with photosensitivity, such as systemic lupus erythematosus, porphyria, and xeroderma pigmentosum, are contraindications, as is a history of melanoma or significant prior phototoxic exposure. Eye and gonadal protection is used, and careful dosimetry reduces the risk of burning; PUVA requires eye protection for the entire day of treatment.

Risks

Short-term effects include erythema, pruritus, and photoaging of the skin. Long-term UVB and PUVA therapy increases the risk of skin cancer, and PUVA is associated with premature photoaging; infant phototherapy may cause loose stools, rash, and hyperthermia.

Aftercare and Recovery

The dermatologic patient uses emollients, avoids sun exposure, and attends scheduled sessions, with doses adjusted to erythema response. In neonates, bilirubin levels are rechecked within hours and phototherapy is continued or intensified until levels fall below treatment thresholds, at which point it is stopped.

Outcome and Follow-up

For skin disease, response is judged by reduction in lesions and symptom control. For neonatal jaundice, a falling bilirubin toward the age-specific treatment threshold indicates effective therapy.

Expected Results

Clearance or significant improvement of skin lesions with dermatologic phototherapy, and bilirubin reduced to a safe level below the phototherapy threshold in the neonate.

Follow-up

Periodic skin examinations are recommended for patients on long-term phototherapy because of skin cancer risk, and bilirubin monitoring continues until safe levels are sustained. Patients who do not respond may be offered phototherapy dose escalation, PUVA, or systemic therapy.

References

1. MedlinePlus. Phototherapy. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 153

Physical Therapy

Overview

Physical therapy is a rehabilitation service that improves movement, strength, balance, endurance, and function after injury, illness, surgery, or disability.

Why It Is Done

It is commonly used for arthritis, back and neck pain, fractures, sports injuries, stroke, balance problems, and recovery after illness or surgery. Treatment is tailored to the person's goals and physical limitations.

How It Is Performed

The therapist evaluates movement, strength, flexibility, balance, and functional tasks. Treatment may include therapeutic exercise, stretching, manual therapy, gait training, balance training, education, and selected modalities such as heat or cold.

Preparation and Safety

Wear comfortable clothing and bring information about symptoms, medicines, and activities that are difficult. Exercises should be performed as instructed and progressed gradually. Report chest pain, severe shortness of breath, new weakness, or worsening pain.

Results and Follow-up

Progress is measured by improvements in mobility, strength, pain, balance, and daily function. The therapist may prescribe a home exercise program and coordinate care with other clinicians.

References

1. MedlinePlus. Physical therapy. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 154

Plasmapheresis

Overview

Plasmapheresis, or therapeutic plasma exchange, is a procedure that removes plasma containing pathogenic antibodies, immune complexes, and toxins from the blood and replaces it with donor plasma or albumin. The cellular components of blood are returned to the patient.

Alternative Names

Therapeutic plasma exchange; TPE; plasma exchange; apheresis

Principle

Blood is withdrawn through a venous catheter and passed through a centrifuge or membrane filter that separates plasma from the cellular elements. The plasma is discarded and replaced with colloid solution, and the cellular components are returned to the circulation, selectively removing disease-causing substances from the blood.

History

Therapeutic apheresis was pioneered in the 1960s and 1970s with the development of continuous-flow centrifugal cell separators, and it became a standard treatment for antibody-mediated and immune-mediated disorders.

Why It Is Done

Ordered to treat conditions caused or worsened by pathogenic antibodies, immune complexes, or paraproteins, including Guillain-Barré syndrome, myasthenia gravis crisis, thrombotic thrombocytopenic purpura (TTP), antibody-mediated rejection of transplanted organs, antiglomerular basement membrane disease, and severe hyperviscosity syndromes.

Is This a Primary or Secondary Test or Procedure

Primary treatment for selected immune-mediated and antibody-mediated disorders; it is a therapeutic rather than a diagnostic procedure.

How It Is Performed

Vascular access is obtained through a large-bore central or peripheral catheter. Blood is drawn into an apheresis machine that separates plasma from cellular components by centrifugation or filtration. The plasma is replaced with albumin or fresh frozen plasma, and the cellular components are reinfused; treatment typically lasts one to two hours.

Preparation

Informed consent, review of the coagulation profile and platelet count, confirmation of adequate vascular access, and baseline laboratory studies are required. Anticoagulant (citrate) is used in the circuit, and the patient should be aware of the required number of sessions.

Contraindications and Precautions

Hemodynamic instability, uncontrolled sepsis, inability to obtain vascular access, and sensitivity to replacement fluids are contraindications. Precautions include hypocalcemia from citrate toxicity, hypotension, and use of fresh frozen plasma in patients with coagulation abnormalities or risk of bleeding.

Risks

Risks include citrate-induced hypocalcemia, hypotension, allergic reactions to plasma, bleeding, catheter-related infection or thrombosis, and rarely fluid overload.

Aftercare and Recovery

The patient is monitored for hypotension, hypocalcemia, and bleeding before discharge or return to the ward. Serial clinical and laboratory monitoring, such as platelet counts in TTP, determines the response and the need for additional sessions.

Outcome and Follow-up

Clinical response is judged by improvement in symptoms and organ function, with laboratory markers such as ADAMTS13 activity, platelet count, antibody titers, or paraprotein levels monitored. A partial or absent response may prompt additional sessions or alternative therapy.

Expected Results

Return of disease-specific laboratory markers toward normal (for example, rising platelet count in TTP, falling antibody titers) indicates an adequate therapeutic response.

Follow-up

Continue immunosuppressive or disease-specific therapy that addresses the underlying disorder, since plasmapheresis alone does not stop antibody production. Serial laboratory monitoring and repeat treatment courses may be required for relapsing disease.

References

1. MedlinePlus. Plasmapheresis. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 155

Platelet-Rich Plasma Injection

Overview

A platelet-rich plasma (PRP) injection is a procedure in which a concentrate of the patient's own platelets is injected into a tendon, ligament, joint, or other tissue to promote healing and reduce pain.

Alternative Names

PRP injection; autologous platelet concentrate injection; platelet-rich plasma therapy

Principle

Platelets release growth factors and signaling proteins that recruit healing cells and stimulate tissue repair. Concentrating the patient's platelets and delivering them to the injury site amplifies this natural healing response.

History

The regenerative potential of platelet concentrates was recognized in the 1970s and popularized in orthopedic medicine in the 2000s as an option for tendinopathy and osteoarthritis.

Why It Is Done

Ordered for chronic tendinopathy, such as lateral epicondylitis or Achilles tendinopathy, selected ligament and muscle injuries, and knee osteoarthritis, when symptoms persist despite conservative treatment.

Is This a Primary or Secondary Test or Procedure

A secondary therapeutic procedure, used after conservative measures have not fully succeeded and before or in place of more invasive treatments.

How It Is Performed

Blood is drawn from the patient and centrifuged to separate platelets from red and white blood cells. The platelet concentrate is then injected into the affected tissue, often with ultrasound guidance, either as a single treatment or a series.

Preparation

Informed consent and review of bleeding risk. Patients are advised to stop anti-inflammatory medications such as NSAIDs before the injection because they may interfere with the healing response.

Contraindications and Precautions

Contraindications include local infection, bleeding disorders, and active systemic infection. Precautions include pregnancy, cancer at the treatment site, and the use of medications that affect platelets.

Risks

Pain and soreness at the injection site, bruising, and rarely infection or nerve injury. There is limited evidence of benefit for some conditions, so expectations should be realistic.

Aftercare and Recovery

The patient rests the treated area briefly and gradually resumes activity. Several weeks may be required before improvement, and more than one injection is sometimes needed.

Outcome and Follow-up

Response is judged by reduction in pain and improvement in function over the weeks after injection. Lack of improvement may indicate the condition will not respond to this treatment.

Expected Results

Reduced pain and improved function with no significant adverse reaction at the injection site.

Follow-up

Patients are reassessed several weeks after the injection; repeat injections may be offered, and persistent symptoms may lead to other treatments such as surgery.

References

1. MedlinePlus. Platelet-rich plasma injection. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 156

Pleurodesis

Overview

Pleurodesis is a procedure that obliterates the pleural space by creating adhesions between the visceral and parietal pleura, preventing recurrent pneumothorax or pleural effusion.

Alternative Names

Chemical pleurodesis; talc pleurodesis; pleural adhesion induction

Principle

A sclerosant (most commonly sterile talc) is instilled into the pleural space, causing inflammation that fuses the two pleural layers together and eliminates the space in which air or fluid can accumulate.

History

Pleurodesis was introduced in the early 20th century and refined with the use of talc poudrage and slurry in the mid-20th century.

Why It Is Done

Performed for recurrent pneumothorax and for recurrent or malignant pleural effusion that recurs despite repeated drainage, in patients whose lungs can re-expand.

Is This a Primary or Secondary Test or Procedure

Secondary treatment for recurrent pleural disease after initial drainage fails.

How It Is Performed

Performed at the bedside or in the operating room. A chest tube drains the fluid or air, the lung is confirmed expanded, and a slurry of talc (or another agent) is instilled through the tube, which is clamped for a period while the patient changes position. In thoracoscopic (VATS) poudrage, talc is sprayed under vision.

Preparation

Chest imaging to confirm lung expansion and absence of a trapped lung, analgesia, and informed consent.

Contraindications and Precautions

An unexpanded (trapped) lung limits the surface contact and success of the procedure. Precaution: significant pain is expected and managed with analgesia.

Risks

Fever and chest pain, infection, empyema, re-expansion pulmonary edema, and the rare risk of acute respiratory distress syndrome with talc.

Aftercare and Recovery

The tube remains on suction until output is low, then is removed. Imaging confirms pleural apposition.

Outcome and Follow-up

Successful pleurodesis is confirmed by lung expansion and lack of re-accumulation of air or fluid after tube removal.

Expected Results

A fused pleural space with no recurrence of pneumothorax or effusion.

Follow-up

Serial chest imaging for recurrence, and oncologic follow-up for malignant effusions.

References

1. MedlinePlus. Pleurodesis. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 157

Prostate Biopsy

Overview

A prostate biopsy removes small tissue cores from the prostate gland using a spring-loaded needle, typically guided by transrectal ultrasound (TRUS) or MRI-fusion.

Alternative Names

TRUS biopsy, Transrectal ultrasound biopsy

Principle

An ultrasound probe in the rectum guides a spring-loaded biopsy needle into the prostate gland, obtaining multiple cores of tissue from different regions, particularly the peripheral zone, where most cancers arise. The cores are examined microscopically for cancer cells.

History

Early biopsies were performed blindly. Transrectal ultrasound guidance was introduced in the 1980s, standardizing the 12-core systematic biopsy.

Why It Is Done

Ordered when prostate cancer is suspected due to an elevated Prostate-Specific Antigen (PSA) level or an abnormal Digital Rectal Exam (DRE).

Is This a Primary or Secondary Test or Procedure

Primary diagnostic test to confirm and grade prostate cancer.

How It Is Performed

Under local anesthesia. An ultrasound probe is placed in the rectum to visualize the prostate. A biopsy needle is advanced through the rectal wall (transrectal) or perineum (transperineal) to take 12 tissue samples.

Preparation

Take prescribed prophylactic antibiotics. Perform an enema on the morning of the procedure. Stop blood thinners.

Contraindications and Precautions

Acute prostatitis, or uncorrected bleeding disorders.

Risks

Hematuria, hematochezia (blood in stool), hematospermia (blood in semen, common for weeks), urinary retention, or severe sepsis (due to rectal bacteria translocation).

Aftercare and Recovery

Complete the full course of antibiotics. Drink plenty of water. Avoid strenuous activity for 2-3 days.

Outcome and Follow-up

Biopsy cores are evaluated under a microscope. Cancer is graded using the Gleason Score (e.g., Gleason 3+3=6 to 5+5=10).

Expected Results

Benign prostatic tissue; no evidence of prostatic adenocarcinoma or high-grade PIN.

Follow-up

Pathology review; discussion of active surveillance, surgery, or radiation if cancer is found.

References

1. MedlinePlus. Prostate Biopsy. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 158

Pulmonary Artery (Swan-Ganz) Catheterization

Overview

Pulmonary artery catheterization is the placement of a balloon-tipped catheter through the right heart into the pulmonary artery to measure hemodynamic pressures and cardiac output. It provides detailed right and left heart hemodynamic information at the bedside.

Alternative Names

Swan-Ganz catheterization, right heart catheterization, pulmonary artery catheter, hemodynamic monitoring

Principle

The catheter, floated through the right atrium and ventricle with an inflated balloon, wedges in a distal pulmonary artery, allowing measurement of pulmonary capillary wedge pressure, a surrogate for left atrial and left ventricular end-diastolic pressure. Pressure waveforms, mixed venous oxygen saturation, and thermodilution cardiac output are used to characterize shock, guide fluid and vasopressor therapy, and assess ventricular function.

History

The flow-directed balloon catheter was introduced by Swan and Ganz in 1970 and revolutionized bedside hemodynamic monitoring in intensive care. Its use declined after trials questioned benefit in unselected patients, but it remains valuable in selected complex cases.

Why It Is Done

Ordered when bedside assessment of hemodynamics is insufficient, including undifferentiated shock, severe heart failure with unclear volume status, pulmonary hypertension, and high-risk cardiac surgery. It helps distinguish cardiogenic, distributive, and hypovolemic shock and guides titration of fluids, inotropes, and vasopressors.

Is This a Primary or Secondary Test or Procedure

A secondary, specialized diagnostic and monitoring procedure reserved for patients in whom standard clinical and noninvasive assessment cannot establish hemodynamics.

How It Is Performed

A central venous sheath is placed, and the catheter is advanced to the right atrium where the balloon is inflated and floated through the right ventricle into the pulmonary artery under continuous waveform and pressure monitoring. The balloon is deflated and the catheter left in place, connected to transducers and a cardiac output computer. Waveforms confirm sequential positioning in the right atrium, right ventricle, and pulmonary artery, then a wedge tracing on inflation.

Preparation

Informed consent, coagulation assessment, and continuous electrocardiographic and blood pressure monitoring. Sterile technique is used for central venous access, and the patient is positioned supine.

Contraindications and Precautions

Relative contraindications include severe coagulopathy, left bundle branch block (risk of complete heart block), recent cardiac surgery with tricuspid valve or right heart pathology, and known right-sided endocarditis. Precautions include avoiding prolonged balloon inflation, monitoring for arrhythmias and catheter coiling, and using ultrasound for venous access.

Risks

Arrhythmias, pulmonary artery rupture or infarction, pneumothorax and bleeding from venous access, infection, catheter knotting, and pulmonary thromboembolism.

Aftercare and Recovery

Hemodynamic parameters are recorded and used to guide therapy, with the catheter generally removed within 24–72 hours. The insertion site is observed for bleeding and infection, and monitoring is continued until the catheter is withdrawn.

Outcome and Follow-up

Cardiogenic shock shows low cardiac output with elevated wedge pressure and filling pressures, while distributive shock shows low or normal wedge pressure with high output. A low wedge pressure suggests hypovolemia warranting fluid resuscitation, whereas a high wedge pressure in a hypotensive patient supports diuresis or inotropic support.

Expected Results

Normal values include cardiac index of 2.5–4.0 $L/min/m^2$, pulmonary artery systolic pressure of 20–30 mm Hg, and mean pulmonary capillary wedge pressure of 6–12 mm Hg.

Follow-up

Repeat cardiac output and pressure measurements during therapy guide further management. Echocardiography and other imaging may be performed to corroborate findings, and the catheter is removed once hemodynamics stabilize or noninvasive monitoring suffices.

References

1. MedlinePlus. Pulmonary artery catheterization. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 159

Pulse Oximetry

Overview

Pulse oximetry is a quick, noninvasive method of estimating blood oxygen saturation and measuring pulse rate. It uses a small sensor, usually placed on a fingertip, toe, or earlobe.

Why It Is Done

It is used to assess oxygenation in people with breathing or heart conditions, monitor patients during illness or procedures, guide oxygen therapy, and screen for low oxygen levels.

How It Is Performed

The sensor shines light through the tissue and detects changes associated with oxygenated and deoxygenated hemoglobin. The device displays oxygen saturation (SpO2) and pulse rate within seconds.

Preparation and Limitations

Remove dark nail polish when possible and keep the hand still and warm. Poor circulation, movement, skin pigmentation, nail products, cold extremities, and carbon monoxide exposure can affect accuracy. Readings should be interpreted with symptoms and the clinical situation.

Risks and Results

Pulse oximetry is painless and generally has no significant risk. A persistently low reading may require clinical assessment, supplemental oxygen, or additional testing such as an arterial blood gas; the device alone does not diagnose the cause of low oxygen.

References

1. MedlinePlus. Pulse oximetry. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 160

Radiation Therapy

Overview

Radiation therapy uses high-energy ionizing radiation (X-rays, gamma rays, or protons) to damage the DNA of cancer cells, preventing them from dividing and growing.

Alternative Names

Radiotherapy, Radiation oncological treatment

Principle

Ionizing radiation is delivered to a tumor target, damaging the DNA of cancer cells directly and through free radicals. Because dividing cancer cells are more vulnerable to unrepaired DNA damage, they die while surrounding normal tissue, spared by fractionation and precise targeting, recovers.

History

Emil Grubbe treated a breast cancer patient with X-rays in 1896, shortly after Roentgen's discovery. Marie Curie pioneered radium treatments in the early 20th century.

Why It Is Done

Ordered to cure localized cancers, shrink tumors before surgery, destroy micro-metastases after surgery, or palliate symptoms (pain, bleeding) in advanced cancer.

Is This a Primary or Secondary Test or Procedure

Primary curative treatment for many localized cancers (e.g., prostate, head and neck, cervical cancer).

How It Is Performed

External beam radiation uses a linear accelerator to direct radiation at the tumor from outside the body. Brachytherapy places radioactive sources directly inside or next to the tumor.

Preparation

A simulation session is performed to design custom molds, masks, and marks on your skin to ensure precise alignment for every treatment session.

Contraindications and Precautions

Prior maximum radiation tolerance to the same tissue, or active connective tissue diseases (relative, due to severe fibrosis risk).

Risks

Skin changes (redness, peeling), fatigue, hair loss in the treated area, and organ-specific side effects (e.g., radiation pneumonitis, cystitis).

Aftercare and Recovery

Keep the treated skin clean and dry. Avoid sun exposure. Side effects typically peak 1-2 weeks after treatment ends and then improve.

Outcome and Follow-up

Evaluated over weeks to months using follow-up imaging (CT, PET) to assess tumor response and regression.

Expected Results

Reduction or resolution of tumor mass; minimal long-term toxicity to surrounding normal tissues.

Follow-up

Regular follow-up with a radiation oncologist, monitoring of side effects, and surveillance scans.

References

1. MedlinePlus. Radiation Therapy. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 161

Radiofrequency Neurotomy

Overview

Radiofrequency neurotomy (rhizotomy) is a minimally invasive procedure that uses heat generated by a radiofrequency current to interrupt the nerve supply to a painful joint or structure. It is performed under fluoroscopic or CT guidance to deliver targeted lesions to specific medial branch nerves or other sensory nerves.

Alternative Names

Radiofrequency ablation of nerves, radiofrequency rhizotomy, medial branch radiofrequency neurotomy, RF neurotomy, facet denervation, pulsed radiofrequency therapy

Principle

A needle electrode is advanced under image guidance to a target nerve, and a radiofrequency current heats the surrounding tissue to denature nerve proteins and produce a thermal lesion. This interrupts transmission of nociceptive signals from the affected joint or structure to the spinal cord. Because the lesion is confined to the target nerve, adjacent motor function is typically preserved. The procedure is diagnostic in concept when combined with temporary blocks that predict pain relief from subsequent denervation.

History

Percutaneous radiofrequency lesioning of spinal nerves was introduced in the 1970s as a development of earlier surgical rhizotomy techniques. It became widely used in pain medicine for cervical and lumbar facet arthropathy after studies validated its efficacy and low complication rate.

Why It Is Done

Radiofrequency neurotomy is ordered for chronic axial neck or back pain attributed to facet joint arthropathy that has not responded to conservative management and that is confirmed by positive responses to diagnostic medial branch blocks. It may also be used for sacroiliac joint pain, occipital neuralgia, and selected cases of discogenic or postsurgical pain. The goal is durable, long-lasting pain relief and functional improvement when injectable therapies provide only temporary benefit.

Is This a Primary or Secondary Test or Procedure

It is a secondary, confirmatory procedure because it follows diagnostic blocks and is reserved for patients with a clear pain generator. It is not a primary test but a definitive therapeutic intervention for selected chronic pain conditions.

How It Is Performed

The patient lies prone, and the target medial branch nerves are identified with fluoroscopy or CT, then anesthetized locally. A radiofrequency cannula with electrode is advanced to each target site, and sensory and motor stimulation is performed to confirm correct placement and avoid motor nerves. After test stimulation, thermal lesions are created at each target nerve for approximately 60–90 seconds each, and the needles are removed.

Preparation

Patients stop anticoagulant and antiplatelet medications before the procedure, and a coagulation panel may be checked. No special fasting is usually required unless sedation is planned. Imaging and a review of prior diagnostic blocks are used to plan the level and number of targets.

Contraindications and Precautions

Uncorrectable coagulopathy, active infection at the site, and pacemakers or implanted devices that may be affected by radiofrequency current are contraindications. The procedure should not be performed without a positive diagnostic block, and patients with widespread or non-mechanical pain may not benefit. Care must be taken near motor nerves, the cervical region, and the vertebral artery.

Risks

Risks are generally low and include transient procedural pain, bruising, bleeding, infection, and neuritis with temporary burning or dysesthesia. Rare complications include incomplete pain relief, motor weakness, and injury to adjacent neurovascular structures, especially at cervical levels.

Aftercare and Recovery

Patients are observed briefly and usually return home the same day with instructions to limit strenuous activity for a few days. Postprocedural soreness is expected and managed with cold packs and analgesics. Pain relief, when it occurs, typically develops over 1–2 weeks as the nerve lesion matures.

Outcome and Follow-up

A successful neurotomy produces significant pain reduction that may last 6–12 months or longer as the treated nerves regenerate slowly. If pain returns, the procedure can be repeated, and recurrence at the same level often responds equally well. Lack of relief after a technically successful neurotomy suggests the pain source is not the targeted nerve.

Expected Results

There is no normal numeric result because neurotomy is a therapeutic intervention; the expected outcome is marked pain reduction and improved function. Success is typically defined as at least 50%–80% pain relief for a sustained period.

Follow-up

Patients are followed clinically to gauge the duration of relief, and repeat neurotomy may be offered when pain recurs. If relief is inadequate, further imaging or additional diagnostic blocks may be considered to reevaluate the pain generator.

References

1. MedlinePlus. Radiofrequency Neurotomy. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 162

Root Canal Treatment

Overview

Root canal treatment, or endodontic therapy, removes infected or damaged pulp from inside a tooth, cleans and seals the root canals, and preserves the natural tooth.

Alternative Names

Endodontic therapy; root canal procedure; endodontics

Principle

After removing the pulp chamber and canals, the dentist shapes and disinfects the canal system and fills it with a biocompatible material (gutta-percha), preventing reinfection and allowing the tooth to be restored.

History

Root canal therapy was developed in the late 19th and early 20th centuries with the use of gutta-percha and root canal instruments, becoming a routine dental procedure.

Why It Is Done

Performed for irreversible pulpitis, pulp necrosis, dental abscess, and trauma or fracture exposing the pulp, to relieve pain and save the tooth.

Is This a Primary or Secondary Test or Procedure

Primary definitive treatment for pulp and periapical disease.

How It Is Performed

Performed under local anesthesia and rubber dam isolation. The tooth is opened, canals are shaped with rotary files, irrigated with disinfectant, and obturated with gutta-percha. A permanent crown or filling follows.

Preparation

Radiographs and vitality testing; antibiotics are reserved for systemic signs of infection.

Contraindications and Precautions

A non-restorable tooth, gross vertical root fracture, and inadequate bone support are contraindications.

Risks

Postoperative pain, instrument fracture, incomplete obturation, reinfection, and tooth fracture without a crown.

Aftercare and Recovery

Mild soreness is expected; the tooth is restored with a crown to prevent fracture.

Outcome and Follow-up

Successful therapy yields a healed apex with no pain, swelling, or fistula at follow-up radiographs.

Expected Results

A symptom-free tooth with intact root filling and no periapical radiolucency.

Follow-up

Clinical and radiographic review; retreatment or extraction if symptoms or lesions persist.

References

1. MedlinePlus. Root canal. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 163

Sacroiliac Joint Injection

Overview

A sacroiliac joint injection places local anesthetic, often with a corticosteroid, into or around the sacroiliac joint to help diagnose and treat pain arising from that joint.

Why It Is Done

It may be considered for persistent lower-back, buttock, or leg pain when examination suggests sacroiliac joint involvement and conservative treatment has not provided adequate relief.

How It Is Performed

The patient lies face down. After the skin is cleaned and numbed, the clinician guides a needle to the joint using appropriate procedural guidance and injects the medication. The patient is observed briefly afterward and may record pain relief over the following hours and days.

Preparation

Tell the clinician about blood thinners, diabetes, infection, allergies, pregnancy, and previous reactions to injection medicines. Medicines may need adjustment only under medical direction. Arrange help with transportation if sedation is used.

Risks and Recovery

Temporary soreness, bleeding, infection, nerve irritation, and a short-lived increase in pain are possible. Corticosteroids may temporarily affect blood glucose and other steroid-sensitive conditions. Most people resume light activity shortly afterward, following the clinician's instructions.

Outcome and Follow-up

Short-term meaningful pain relief supports the possibility that the sacroiliac joint contributes to symptoms. Lack of relief suggests that another pain source may need evaluation. Repeated injections are limited according to clinical response and safety considerations.

References

1. MedlinePlus. Sacroiliac joint injection. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 164

Sclerotherapy

Overview

Sclerotherapy is a procedure that treats varicose and spider veins by injecting a solution that causes the vein to collapse and fade.

Alternative Names

Varicose vein injection therapy; sclerosing injection; spider vein treatment

Principle

A sclerosing agent is injected into a vein, irritating its lining and causing the vein to close and scar, which redirects blood to healthier veins. The treated vein gradually fades and is reabsorbed.

History

Sclerotherapy has been practiced in various forms since the 19th century and remains a first-line treatment for small varicose and spider veins.

Why It Is Done

Performed for symptomatic spider veins and small varicose veins causing aching, burning, or swelling, and for cosmetic improvement.

Is This a Primary or Secondary Test or Procedure

Primary treatment for spider veins and small varicose veins.

How It Is Performed

The vein is injected with a liquid or foam sclerosant. Foam sclerosant is often used for larger veins. Compression is applied afterward. Several sessions may be needed.

Preparation

A venous ultrasound may precede treatment; sun exposure, certain medications, and compression hosiery recommendations are reviewed.

Contraindications and Precautions

Deep vein thrombosis, severe peripheral arterial disease, allergy to the sclerosant, and pregnancy are contraindications.

Risks

Pain, bruising, skin discoloration, blood clots, ulceration, and rarely allergic reactions.

Aftercare and Recovery

Compression stockings are worn for days to weeks; walking is encouraged, and treated veins fade over several weeks.

Outcome and Follow-up

Improvement is judged by fading of veins and symptom relief; persistent vessels may require repeat injections.

Expected Results

Fading or disappearance of treated veins with no complications.

Follow-up

Follow-up visits and ultrasound if large truncal veins are present.

References

1. MedlinePlus. Sclerotherapy. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 165

Sentinel Node Biopsy

Overview

Sentinel node biopsy is a procedure that identifies and removes the first lymph node or nodes that drain a cancer, most often in breast cancer and melanoma, to stage the disease.

Alternative Names

Sentinel lymph node biopsy; SLNB; sentinel node mapping

Principle

A tracer (radioactive colloid, blue dye, or both) is injected near the tumor and travels to the sentinel node, which represents the status of the whole nodal basin. Removing it predicts whether cancer has spread.

History

The sentinel node concept was introduced for melanoma by Morton in 1992 and rapidly extended to breast cancer, replacing routine axillary dissection for staging.

Why It Is Done

Performed for early breast cancer and melanoma without clinically involved nodes, to stage the axilla or groin accurately with minimal surgery.

Is This a Primary or Secondary Test or Procedure

Primary staging procedure for node-negative-appearing solid tumors.

How It Is Performed

A radioactive tracer or dye is injected before or during surgery, and a gamma probe and visual inspection identify the sentinel node, which is removed and sent for pathology.

Preparation

Injection of tracer, and discussion of the implications of a positive result.

Contraindications and Precautions

Clinically positive nodes and prior surgery in the draining basin may preclude the procedure.

Risks

Seroma, infection, allergic reaction to dye, and a small risk of lymphedema.

Aftercare and Recovery

Pathology determines whether further node dissection or axillary radiation is needed.

Outcome and Follow-up

A negative sentinel node predicts a very low chance of cancer in remaining nodes; a positive node changes treatment.

Expected Results

No cancer in the sentinel lymph node.

Follow-up

Further treatment or surveillance based on nodal status.

References

1. MedlinePlus. Sentinel node biopsy. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 166

Sigmoidoscopy

Overview

Sigmoidoscopy uses a flexible scope to examine only the rectum and the lower part of the colon (sigmoid colon), which constitutes the distal third of the large intestine.

Alternative Names

Flexible sigmoidoscopy

Principle

A flexible endoscope is passed through the anus into the rectum and sigmoid colon, which is inflated to allow inspection of the lining. Biopsies and treatment of lesions in this distal segment can be performed through the scope.

History

Rigid sigmoidoscopes were used in the early 20th century. Flexible fiberoptic sigmoidoscopes were introduced in the 1970s, providing a more comfortable screening option.

Why It Is Done

Ordered for colorectal cancer screening, evaluating rectal bleeding, monitoring inflammatory bowel disease, or investigating changes in bowel habits.

Is This a Primary or Secondary Test or Procedure

Alternative screening option for colorectal cancer, often combined with annual fecal tests.

How It Is Performed

You lie on your left side. Sedation is usually not required. The scope is inserted and advanced up to 60 cm into the lower colon.

Preparation

Cleanse the lower bowel using one or two enemas on the morning of the procedure. A clear liquid diet may be recommended.

Contraindications and Precautions

Acute severe diverticulitis, suspected colonic perforation, or toxic megacolon.

Risks

Extremely low risk of perforation (<1 in 10,000) or bleeding (<1 in 2,000 if biopsy is performed).

Aftercare and Recovery

You can immediately drive and return to normal activities. You may pass gas to relieve mild bloating.

Outcome and Follow-up

Allows visualization of hemorrhoids, fissures, polyps, or inflammation in the distal colon. Biopsies can be performed.

Expected Results

Normal mucosal lining of the rectum and sigmoid colon; no polyps, inflammation, or lesions.

Follow-up

Repeat sigmoidoscopy in 5 years, or full colonoscopy if polyps or lesions are identified.

References

1. MedlinePlus. Sigmoidoscopy. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 167

Skin Biopsy

Overview

A skin biopsy removes a small sample of skin tissue for microscopic examination to diagnose skin conditions, including skin cancers, inflammatory dermatoses, or infections.

Alternative Names

Dermatological biopsy, Punch biopsy, Shave biopsy

Principle

A small sample of skin is removed by shaving the surface, punching a core, or excising the lesion and is submitted for histopathologic examination. Microscopic analysis provides a definitive diagnosis of skin conditions and tumors.

History

Developed in the 19th century. The invention of the circular punch biopsy instrument in the 20th century standardized the collection of full-thickness skin cores.

Why It Is Done

Ordered to evaluate suspicious moles, diagnose skin cancers (basal cell, squamous cell, melanoma), or identify chronic inflammatory skin disorders (psoriasis, eczema).

Is This a Primary or Secondary Test or Procedure

Primary diagnostic test for suspected skin cancer and atypical dermatoses.

How It Is Performed

The area is cleaned, and local anesthetic is injected. Depending on the technique: shave biopsy scrapes a superficial sample; punch biopsy punches a circular core; excisional biopsy cuts out the entire lesion.

Preparation

Clean the skin. Tell the doctor if you take blood thinners or have a history of keloid scarring.

Contraindications and Precautions

None. Active local skin infection is a relative contraindication (select a different site if possible).

Risks

Minor bleeding, bruising, local infection, or scar formation (keloids).

Aftercare and Recovery

Keep the biopsy site clean and covered with petroleum jelly and a bandage. If sutures were placed, they will be removed in 5-14 days.

Outcome and Follow-up

The sample is analyzed by a dermatopathologist. Results describe the histopathology (e.g., benign nevus, basal cell carcinoma, or specific dermatitis).

Expected Results

Normal epidermis and dermis; no cellular atypia, malignancy, or specific inflammatory pathology.

Follow-up

Suture removal; review of pathology; surgical excision if malignant margins are positive.

References

1. MedlinePlus. Skin Biopsy. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 168

Splint Application

Overview

Splint application supports and immobilizes an injured body part while allowing room for swelling. It is commonly used for fractures, sprains, dislocations, tendon injuries, and painful joints.

How It Is Performed

The clinician checks the skin, circulation, sensation, and movement before applying the splint. Padding and a rigid or flexible support are placed along the injured area and secured without excessive tightness. The limb is rechecked afterward.

Common Types

Examples include short-arm, sugar-tong, thumb-spica, ulnar-gutter, volar, posterior ankle, and finger splints. The choice depends on the injury and the need to control movement while preserving swelling room.

Preparation and Aftercare

Remove rings and tight jewelry when swelling is possible. Keep the splint dry and elevated as instructed. Do not alter the splint without medical advice. Move any joints that are not immobilized when permitted.

Risks and Warning Signs

Pressure injury, skin irritation, stiffness, nerve compression, and reduced circulation can occur if a splint is too tight or poorly fitted. Increasing pain, numbness, tingling, pale or blue fingers or toes, severe swelling, or inability to move the digits requires prompt medical assessment.

Follow-up

Follow-up may include repeat examination, imaging when needed, adjustment or replacement of the splint, and transition to a cast, brace, or rehabilitation program as healing progresses.

References

1. MedlinePlus. Splints. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 169

Suprapubic Catheterization

Overview

Suprapubic catheterization is the placement of a drainage catheter into the bladder through a puncture in the lower abdominal wall, above the pubic symphysis. It provides bladder drainage when urethral catheterization is impossible, contraindicated, or undesirable.

Alternative Names

Suprapubic cystostomy; suprapubic bladder drainage; punch suprapubic catheter insertion

Principle

With the bladder distended, a catheter is passed through the skin, rectus sheath, and bladder wall into the bladder lumen. The inflated balloon holds the catheter in place, and drainage is achieved by gravity or intermittent clamping.

History

Suprapubic bladder drainage was practiced in antiquity as a treatment for urinary retention. The modern method, using a percutaneous trocar and a self-retaining catheter, was popularized in the mid-20th century as a safer alternative to repeated urethral instrumentation.

Why It Is Done

Ordered for urinary retention when urethral catheterization fails or is contraindicated, for long-term bladder drainage in men to reduce urethral complications, and for patients with urethral injury, prostatic surgery, or recurrent urethral catheter problems.

Is This a Primary or Secondary Test or Procedure

Primary therapeutic procedure for bladder drainage when the urethral route is unsuitable.

How It Is Performed

The lower abdomen is prepared, and the bladder is confirmed distended by palpation or ultrasound. The puncture site is marked 2-3 cm above the pubic symphysis and anesthetized. A trocar and catheter are advanced through the abdominal wall into the bladder, the catheter is advanced, the trocar is removed, and the balloon is inflated and secured to the abdominal wall.

Preparation

Informed consent, confirmation of a distended bladder, and imaging guidance when the bladder cannot be clearly palpated or there is doubt about anatomy. Anticoagulant use and bleeding risk are reviewed.

Contraindications and Precautions

Contraindications include an empty or non-distended bladder, severe bleeding diathesis, known bladder cancer or previous lower abdominal surgery (potential bowel adhesions), and skin infection at the site. Caution is required if bowel overlies the puncture path.

Risks

Hematuria, infection, bowel injury or perforation, puncture of abdominal vessels, catheter displacement, and leakage around the catheter site.

Aftercare and Recovery

The catheter is connected to a drainage bag and the site is kept clean and dry. The urine is observed for hematuria and the patient is monitored for abdominal pain or signs of infection.

Outcome and Follow-up

A free flow of urine confirms correct placement, and clear urine with no abdominal tenderness suggests an uncomplicated procedure. Difficult insertion, blood, or fever may indicate complications.

Expected Results

A patent suprapubic catheter draining clear urine with a clean, non-inflamed exit site and no leakage or infection.

Follow-up

The catheter site and output are monitored, the catheter is changed at scheduled intervals, and a urinalysis or culture is performed if infection is suspected. Trial of clamping or voiding is attempted before eventual removal.

References

1. MedlinePlus. Suprapubic catheter care. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 170

Temporary (Transvenous) Cardiac Pacing

Overview

Temporary transvenous cardiac pacing is the placement of an electrode catheter into the right ventricle (or atrium) through a central vein to provide electrical pacing of the heart. It is used when the native conduction system fails or heart rate is dangerously slow.

Alternative Names

Transvenous temporary pacemaker, emergency pacing, ventricular pacing wire placement

Principle

An electrode catheter positioned against the endocardium delivers a small electrical impulse that depolarizes the myocardium, producing a paced rhythm. The pacemaker fires at a programmed rate when the intrinsic heart rate falls below a set threshold, maintaining cardiac output and preventing asystole.

History

Temporary transvenous pacing was developed in the 1950s, shortly after the first external pacemakers, and became a standard tool for managing bradyarrhythmias in acute care. Alternative methods such as transcutaneous pacing are often used as a bridge to transvenous pacing.

Why It Is Done

Ordered for symptomatic bradycardia, complete heart block with hemodynamic instability, and bradyarrhythmias following myocardial infarction or cardiac surgery. It is also used to overdrive pace certain tachyarrhythmias and to provide back-up pacing during high-risk procedures or temporary interruption of a permanent pacemaker.

Is This a Primary or Secondary Test or Procedure

A primary therapeutic procedure for acute, potentially reversible bradyarrhythmias until a permanent pacemaker can be placed.

How It Is Performed

Access is obtained through a central vein, most often the internal jugular, subclavian, or femoral, and a balloon-tipped pacing catheter is advanced under fluoroscopic or electrocardiographic guidance into the right ventricle. Pacing is initiated at a set rate and output, and capture is confirmed by a widened QRS followed by a T wave on the monitor. The catheter is secured and connected to an external pulse generator.

Preparation

Informed consent when feasible, continuous electrocardiographic and blood pressure monitoring, and sterile technique for central venous access. Transcutaneous pacing pads may be placed as a bridge while the transvenous catheter is being inserted.

Contraindications and Precautions

Contraindications include profound hypothermia with cold myocardial capture failure and severe hyperkalemia or drug toxicity where pacing may be ineffective; a reversible cause should be treated concurrently. Precautions include avoiding the subclavian approach in patients with coagulopathy or during thrombolysis, and confirming capture and sensing thresholds regularly.

Risks

Arrhythmias, ventricular perforation with tamponade, infection, pneumothorax or bleeding from central venous access, and failure to capture or sense leading to inadequate rhythm support.

Aftercare and Recovery

The patient is monitored in a critical care unit with continuous electrocardiography, and capture, sensing, and thresholds are checked daily. The temporary pacemaker is removed once the underlying rhythm stabilizes or a permanent pacemaker is implanted.

Outcome and Follow-up

A paced rhythm with capture (pacing spike followed by a wide QRS and mechanical pulse) and appropriate sensing indicate correct function. Loss of capture, undersensing, or a low output threshold requiring escalation suggests catheter dislodgement, threshold rise, or generator failure.

Expected Results

Stable pacing at the programmed rate with consistent capture, appropriate sensing of intrinsic beats, and hemodynamic stability.

Follow-up

Continuous electrocardiographic monitoring, daily pacing threshold checks, and chest radiography to confirm catheter position. The patient is referred for permanent pacemaker implantation if the bradyarrhythmia persists.

References

1. MedlinePlus. Pacemaker placement. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 171

Thoracentesis

Overview

Thoracentesis is a procedure that removes fluid from the space between the lung and chest wall (pleural space) through a needle or catheter, for diagnosis and symptom relief.

Alternative Names

Pleural tap; pleural fluid aspiration; chest tap

Principle

A needle is advanced into the pleural space, usually under ultrasound guidance, to aspirate fluid. Fluid analysis distinguishes transudates from exudates and may identify infection, malignancy, or other causes.

History

Thoracentesis has been performed for over a century, and ultrasound guidance, introduced in the late 20th century, greatly reduced complication rates.

Why It Is Done

Performed for new or unexplained pleural effusion to determine its cause, and to relieve dyspnea from large effusions.

Is This a Primary or Secondary Test or Procedure

Primary diagnostic and therapeutic procedure for pleural effusion.

How It Is Performed

Performed at the bedside with the patient sitting upright. Ultrasound locates the fluid, the site is prepped and anesthetized, and a needle with or without a catheter is advanced above the rib over the fluid. Fluid is aspirated and sent for analysis.

Preparation

Chest imaging (radiograph or ultrasound) to confirm fluid, coagulation assessment, and informed consent. Coughing or a ventilator may complicate the procedure.

Contraindications and Precautions

Severe coagulopathy, thrombocytopenia, and overlying skin infection are relative contraindications. Precaution: avoid the diaphragm, and drain large volumes slowly to prevent re-expansion pulmonary edema.

Risks

Pneumothorax, bleeding, infection, injury to the lung, liver, or spleen, and re-expansion pulmonary edema.

Aftercare and Recovery

A chest radiograph is often performed to exclude pneumothorax, and the patient is observed briefly.

Outcome and Follow-up

Pleural fluid protein, LDH, cell count, pH, cytology, and culture distinguish transudate (heart failure, cirrhosis) from exudate (infection, malignancy, inflammation).

Expected Results

A small amount of clear fluid with low protein and cell counts, and relief of dyspnea.

Follow-up

Results guide treatment; recurrent effusion may require repeat drainage, pleurodesis, or a chest tube.

References

1. MedlinePlus. Thoracentesis. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 172

Tilt-Table Test

Overview

A tilt-table test evaluates how the heart rate and blood pressure respond when the body changes from lying flat to an upright position. It is used mainly to investigate fainting or unexplained lightheadedness.

Why It Is Done

The test may help evaluate suspected vasovagal syncope, orthostatic intolerance, or other disorders of autonomic blood-pressure control when the cause is not clear from the history and routine examination.

How It Is Performed

The patient lies on a motorized table secured with straps. Heart rhythm, blood pressure, and symptoms are monitored while the table is slowly tilted upright for a set period. In some protocols, a medicine is used to increase the likelihood of reproducing symptoms.

Preparation and Risks

Fasting and temporary changes to selected medicines may be recommended. Follow the testing center's instructions. Dizziness, nausea, sweating, fainting, palpitations, or a temporary blood-pressure change may occur; monitoring and staff support are provided throughout.

Outcome and Follow-up

The clinician compares symptoms with changes in blood pressure and heart rate. A positive pattern may support a diagnosis such as vasovagal syncope or orthostatic hypotension, but results must be interpreted with the full clinical history.

References

1. MedlinePlus. Tilt-table test. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 173

Tonometry

Overview

Tonometry measures intraocular pressure, the pressure of fluid inside the eye. It is an important part of evaluating and monitoring glaucoma and other eye conditions.

How It Is Performed

In applanation tonometry, numbing drops are used and a small instrument briefly contacts the cornea to measure the force needed to flatten it. Noncontact tonometry uses a brief puff of air. Other handheld instruments may be used when contact with the eye is difficult.

Preparation

Remove contact lenses as directed and tell the eye-care professional about eye surgery, injuries, infections, or medicines. Avoid rubbing the eyes after numbing drops until sensation returns.

Risks and Results

Tonometry is usually quick and painless. Contact methods can cause brief irritation or, rarely, a corneal abrasion or infection. The result is interpreted with the optic-nerve examination, visual-field testing, corneal measurements, and other clinical findings; a single pressure reading does not diagnose or exclude glaucoma.

References

1. MedlinePlus. Tonometry. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 174

Total Parenteral Nutrition

Overview

Total parenteral nutrition (TPN) is the intravenous delivery of complete nutrition, including carbohydrates, protein, lipids, electrolytes, vitamins, and trace elements, bypassing the gastrointestinal tract. It is infused through a central venous catheter when oral or enteral feeding is impossible or insufficient.

Alternative Names

TPN, Intravenous hyperalimentation, Parenteral nutrition, Central parenteral nutrition

Principle

TPN delivers a sterile, individually compounded nutrient solution directly into the bloodstream, meeting caloric and nutritional needs without enteral absorption. Because the solution is hypertonic, it must be infused through a central venous catheter. The content of macronutrients, electrolytes, and micronutrients is individualized to the patient's metabolic and fluid requirements.

History

Total parenteral nutrition was pioneered by Dudrick and Wilmore in the late 1960s, establishing a life-sustaining therapy for patients with intestinal failure.

Why It Is Done

TPN is ordered for patients with intestinal failure, short bowel syndrome, prolonged ileus or bowel obstruction, severe pancreatitis, or other conditions in which the gut cannot be used for more than several days. It is also used in severe malnutrition when oral or enteral feeding is not feasible.

Is This a Primary or Secondary Test or Procedure

Primary therapy for intestinal failure and a definitive source of nutrition when enteral feeding is not possible.

How It Is Performed

A central venous catheter is placed under sterile technique, typically in the subclavian or internal jugular vein or as a PICC line. The pharmacy-compounded TPN solution is infused continuously or cyclically through a dedicated catheter lumen. Glucose, electrolytes, fluid balance, and clinical status are monitored throughout the infusion.

Preparation

Nutritional assessment and baseline laboratory studies, including electrolytes and renal and liver function tests, are obtained. Central venous access is inserted with aseptic precautions, and informed consent is obtained.

Contraindications and Precautions

TPN should be avoided when the gastrointestinal tract is usable or in patients with hemodynamic instability who cannot tolerate volume loading. Use caution in hyperglycemia, hypertriglyceridemia, severe electrolyte disturbances, or fluid overload, and treat these before initiating therapy. Central line insertion is contraindicated with untreated coagulopathy or local infection.

Risks

Catheter-related bloodstream infection, central venous thrombosis, pneumothorax at insertion, hyperglycemia, refeeding syndrome, and electrolyte imbalances. Long-term use can cause hepatic steatosis, cholestasis, and metabolic bone disease.

Aftercare and Recovery

Glucose, electrolytes, and liver function are monitored regularly, and the solution is adjusted to maintain metabolic stability. TPN is tapered before discontinuation to avoid hypoglycemia, and oral or enteral feeding resumes as soon as gut function allows.

Outcome and Follow-up

Response is assessed through laboratory values such as glucose, electrolytes, liver function tests, triglycerides, and nitrogen balance rather than a single numeric result. The formulation is adjusted to keep these parameters stable.

Expected Results

Stable glucose, electrolytes, and fluid balance with adequate nutritional parameters and weight maintenance in the absence of metabolic complications.

Follow-up

Periodic metabolic and nutritional laboratory monitoring, with surveillance of liver function and bone mineral density in long-term patients. Patients may be discharged on home TPN with intestinal rehabilitation plans.

References

1. MedlinePlus. Total parenteral nutrition. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 175

Tracheostomy Management

Overview

Tracheostomy management is the ongoing care of a tracheostomy tube, including cleaning, suctioning, cuff management, tube changes, and prevention of complications. It is required for patients with a surgical airway for prolonged ventilation, airway obstruction, or secretion clearance.

Alternative Names

Tracheostomy care; trach care; tracheostomy tube management; stoma care

Principle

A tracheostomy tube bypasses the upper airway, delivering air directly to the trachea. Management maintains tube patency by removing secretions, secures the tube in place, provides humidification, and monitors the stoma and cuff pressures to prevent tissue injury and infection.

History

Tracheostomy has been performed since antiquity, but modern indwelling tracheostomy tubes and structured management protocols developed during the 20th century, particularly with the growth of intensive care and prolonged mechanical ventilation.

Why It Is Done

Required for patients who cannot tolerate decannulation, those on long-term mechanical ventilation, patients with upper airway obstruction or aspiration, and as part of routine care to prevent tube occlusion and infection.

Is This a Primary or Secondary Test or Procedure

A primary supportive procedure that maintains a patent airway and is essential to the safe ongoing use of a tracheostomy.

How It Is Performed

Routine care includes humidified oxygen or air, regular suctioning with a sterile catheter, cleaning of the inner cannula, and skin care around the stoma. The tube ties are changed and the tube is secured, the cuff is deflated periodically and its pressure checked, and tube changes are performed by trained staff when needed.

Preparation

Gathering suction, cleaning supplies, a replacement tube of the same size, ties, and saline before any procedure. The patient is positioned with the head slightly extended, and the assistant is prepared for emergency tube reinsertion.

Contraindications and Precautions

Avoid disconnecting humidification for prolonged periods, and never advance suction beyond the pre-measured length. Precautions include confirming a mature stoma before tube changes, keeping a spare tube and obturator at the bedside, and having an emergency plan for accidental decannulation.

Risks

Tube occlusion, accidental decannulation, bleeding, infection, tracheal stenosis, and tracheal wall injury from high cuff pressures.

Aftercare and Recovery

The patient's oxygen saturation and breathing are monitored after suctioning or tube changes, and the tube position is confirmed. Any change in ventilation, bleeding, or signs of infection is reported promptly.

Outcome and Follow-up

A patent airway with clear breath sounds, stable oxygenation, and a clean stoma indicates effective management. Increasing suction requirements, fever, or bleeding may indicate a complication.

Expected Results

A clean, dry, non-inflamed stoma with a secure tracheostomy tube, clear airways, and stable respiratory status.

Follow-up

The tracheostomy is reviewed by the clinical team, and the patient is assessed for readiness for decannulation with a trial of capping when the underlying indication has resolved. Speech and swallowing assessment and bronchoscopy may be needed.

References

1. MedlinePlus. Tracheostomy - aftercare. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 176

Transcranial Magnetic Stimulation

Overview

Transcranial magnetic stimulation (TMS) is a noninvasive technique that uses magnetic fields to stimulate neurons in targeted regions of the brain. It is used both as a diagnostic tool to assess cortical excitability and as a therapeutic treatment, most commonly for depression.

Alternative Names

TMS, repetitive transcranial magnetic stimulation (rTMS), theta burst stimulation (TBS), motor evoked potential testing

Principle

A rapidly changing magnetic field passes through the skull and induces an electrical current in underlying cortical neurons, depolarizing them. Repetitive stimulation (rTMS) at low frequencies tends to suppress cortical activity, whereas high frequencies increase it. Repeated sessions of stimulation to the dorsolateral prefrontal cortex are thought to modulate dysfunctional circuits underlying mood disorders.

History

Barker and colleagues developed modern TMS at the University of Sheffield in 1985. In 2008, rTMS received FDA approval for the treatment of medication-resistant major depressive disorder.

Why It Is Done

Therapeutically ordered for major depressive disorder that has not responded adequately to one or more antidepressant trials, and for several other conditions including obsessivecompulsive disorder and migraine. Diagnostically, it is used to map motor cortex function and to test corticospinal tract integrity.

Is This a Primary or Secondary Test or Procedure

A secondary treatment used after inadequate response to pharmacotherapy; a diagnostic adjunct to routine neurologic evaluation.

How It Is Performed

The patient is awake and seated in a chair while a figure-of-eight coil is placed against the scalp. The resting motor threshold is first determined by observing thumb or hand twitch. Treatment delivers trains of repeated pulses to the target region in sessions lasting 20-40 minutes, typically five days per week for four to six weeks.

Preparation

Screening for metallic implants and history of seizures, and removal of metal objects, hearing protection, and jewelry. No sedation is required, and the patient may eat and take regular medications.

Contraindications and Precautions

Contraindicated in patients with ferromagnetic metal in the head (aneurysm clips, cochlear implants) or implanted devices affected by magnetic fields, such as vagus nerve or deep brain stimulators. Precautions include seizure history, epilepsy, and pregnancy.

Risks

The most common side effects are mild headache, scalp discomfort, and facial twitching that resolve rapidly. The most serious risk is seizure, which is rare, along with transient hearing changes or mania.

Aftercare and Recovery

Patients may immediately resume normal activities and driving. Response is assessed over the treatment course with symptom rating scales, typically evident after two to four weeks.

Outcome and Follow-up

Therapeutic response is measured clinically rather than by imaging or laboratory values, using standardized depression scores. Maintenance sessions may be scheduled for patients who respond, to sustain the benefit.

Expected Results

Reduction in depression severity scores, such as a 50% or greater drop on the Hamilton Depression Rating Scale. Recovery of motor evoked potentials with normal latencies reflects intact corticospinal conduction.

Follow-up

Serial mood rating scales during and after the treatment course to document response. Responders may require tapering maintenance rTMS or continued antidepressant therapy to prevent relapse.

References

1. MedlinePlus. Transcranial magnetic stimulation. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 177

Transcutaneous Electrical Nerve Stimulation (TENS)

Overview

TENS applies low-voltage electrical impulses through electrodes placed on the skin to modulate pain signals and support comfort during activity or rehabilitation.

Why It Is Done

It may be used as an adjunct for selected acute or chronic musculoskeletal, neuropathic, or postoperative pain. Benefit varies and it does not treat the underlying cause of pain.

How It Is Performed

Adhesive electrodes are placed around, but not directly over, the painful area. A battery-powered device delivers adjustable pulses; intensity is increased only to a strong but comfortable sensation.

Precautions and Risks

Skin irritation and discomfort can occur. Do not place electrodes over an implanted electronic device, the front of the neck, open wounds, or areas with absent sensation unless specifically directed by a clinician. Avoid use during bathing or driving.

References

1. MedlinePlus. Transcutaneous Electrical Nerve Stimulation. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 178

Transesophageal Echocardiography

Overview

Transesophageal echocardiography (TEE) is an ultrasound examination of the heart performed with a probe passed into the esophagus and stomach, allowing imaging of the heart from directly behind it. It provides high-resolution images of the cardiac structures, valves, and great vessels.

Alternative Names

TEE; esophageal echocardiogram; transesophageal echo

Principle

A transducer at the tip of a flexible endoscope emits and receives ultrasound waves from within the esophagus, which lies immediately posterior to the left atrium and aorta. The short distance between the probe and the heart avoids interference from the chest wall and lungs, producing superior image quality compared with transthoracic imaging.

History

Transesophageal echocardiography was developed in the 1970s and entered widespread clinical use in the 1980s to 1990s, becoming a standard tool for evaluating structures that are difficult to visualize from the chest wall.

Why It Is Done

Ordered to evaluate infective endocarditis and its complications, atrial thrombi before cardioversion or ablation, prosthetic valve function, the source of cardioembolic stroke, aortic dissection, congenital heart defects, and to guide cardiac surgery and interventional procedures.

Is This a Primary or Secondary Test or Procedure

Secondary to transthoracic echocardiography for most indications; it is the primary modality for infective endocarditis assessment, prosthetic valve evaluation, and intraoperative cardiac monitoring.

How It Is Performed

The patient fasts and is placed in the left lateral position after topical anesthesia and conscious sedation. A lubricated transesophageal probe is passed into the esophagus under guidance, and multiple two- and three-dimensional images, Doppler measurements, and, when indicated, three-dimensional reconstructions are acquired.

Preparation

Fasting for at least four to six hours, informed consent, review of swallowing disorders and anticoagulation, and removal of dentures are needed. The patient should arrange for someone to accompany them home after sedation.

Contraindications and Precautions

Esophageal stricture, esophageal or gastric varices, recent esophageal or gastric surgery, esophageal perforation, and active upper gastrointestinal bleeding are contraindications. Precautions include monitoring oxygen saturation and blood pressure during the procedure and avoiding the study in patients with suspected esophageal pathology.

Risks

Risks include transient throat discomfort, mucosal injury, and rarely esophageal perforation, bleeding, arrhythmia, or respiratory compromise from sedation.

Aftercare and Recovery

The patient is observed until sedation wears off, and the throat may be sore for a short period. Food and drink are resumed after sensation returns and the patient is confirmed to be able to swallow safely.

Outcome and Follow-up

Findings such as valvular regurgitation, vegetations, thrombus, or dissection are interpreted by the cardiologist and integrated with clinical and laboratory data. Abnormalities guide decisions on surgery, anticoagulation, or further imaging.

Expected Results

Normal findings show intact cardiac structures, normally functioning valves without vegetations or regurgitation, no thrombus, and normal chamber sizes and ventricular function.

Follow-up

Abnormal findings may lead to cardiac surgery, repeat imaging, or blood cultures in suspected endocarditis. Follow-up TEE is used to monitor prosthetic valves, assess therapy response, or reevaluate thrombus resolution before cardioversion.

References

1. MedlinePlus. Transesophageal echocardiography (TEE). U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 179

Transjugular Intrahepatic Portosystemic Shunt

Overview

The transjugular intrahepatic portosystemic shunt (TIPS) is a procedure that creates a channel between the portal and hepatic veins to lower portal pressure.

Alternative Names

TIPS; TIPS procedure; transjugular intrahepatic portosystemic shunt

Principle

A stent is placed through the liver to connect a hepatic vein to the portal vein, redirecting portal blood flow and reducing the pressure that drives variceal bleeding and ascites.

History

TIPS was developed in the late 1980s and, with the advent of covered stents, became a standard treatment for refractory complications of portal hypertension.

Why It Is Done

Performed for variceal bleeding that fails medical or endoscopic therapy, refractory ascites, hepatic hydrothorax, and Budd-Chiari syndrome.

Is This a Primary or Secondary Test or Procedure

Secondary (salvage) therapy for complications of portal hypertension.

How It Is Performed

Performed under sedation or anesthesia by an interventional radiologist. Through the jugular vein, a needle and catheter cross the liver from a hepatic vein to the portal vein, and a covered stent is deployed to create the shunt.

Preparation

Pre-procedure imaging, correction of coagulopathy, and antibiotic prophylaxis.

Contraindications and Precautions

Severe heart failure, uncontrolled infection, and advanced liver failure (high MELD) are relative contraindications. Precaution: assess for hepatic encephalopathy risk.

Risks

Hepatic encephalopathy, bleeding, infection, shunt stenosis or thrombosis, and heart failure exacerbation.

Aftercare and Recovery

Portal pressure is measured; the patient is monitored, and medications target encephalopathy.

Outcome and Follow-up

A pressure gradient below 12 mmHg predicts low variceal rebleeding risk.

Expected Results

Reduced portosystemic gradient with control of bleeding or ascites.

Follow-up

Doppler ultrasound surveillance of shunt patency; interventions for stenosis.

References

1. MedlinePlus. Transjugular intrahepatic portosystemic shunt (TIPS). U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 180

Trigger Point Injection

Overview

Trigger point injection is a procedure in which a local anesthetic, often with a corticosteroid, is injected directly into a myofascial trigger point. It is used to relieve pain, muscle spasm, and referred tenderness associated with myofascial pain syndrome.

Alternative Names

Myofascial trigger point injection; dry needling; TP injection; trigger point block

Principle

A trigger point is a hyperirritable taut band of skeletal muscle that generates local and referred pain. Injecting anesthetic into the point mechanically disrupts the taut band, releases muscle spasm, and blocks local nociceptive signaling, with corticosteroid added when inflammation is thought to contribute.

History

Myofascial trigger points were described by Janet Travell in the mid-20th century, and injection-based treatment became a widely used technique for myofascial pain.

Why It Is Done

Ordered for patients with myofascial pain syndrome, fibromyalgia-related focal pain, chronic neck and back pain, tension headache, and localized muscle spasm that has not responded to physical therapy, heat, massage, or analgesics.

Is This a Primary or Secondary Test or Procedure

A secondary, adjunctive treatment for myofascial pain; it is therapeutic rather than diagnostic.

How It Is Performed

The trigger point is identified by palpation of a taut band with a localized twitch response. The skin is cleaned, and a small-gauge needle is inserted into the trigger point, with the needle withdrawn slightly and redirected several times to elicit twitch responses before the anesthetic is injected.

Preparation

Informed consent, review of anticoagulant use and bleeding risk, and identification of the target trigger point are needed. The patient should be positioned comfortably and should report if the injection causes distant or radiating pain.

Contraindications and Precautions

Local skin infection, anticoagulation or bleeding disorder, and known allergy to the injected agents are contraindications. Precautions include avoiding direct injection into the lung, pleura, or major neurovascular structures and limiting corticosteroid use in patients with diabetes or immune compromise.

Risks

Risks are generally low and include transient soreness, bleeding or bruising, infection, nerve injury, and pneumothorax when injecting near the chest wall.

Aftercare and Recovery

The patient is observed briefly and advised that transient soreness is common. Physical therapy, stretching, and posture correction are encouraged to prevent recurrence, since injection alone provides temporary relief.

Outcome and Follow-up

Immediate relief of the local and referred pain with improved muscle relaxation indicates a successful injection. Persistence or early recurrence of pain suggests the need for repeat injection combined with rehabilitative therapy or further evaluation.

Expected Results

Complete or substantial relief of the injected trigger point and its referred pain, with normal muscle function, indicates a good response.

Follow-up

Repeat injections are spaced to limit steroid exposure, and physical therapy with stretching exercises is continued. Imaging or electrodiagnostic studies may be considered if pain persists or if an alternative diagnosis is suspected.

References

1. MedlinePlus. Trigger point injection. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 181

Tuberculosis Skin Test (PPD or Mantoux Test)

Overview

The tuberculosis skin test is an intradermal test that checks whether the immune system has reacted to proteins associated with *Mycobacterium tuberculosis*.

Why It Is Done

It is used to screen people at risk for tuberculosis infection, including some people with exposure, healthcare work, residence in high-risk settings, or immune suppression. It does not by itself distinguish latent infection from active tuberculosis.

How It Is Performed

A small amount of purified protein derivative (PPD) is injected just under the skin of the forearm, creating a small raised area. The injection site must be examined by a trained clinician 48 to 72 hours later; the diameter of firm swelling, not redness alone, is measured.

Preparation and Limitations

No fasting or special preparation is needed. Tell the clinician about prior tuberculosis infection, BCG vaccination, immune suppression, or a previous severe reaction. A positive result may require further medical evaluation, and a negative result does not always exclude infection.

Risks and Results

The test may cause temporary itching or soreness. Rarely, a stronger local reaction occurs. Interpretation depends on the size of induration and the person's risk factors; chest imaging and other evaluations may be needed when clinically indicated.

References

1. Centers for Disease Control and Prevention. Tuberculin skin testing. 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 182

Umbilical Catheterization

Overview

Umbilical catheterization is the placement of a catheter into the umbilical artery or vein of a newborn for vascular access. It provides reliable arterial blood pressure monitoring and a route for blood sampling and the delivery of fluids and medications in critically ill neonates.

Alternative Names

Umbilical arterial catheter; UAC; umbilical venous catheter; UVC; umbilical line

Principle

The umbilical vessels remain patent in the newborn and provide direct access to the central circulation. The catheter tip is positioned in the descending aorta (arterial) or inferior vena cava (venous) under sterile technique, with tip location confirmed radiographically before use.

History

Umbilical catheterization was introduced in the 1940s to 1960s and became an essential component of neonatal intensive care, enabling continuous arterial monitoring and safe administration of therapies to preterm and critically ill newborns.

Why It Is Done

Ordered in sick newborns requiring continuous arterial blood pressure monitoring, frequent arterial blood gas and laboratory sampling, exchange transfusion, parenteral nutrition, or the infusion of vasoactive medications and fluids when peripheral access is inadequate.

Is This a Primary or Secondary Test or Procedure

A primary vascular access procedure in critically ill neonates and a secondary supportive measure when peripheral access is not feasible.

How It Is Performed

The umbilical stump is cleaned and draped sterilely, and the catheter is inserted into the umbilical artery or vein using sterile technique. The catheter is advanced to the predetermined depth, secured in place, and its position confirmed by radiography before infusion.

Preparation

Informed consent from the parents or guardians, sterile preparation of the umbilical stump, and determination of appropriate insertion depth based on birth weight or shoulder-to-umbilicus length are needed before placement.

Contraindications and Precautions

Necrotizing enterocolitis, omphalitis or infection of the umbilical stump, peritonitis, and lower-extremity vascular compromise are contraindications. Precautions include strict sterile technique, confirming catheter tip position, and discontinuing the line as soon as possible to reduce thrombotic and infectious risk.

Risks

Risks include thrombosis or embolism, catheter-related bloodstream infection, vessel perforation with hemorrhage, air embolism, and extremity ischemia.

Aftercare and Recovery

The catheter is used for monitoring and infusion while in place, with limb perfusion, line position, and signs of infection monitored regularly. It is removed as soon as the infant's condition permits, typically after a few days.

Outcome and Follow-up

Arterial catheter readings and samples guide respiratory and hemodynamic management, with results such as blood gases, glucose, and blood pressure interpreted against gestational-age-specific norms. Abnormal values are corrected by ventilation, fluids, and medications delivered through the line.

Expected Results

Blood pressure, blood gas, glucose, and electrolyte values within the expected ranges for the infant's gestational and postnatal age indicate stable neonatal physiology.

Follow-up

Peripheral intravenous or enteral access replaces the catheter once the infant is stable, and the site is observed for bleeding or infection after removal. Ongoing surveillance for complications such as hypertension or limb ischemia is warranted in selected cases.

References

1. MedlinePlus. Umbilical catheters. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 183

Ureteral Stent Placement

Overview

Ureteral stent placement is the insertion of a thin, flexible tube into the ureter to maintain drainage of urine from the kidney to the bladder. It is placed when a ureter is obstructed or when drainage is needed after urologic procedures.

Alternative Names

Ureteral stent insertion; double-J stent; JJ stent; retrograde ureteral stent

Principle

A stent with curled ends (double-J) is passed through the ureter so that its coils rest in the renal pelvis and bladder, holding the ureter open. Urine then drains by gravity and peristalsis around and through the stent.

History

Temporary ureteral drainage was refined through the 20th century, and the double-J stent introduced by Finney in 1978 became the standard device for maintaining ureteral patency.

Why It Is Done

Ordered for ureteral obstruction from stones, tumors, or strictures, to protect the kidney during and after ureteroscopy or lithotripsy, to bypass an injured ureter, and to relieve hydronephrosis when urgent drainage is required.

Is This a Primary or Secondary Test or Procedure

Primary therapeutic procedure for ureteral obstruction and a supportive measure during other urologic procedures.

How It Is Performed

Under fluoroscopic guidance, a guidewire is passed through a cystoscope into the ureter, and the stent is advanced over the wire until its proximal coil is in the renal pelvis and its distal coil in the bladder. The wire is removed and correct position is confirmed by imaging.

Preparation

Informed consent, review of anticoagulants and bleeding risk, and evaluation of renal function and urine culture before the procedure.

Contraindications and Precautions

Contraindications include untreated urinary tract infection and severe bleeding diathesis. Precautions include reviewing the patient for contrast allergy if imaging with contrast is used and using caution in pregnancy.

Risks

Pain and hematuria, bladder irritation and urinary frequency, stent migration or obstruction, infection, and rarely ureteral injury or perforation.

Aftercare and Recovery

The patient is monitored for hematuria and voiding symptoms and given analgesia as needed. The stent is left in place for the planned duration and removed endoscopically.

Outcome and Follow-up

Good urine drainage with resolution of hydronephrosis and relief of obstruction indicates success. Persistent pain, fever, or poor output may indicate stent dysfunction or infection.

Expected Results

A correctly positioned stent with free drainage of urine, no infection, and resolution of obstruction.

Follow-up

The stent is removed or exchanged at the scheduled interval, typically within weeks to months, and imaging is repeated to confirm drainage of the kidney.

References

1. MedlinePlus. Ureteral stent. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 184

Ureteroscopy

Overview

Ureteroscopy involves passing a thin, rigid or flexible fiberoptic scope through the urethra and bladder into the ureter to diagnose and treat ureteral stones or tumors.

Alternative Names

URS, Ureteral stone scope

Principle

A thin endoscope is passed through the urethra, bladder, and ureter, giving direct visualization of the ureter and renal pelvis. Lasers and baskets passed through the scope fragment and extract stones, and strictures can be dilated or incised.

History

Hugh Hampton Young performed the first ureteroscopy in a patient with valves in 1912. The technique advanced in the 1980s with flexible scopes and laser fibers.

Why It Is Done

Ordered to treat ureteral stones (especially in the middle or lower ureter) that cannot be treated by lithotripsy, or to diagnose ureteral strictures or tumors.

Is This a Primary or Secondary Test or Procedure

Primary endoscopic treatment for ureteral stones and upper tract urothelial carcinoma biopsy.

How It Is Performed

Performed under general anesthesia. The ureteroscope is guided through the urinary tract. A holmium laser fiber is used to fragment the stone, and fragments are removed with a basket. A ureteral stent is often placed.

Preparation

Fast for 8 hours. Obtain a negative urine culture. Stop anticoagulants.

Contraindications and Precautions

Untreated urinary tract infection, or severe urethral stricture.

Risks

Ureteral injury (perforation, avulsion), hematuria, UTI, or stent pain/cramping.

Aftercare and Recovery

Drink plenty of water. If a ureteral stent was placed, you will feel a frequent urge to urinate and mild flank pain during urination. The stent is removed in 3-10 days.

Outcome and Follow-up

Direct visualization confirms complete stone clearance and normal ureteral mucosa.

Expected Results

Complete clearance of ureteral stone; normal ureteral mucosa; patent ureteral lumen.

Follow-up

Stent removal; follow-up ultrasound or X-ray in 2-4 weeks to check for hydronephrosis.

References

1. MedlinePlus. Ureteroscopy. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 185

Urethrotomy

Overview

Urethrotomy is a procedure that cuts a urethral stricture to widen the urethra and restore the flow of urine.

Alternative Names

Direct vision internal urethrotomy; optical urethrotomy; urethral stricture surgery

Principle

Under direct vision, an instrument with a blade cuts through the stricture, allowing the urethra to heal at a wider caliber.

History

Optical urethrotomy was introduced in the 1970s and became the first-line endoscopic treatment for short urethral strictures.

Why It Is Done

Performed for urethral strictures causing obstructed voiding, urinary tract infection, or acute urinary retention.

Is This a Primary or Secondary Test or Procedure

Primary endoscopic treatment for short strictures.

How It Is Performed

Performed under general or regional anesthesia. A urethrotome is passed to the stricture, and the blade cuts it, after which a catheter is placed.

Preparation

Imaging (retrograde urethrogram) defines the stricture.

Contraindications and Precautions

Long or dense strictures may require open urethroplasty instead.

Risks

Recurrence, bleeding, infection, and injury to the urethra.

Aftercare and Recovery

A urinary catheter remains for days; voiding is reassessed.

Outcome and Follow-up

Successful voiding and imaging confirm a widened urethra.

Expected Results

A patent urethra with normal urinary flow.

Follow-up

Uroflowmetry and repeat procedures if the stricture recurs.

References

1. MedlinePlus. Urethrotomy. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 186

Uterine Artery Embolization

Overview

Uterine artery embolization (UAE), also called uterine fibroid embolization (UFE), is a minimally invasive procedure that blocks blood flow to uterine fibroids to shrink them and relieve symptoms.

Alternative Names

Uterine fibroid embolization; UAE; UFE; fibroid embolization

Principle

Small particles are injected into the uterine arteries, blocking the blood supply to fibroids while sparing the normal uterus. Ischemic fibroids shrink over months.

History

Uterine artery embolization was introduced in the 1990s as a uterine-sparing alternative to hysterectomy and myomectomy for symptomatic fibroids.

Why It Is Done

Performed for heavy menstrual bleeding, pelvic pressure, or pain caused by fibroids in women who wish to avoid surgery or preserve the uterus.

Is This a Primary or Secondary Test or Procedure

Primary treatment for symptomatic fibroids in appropriate candidates.

How It Is Performed

Performed by an interventional radiologist. A catheter is threaded from the groin artery into each uterine artery, and embolic particles are injected under fluoroscopy until fibroid flow is blocked.

Preparation

Imaging (ultrasound or MRI) to map fibroids, and discussion of fertility implications.

Contraindications and Precautions

Pregnancy, active infection, and suspected malignancy are contraindications. Precaution: the procedure can affect future fertility.

Risks

Post-embolization pain and fever, infection, damage to the uterus or ovaries, and passage of fibroid tissue.

Aftercare and Recovery

Moderate cramping and pain are managed in the hospital for a day; symptoms improve over months as fibroids shrink.

Outcome and Follow-up

Follow-up imaging shows fibroid shrinkage; symptoms decrease over 3-6 months.

Expected Results

Significant fibroid volume reduction with improvement in bleeding and pressure.

Follow-up

Ultrasound or MRI at 3-6 months to document response.

References

1. MedlinePlus. Uterine artery embolization. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 187

Vaccination

Overview

Vaccination involves administering a vaccine (antigenic material) to stimulate an individual's immune system to develop adaptive immunity to a pathogen, preventing infectious disease.

Alternative Names

Immunization, Vaccine injection

Principle

A vaccine introduces a killed or weakened organism, a subunit, or genetic material coding for an antigen without causing disease. The immune system mounts a primary response and produces memory cells, so later exposure to the real pathogen triggers a rapid, protective secondary response.

History

Edward Jenner developed the smallpox vaccine in 1796. Louis Pasteur developed vaccines for rabies and anthrax. Jonas Salk developed the polio vaccine in 1952.

Why It Is Done

Ordered to prevent infectious diseases (like influenza, tetanus, measles, COVID-19), protect public health, and satisfy travel or employment requirements.

Is This a Primary or Secondary Test or Procedure

Primary preventive intervention for infectious diseases.

How It Is Performed

Usually injected into a muscle (e.g., deltoid or anterolateral thigh) or given orally/nasally. The injection site is cleaned with alcohol beforehand.

Preparation

Verify the patient's vaccination record and check for allergies (e.g., egg allergy for certain flu vaccines).

Contraindications and Precautions

Severe allergic reaction (anaphylaxis) to a prior dose of the same vaccine. Live vaccines are contraindicated in pregnancy and immunodeficiency.

Risks

Common side effects include pain/redness at the injection site, mild fever, or muscle aches. Anaphylaxis is extremely rare (1 in 1 million).

Aftercare and Recovery

Monitor the patient for 15 minutes after the injection for any immediate allergic reaction or vasovagal syncope.

Outcome and Follow-up

Stimulates the production of antibodies, providing immunity. Some vaccines require checking antibody titers later.

Expected Results

Successful administration of the vaccine; expected mild localized immune response without systemic complications.

Follow-up

Documentation in immunization registry; schedule subsequent booster doses.

References

1. MedlinePlus. Vaccination. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 188

Vasectomy

Overview

A vasectomy is a surgical procedure for male sterilization or permanent contraception. The vas deferens are severed and sealed to prevent sperm from entering the seminal fluid.

Alternative Names

Male sterilization

Principle

Each vas deferens is identified, isolated through a small scrotal incision, and a segment is cut and sealed with ligation or cautery. Blocking the sperm transport pathway means the ejaculate contains seminal fluid without sperm.

History

First performed on dogs by Sir Astley Cooper in 1830. Regarded as a sterilization method in the early 20th century. The 'no-scalpel' technique was developed in China in 1974.

Why It Is Done

Ordered by men seeking permanent, highly effective contraception.

Is This a Primary or Secondary Test or Procedure

Primary, highly effective method of permanent male contraception.

How It Is Performed

Usually performed under local anesthesia in an office setting. The clinician makes small punctures in the scrotum, isolates the vas deferens, cuts them, and seals the ends with clips or cautery.

Preparation

Shave the scrotum and clean the area on the day of the procedure. Wear tight-fitting underwear or a jockstrap to support the scrotum afterward.

Contraindications and Precautions

Active scrotal infection, local anatomical abnormalities (large varicocele, inguinal hernia), or uncertainty about permanent sterilization.

Risks

Hematoma, infection, sperm granuloma, chronic post-vasectomy pain syndrome (<1

Aftercare and Recovery

Apply ice packs to the scrotum. Rest for 24-48 hours. Avoid heavy lifting and sexual activity for 1 week. Use back-up contraception until cleared.

Outcome and Follow-up

A semen analysis is performed 8-12 weeks post-procedure. Sterilization is confirmed only when the semen shows azoospermia.

Expected Results

Successful occlusion of the vas deferens; post-vasectomy semen analysis demonstrating zero sperm (azoospermia).

Follow-up

Semen analysis at 12 weeks or after 20 ejaculations.

References

1. MedlinePlus. Vasectomy. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 189

Venepuncture

Overview

Venepuncture is the puncturing of a peripheral vein with a needle to obtain a blood sample for laboratory testing. It is one of the most frequently performed procedures in health care, providing blood for hematologic, biochemical, and microbiologic analysis.

Alternative Names

Phlebotomy; venipuncture; blood draw; venous blood sampling

Principle

A tourniquet distends a superficial peripheral vein, and a beveled needle or vacuum tube system pierces the vein wall to allow blood to flow into collection tubes. Additives in the tubes preserve the sample, and the order of draw minimizes contamination between tubes.

History

The practice of bloodletting by venesection dates to antiquity, but modern venepuncture for diagnostic testing developed with laboratory medicine in the 19th and 20th centuries, replacing crude lancets with the vacuum tube system introduced in the 1940s.

Why It Is Done

Ordered to obtain blood for complete blood counts, metabolic panels, cultures, serology, coagulation studies, therapeutic drug monitoring, and countless other laboratory assays, and to screen, diagnose, or monitor disease.

Is This a Primary or Secondary Test or Procedure

A primary procedure whose purpose is specimen collection; it is supportive rather than therapeutic and is the gateway to laboratory testing.

How It Is Performed

The patient is seated or supine with the arm extended. A tourniquet is applied, and a suitable vein in the antecubital fossa is selected by inspection and palpation. The site is cleaned with alcohol, the vein is anchored, and the needle is advanced at a shallow angle. Blood is drawn into evacuated tubes in the correct order, the tourniquet is released, and the needle is withdrawn with pressure applied to the site.

Preparation

Informed consent and, for some tests, fasting. The collector should confirm patient identification, test requirements, and collection tubes before the draw.

Contraindications and Precautions

Avoid the limb with an arteriovenous fistula, lymphedema, infection, or a previous mastectomy on that side. Use caution with anticoagulated patients, coagulopathies, and children, and avoid repeated probing of a site.

Risks

Pain, bruising, hematoma, bleeding, nerve injury from deep probing, arterial puncture, infection, and vasovagal reactions or fainting. Hemolyzed or clotted samples may require repeat collection.

Aftercare and Recovery

Pressure is held over the site for a few minutes and a small dressing is applied. The samples are labeled, transported to the laboratory, and analyzed according to the ordered tests.

Outcome and Follow-up

Results depend on the tests ordered and are interpreted by the clinician in the context of the patient's history and examination. The quality of collection directly affects result reliability.

Expected Results

Reference ranges vary by test, age, and sex; normal results fall within the laboratory's published reference intervals for the ordered panel.

Follow-up

Abnormal results may prompt repeat testing, additional confirmatory or reflex testing, and further investigations directed by the clinical picture.

References

1. MedlinePlus. Venipuncture. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 190

Vertebroplasty

Overview

Vertebroplasty is a minimally invasive procedure where medical-grade bone cement (PMMA) is injected into a fractured vertebra to stabilize it and relieve pain.

Alternative Names

Spinal cement injection, Percutaneous vertebroplasty

Principle

Bone cement is injected under imaging guidance through a needle directly into a fractured vertebral body. The cement hardens within minutes, stabilizing the vertebra and reducing the pain caused by movement of the fracture fragments.

History

First performed in France in 1984 by Galibert and Deramond to treat an aggressive vertebral hemangioma.

Why It Is Done

Ordered for painful vertebral compression fractures caused by osteoporosis or bone metastasis that do not respond to conservative care.

Is This a Primary or Secondary Test or Procedure

Secondary treatment option for painful vertebral compression fractures when conservative treatment fails.

How It Is Performed

Under local anesthesia and conscious sedation. Under continuous fluoroscopic guidance, a needle is placed into the vertebral body, and liquid PMMA is injected.

Preparation

Ensure coagulation profile is normal; stop blood thinners; obtain spinal MRI to confirm fracture acuity.

Contraindications and Precautions

Active systemic or local osteomyelitis; uncorrected coagulopathy; fracture causing severe spinal cord compression.

Risks

Cement leakage (into spinal canal or veins), pulmonary cement embolism, nerve root compression, or adjacent level fractures.

Aftercare and Recovery

Lie flat for 1-2 hours until cement hardens; monitor vital signs; assess pain relief (often immediate).

Outcome and Follow-up

Success is marked by immediate reduction in back pain and stabilized vertebral height.

Expected Results

Stable vertebral body with uniform cement distribution; significant reduction in pain; no cement leakage.

Follow-up

Re-evaluation in 1-2 weeks; bone density management (osteoporosis therapy).

References

1. MedlinePlus. Vertebroplasty. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 191

Viscosupplementation

Overview

Viscosupplementation is the injection of hyaluronic acid, a gel-like lubricant, into a joint to relieve pain and improve function in osteoarthritis. It is most commonly performed in the knee.

Alternative Names

Hyaluronic acid injection; hyaluronan injection; knee joint lubricant injection; hylan G-F 20 injection

Principle

Injected hyaluronic acid supplements the joint's naturally occurring synovial fluid, restoring its viscoelastic and shock-absorbing properties and reducing friction in the arthritic joint. It may also have anti-inflammatory and cartilage-protective effects.

History

The concept of replacing deficient joint fluid developed from research on synovial fluid in the mid-20th century, and hyaluronic acid injections for knee osteoarthritis entered clinical use in the 1980s.

Why It Is Done

Ordered for symptomatic knee osteoarthritis in patients with persistent pain who have not responded adequately to conservative measures such as exercise, weight loss, and analgesics, and who wish to delay or avoid surgery.

Is This a Primary or Secondary Test or Procedure

A secondary or adjunctive therapeutic procedure, used when first-line osteoarthritis management has been insufficient.

How It Is Performed

The joint is identified and the skin is cleaned. Under sterile technique, a needle is inserted into the joint, and one or more injections of hyaluronic acid are given, either as a single dose or as a series separated by a week, depending on the product.

Preparation

Informed consent, and the procedure is avoided if the joint shows signs of active infection or significant effusion. Aspiration of joint fluid may be performed first if an effusion is present.

Contraindications and Precautions

Contraindications include infection or skin disease over the injection site, known allergy to hyaluronan preparations, and bleeding disorders. Precautions include pregnancy, recent joint injury, and systemic anticoagulation.

Risks

Injection-site pain and swelling, transient post-injection joint pain or flare, and rarely infection or bleeding within the joint.

Aftercare and Recovery

The patient is advised to rest the joint briefly and avoid strenuous activity for a day or two. Benefit, if achieved, develops over several weeks and may last several months to a year.

Outcome and Follow-up

Clinical response is judged by pain reduction and improved function. Patients who do not improve may not benefit from further courses and may be candidates for other treatments.

Expected Results

Reduced joint pain and stiffness with improved mobility and no significant adverse reaction to the injection.

Follow-up

Patients are reassessed after the injection series, and repeated courses may be offered when symptoms recur. Failure to respond prompts consideration of corticosteroid injection, physical therapy, or joint replacement.

References

1. MedlinePlus. Viscosupplementation for the knee. U.S. National Library of Medicine; 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 192

Visual Field Test (Perimetry)

Overview

A visual field test measures how much a person can see to the sides, above, and below while looking at a fixed central target. It detects areas of reduced or missing vision that may not be noticed in daily life.

Why It Is Done

It is commonly used to diagnose and monitor glaucoma, optic-nerve disease, retinal disorders, and neurological conditions affecting the visual pathways. It may also document changes caused by medicines or other eye disease.

How It Is Performed

One eye is tested at a time while the patient looks at a central point inside a bowl-shaped instrument. Small lights appear in different locations, and the patient presses a button whenever a light is seen. The test is repeated or adjusted if attention, fatigue, or learning affects reliability.

Preparation and Results

Wear usual glasses unless instructed otherwise and bring information about eye conditions and medicines. Results are shown as a map of sensitivity and are interpreted with the eye examination, intraocular pressure, and previous tests. A reliable test may require good attention and steady fixation.

Risks and Limitations

The test is noninvasive and painless. Temporary eye fatigue or frustration can occur. Poor concentration, blurred vision, uncorrected refractive error, or eyelid obstruction can produce an apparently abnormal result.

References

1. National Eye Institute. Visual field test. 2025.

2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 193

Wound Debridement

Overview

Wound debridement is the medical removal of dead, damaged, or infected tissue to improve the healing potential of the remaining healthy tissue.

Alternative Names

Surgical debridement, Necrotic tissue removal

Principle

Dead, infected, or foreign tissue is removed from a wound by surgical cutting, enzymes, or other means, leaving a clean viable bed. Removing necrotic tissue eliminates the medium for bacterial growth and allows granulation and healing to proceed.

History

Developed in military surgery. Ambroise Pare in the 16th century advocated against cauterizing wounds and for cleaning devitalized tissue.

Why It Is Done

Ordered for chronic, non-healing wounds, diabetic foot ulcers, pressure sores, or heavily contaminated traumatic wounds.

Is This a Primary or Secondary Test or Procedure

Primary treatment for chronic wounds and necrotizing soft-tissue infections.

How It Is Performed

Can be performed surgically (using scalpel/scissors to cut away dead tissue), enzymatically (using chemical ointments), or autolytically (using special dressings).

Preparation

Administer local or systemic pain medication. Irrigate the wound.

Contraindications and Precautions

Severe arterial insufficiency (where dry gangrene should be left intact until revascularization is achieved).

Risks

Pain, bleeding, damage to underlying structures (tendons, nerves), or introducing infection if sterile technique is breached.

Aftercare and Recovery

Apply a sterile dressing designed for wound healing (alginate, hydrogel). Monitor wound dimensions and drainage.

Outcome and Follow-up

Success is marked by a clean, red wound bed composed of healthy granulation tissue with no slough or eschar.

Expected Results

Clean wound bed containing healthy red granulation tissue; no purulent odor or necrotic tissue.

Follow-up

Frequent dressing changes, regular wound measurements, and weekly debridement sessions as needed.

References

1. MedlinePlus. Wound Debridement. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.

Chapter 194

Wound Suturing

Overview

Wound suturing involves stitching the edges of a laceration or surgical incision together to promote healing by primary intention, minimize scarring, and reduce infection risk.

Alternative Names

Suture repair, Stitches

Principle

The skin edges of a clean wound are approximated with sutures that hold the tissue layers in alignment during healing. Closure reduces wound tension, controls bleeding, and restores tissue continuity, promoting healing by primary intention while minimizing scarring and infection risk.

History

Suturing was practiced in ancient Egypt using plant fibers and hair. Joseph Lister introduced sterile catgut sutures in the 1860s. Synthetic polymers were introduced in the 20th century.

Why It Is Done

Ordered to close open lacerations or wounds that are deep, have gaping edges, or are bleeding persistently.

Is This a Primary or Secondary Test or Procedure

Primary definitive treatment for skin lacerations requiring primary closure.

How It Is Performed

The wound is irrigated and cleaned. Local anesthetic is injected. The clinician uses a needle holder, suture material, and tissue forceps to sew the wound edges together.

Preparation

Irrigate the wound thoroughly with normal saline. Debride devitalized tissue if necessary.

Contraindications and Precautions

Wounds that are heavily contaminated, showing signs of active infection, or presenting >12-24 hours after injury (which should heal by secondary intention).

Risks

Wound infection, dehiscence (opening of the wound), scarring, or local nerve injury.

Aftercare and Recovery

Apply antibiotic ointment and a sterile dressing. Keep the wound clean and dry for 24-48 hours.

Outcome and Follow-up

Successful closure shows aligned skin edges, minimal tension, and absence of active bleeding or infection.

Expected Results

Well-approximated wound edges; no active bleeding; minimal swelling.

Follow-up

Suture removal in 5-7 days (face), 7-10 days (scalp/trunk), or 10-14 days (extremities).

References

1. MedlinePlus. Wound Suturing. U.S. National Library of Medicine; 2025.
2. Current Medical Diagnosis and Treatment. McGraw-Hill; 2025.